Before and after

The Discrete Algebra of the Fourier Transform

Gabriel Peyré — September 9, 2026

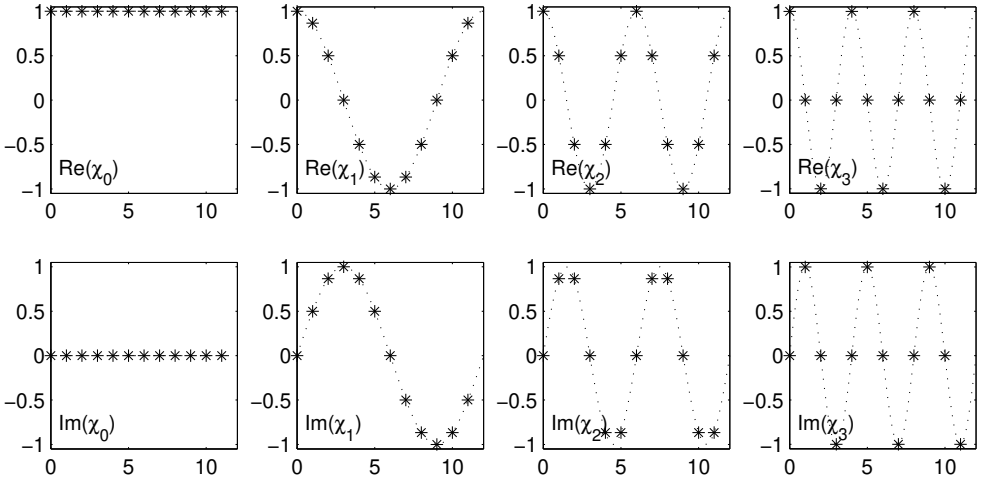
The left panels reproduce the previous English edition. The right panels are regenerated from Python or TikZ and used in the revised book. Figure numbers and captions below are read from the compiled book and its source. The original number is also recorded when needed.

Figure 1.1

Real and imaginary parts of the first four characters of $\mathbb{Z}/12\mathbb{Z}$.

Book page 3; original figure 1.1.

Previous edition



Revised edition

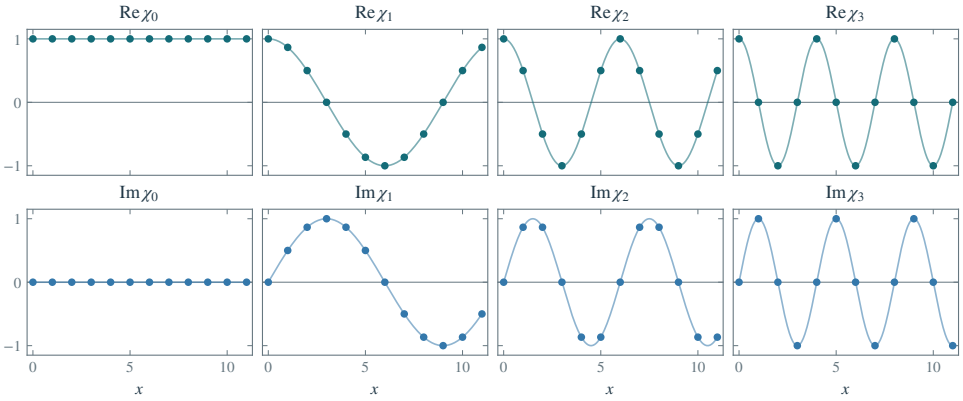
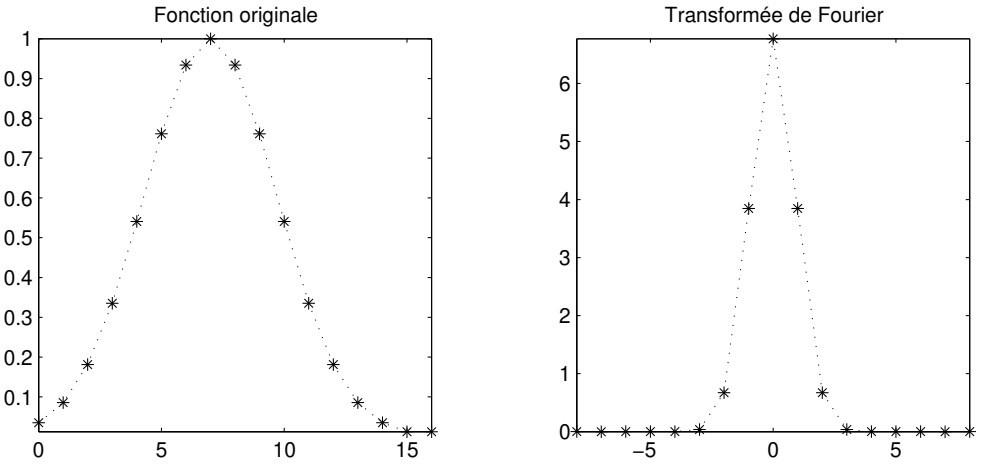


Figure 1.2

An even bell-shaped function on $\mathbb{Z}/17\mathbb{Z}$ and its real Fourier coefficients, with centered indices.

Book page 9; original figure 1.2.

Previous edition



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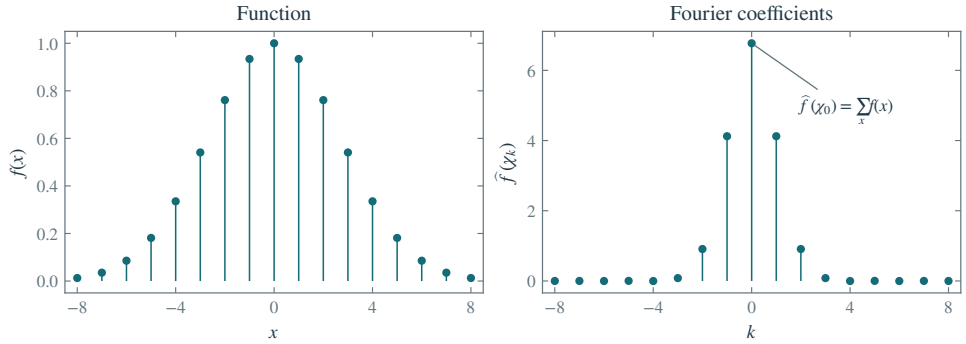
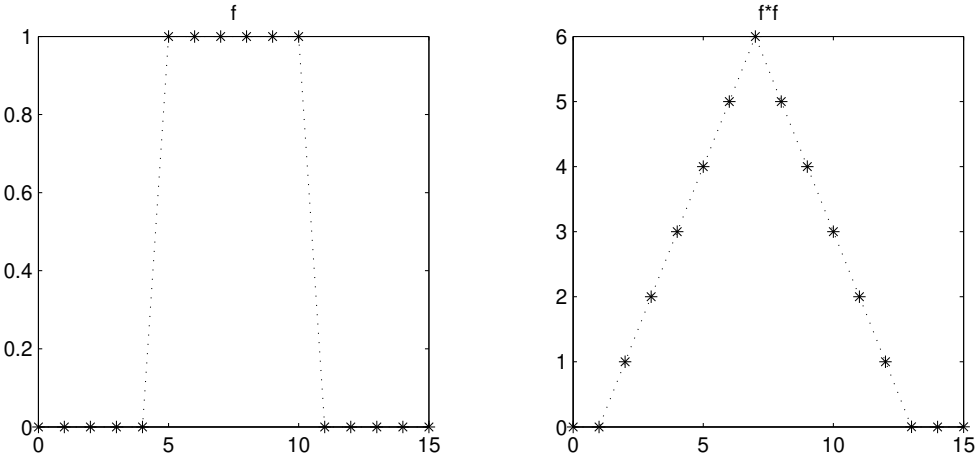


Figure 1.3

Circular self-convolution of $f = \mathbf{1}_{\{5,\ldots,10\}}$ on $\mathbb{Z}/16\mathbb{Z}$. The maximum occurs at 15 modulo 16; the remaining values wrap around at 0.

Book page 10; original figure 1.3.

Previous edition



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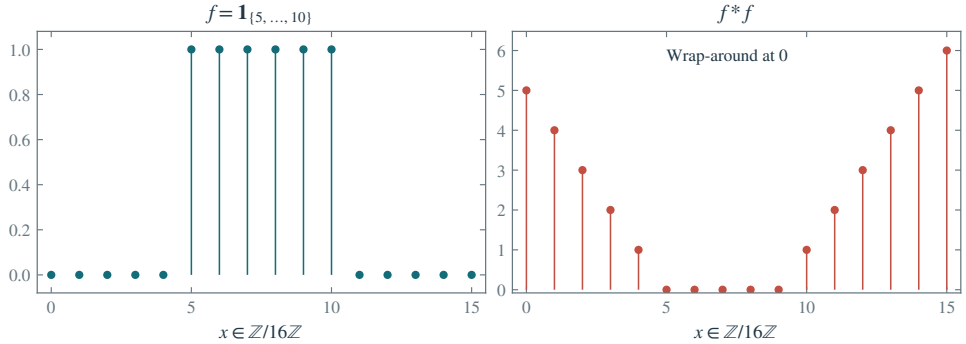
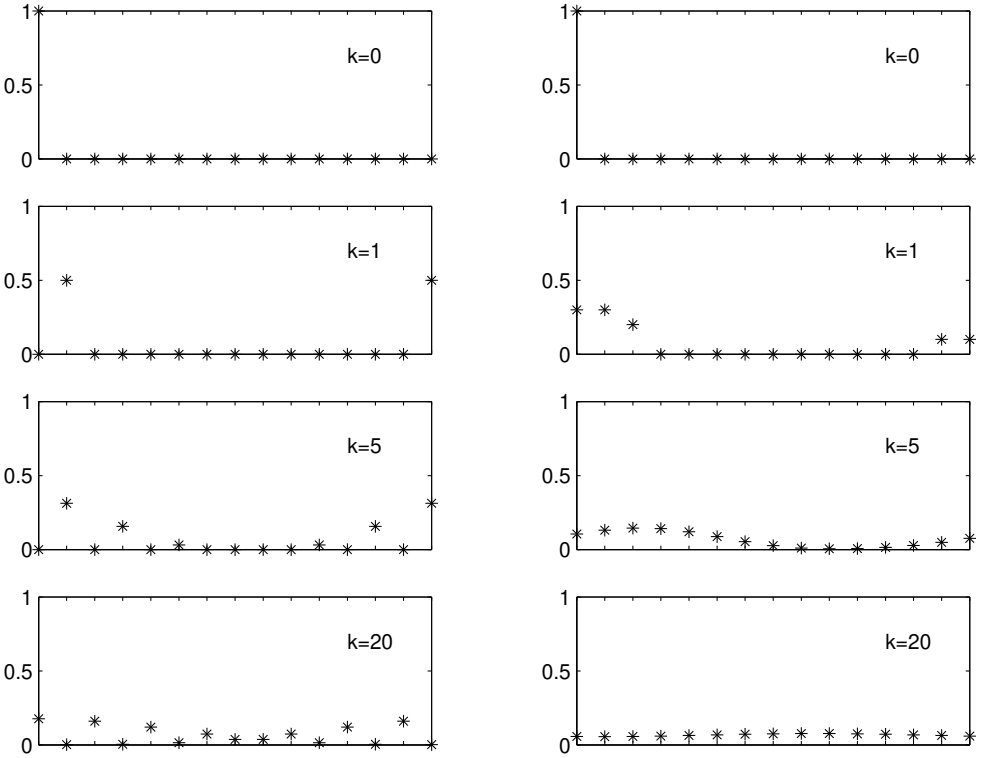


Figure 1.4

Random walks on $\mathbb{Z}/15\mathbb{Z}$ with increment laws $(0, 1/2, 0, \ldots, 0, 1/2)$ and $(0.3, 0.3, 0.2, 0, \ldots, 0, 0.1, 0.1)$. Dashed lines show the uniform distribution.

Book page 15; original figure 1.4.

Previous edition



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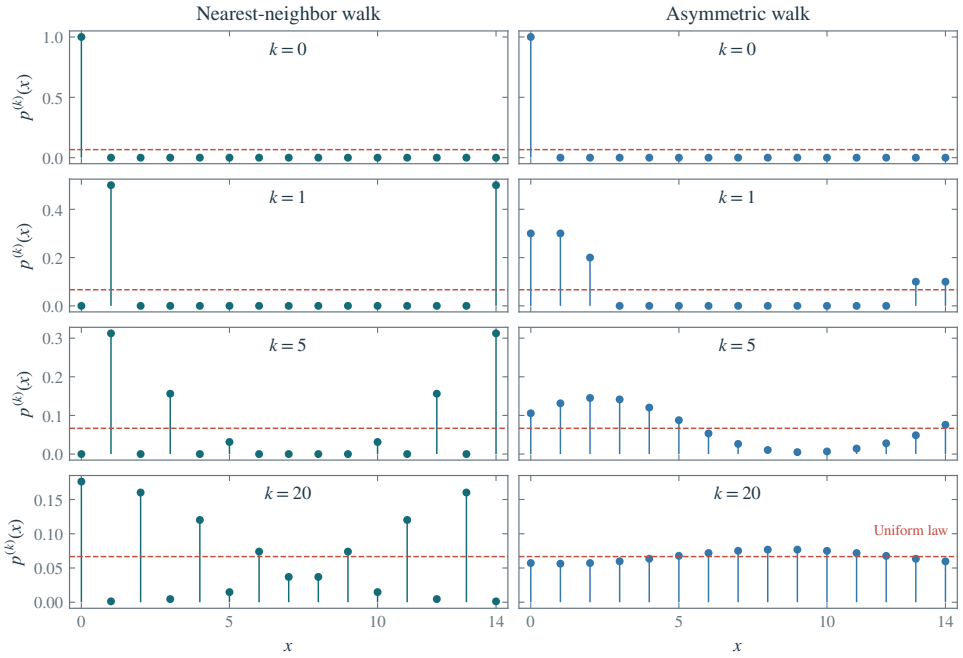
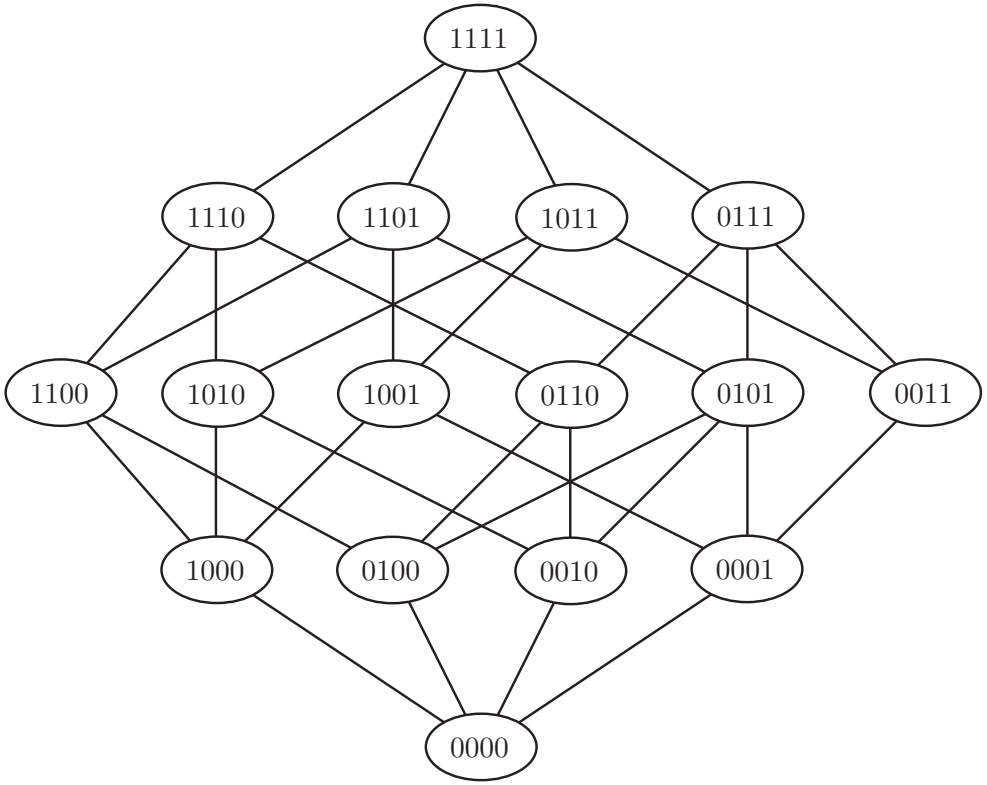


Figure 2.1

The Boolean cube $\mathbb{F}_2^4$, arranged by Hamming weight. Each edge changes exactly one coordinate.

Book page 27; original figure 2.1.

Previous edition



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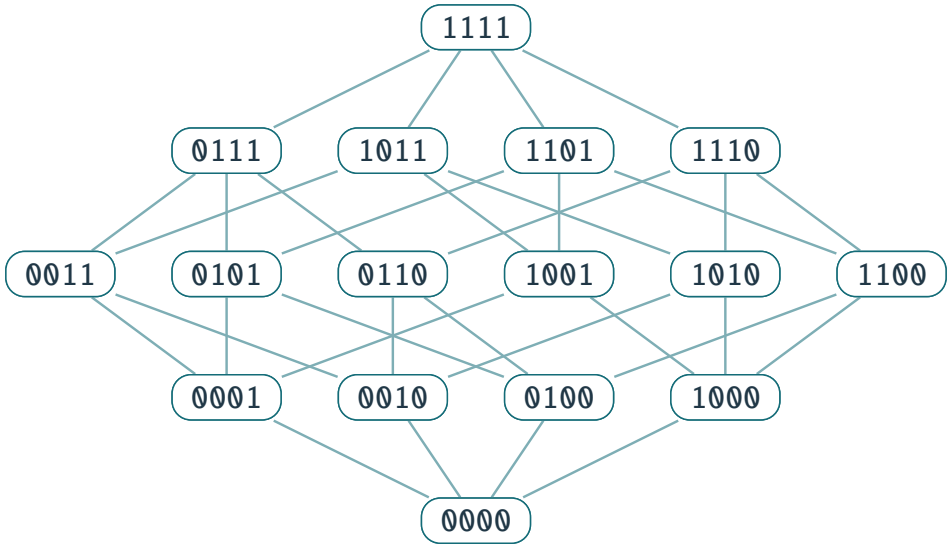
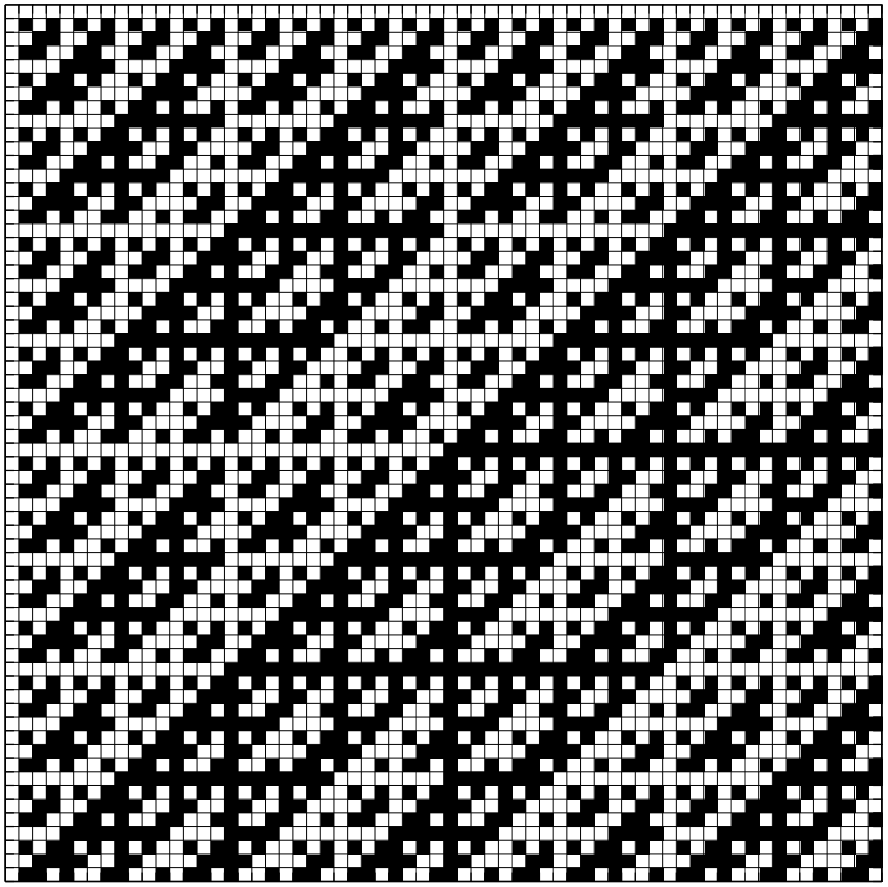


Figure 2.2

The Walsh matrix W_{64} in natural order. White represents +1 and dark teal represents −1.

Book page 28; original figure 2.2.

Previous edition



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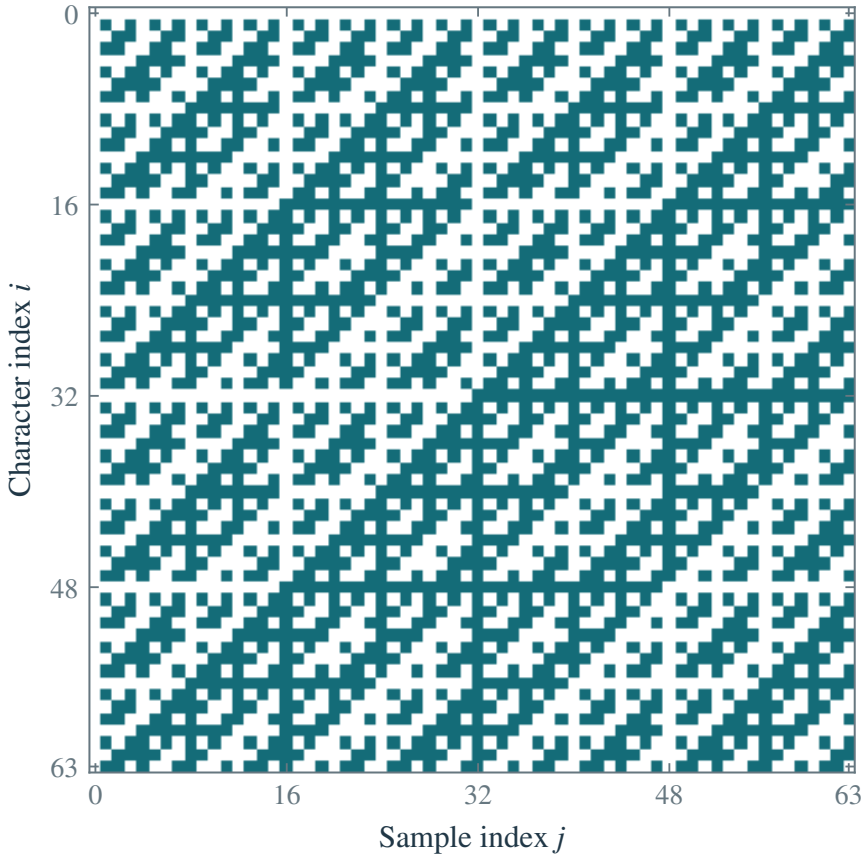
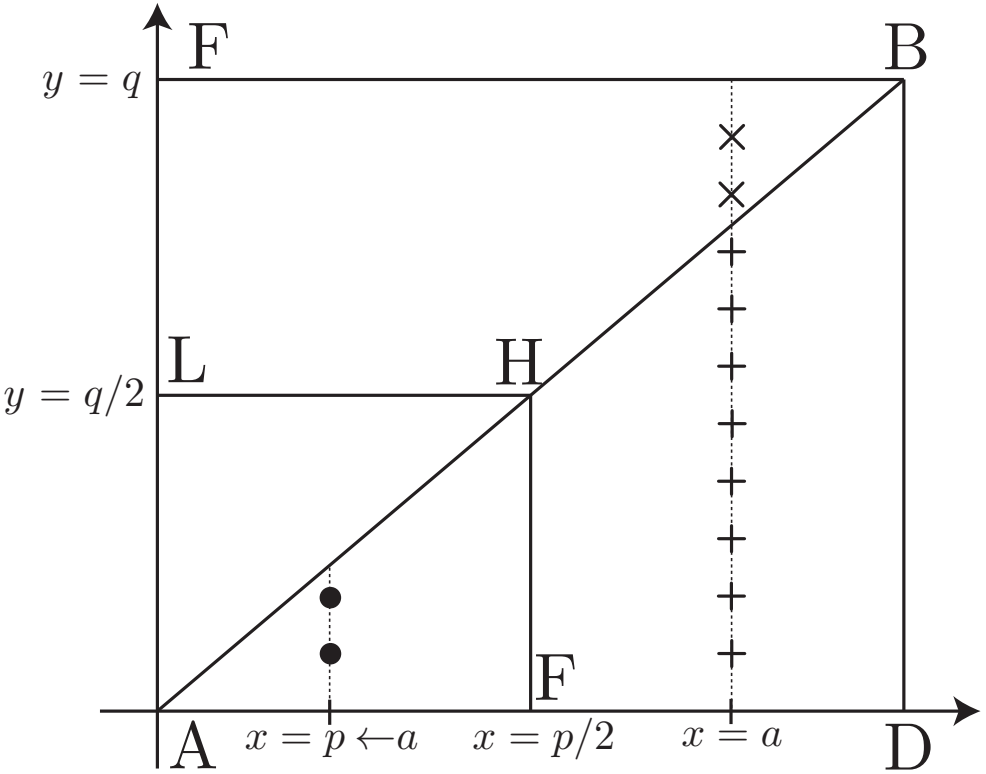


Figure 2.3

Lattice-point pairing in the geometric proof of quadratic reciprocity, illustrated with $p = 13$, $r = 11$, and $a = 10$.

Book page 34; original figure 2.3.

Previous edition



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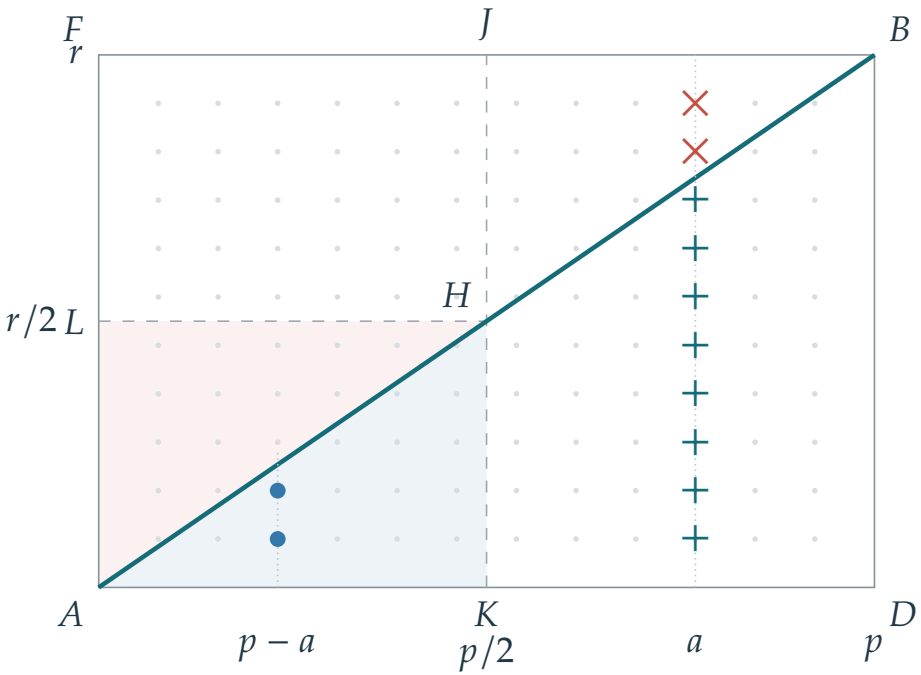
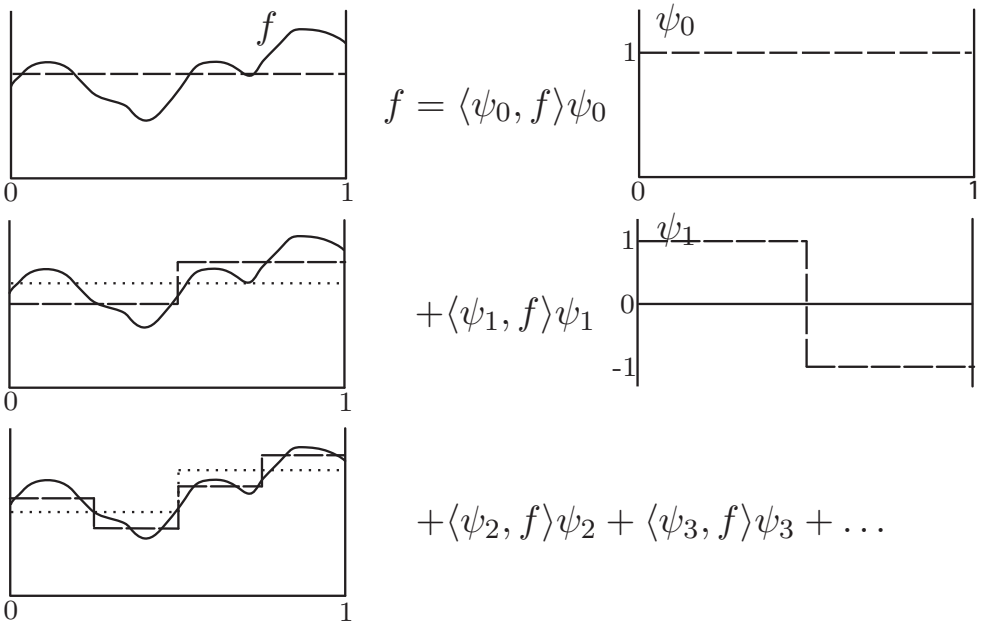


Figure 2.4

Successive Haar projections onto F_0, F_1, and F_2 (left), and the orthonormal basis functions added at each stage (right).

Book page 35; original figure 2.4.

Previous edition



Revised edition

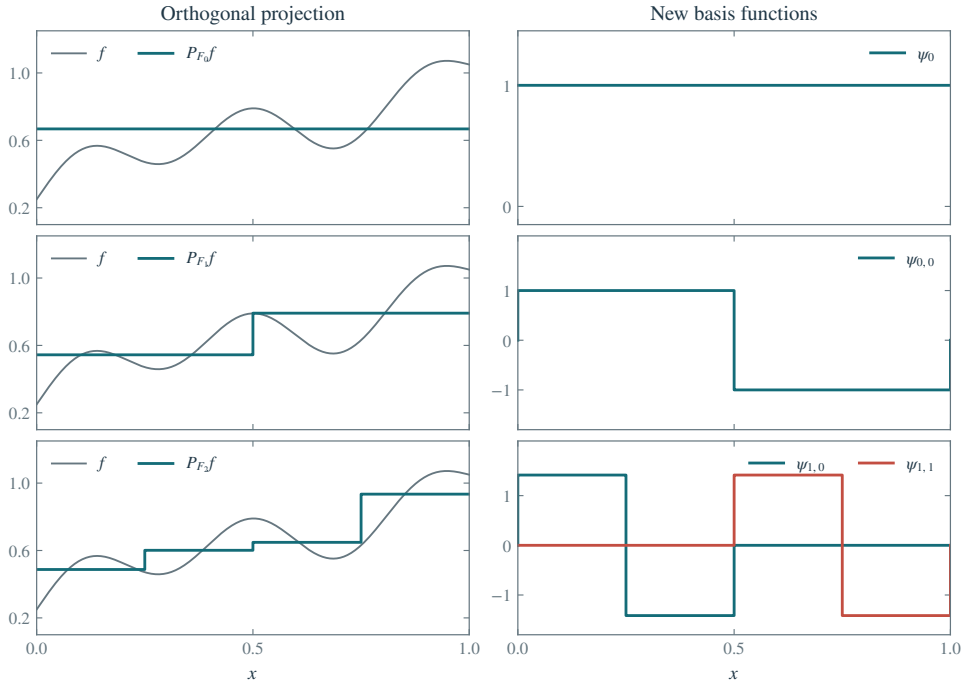
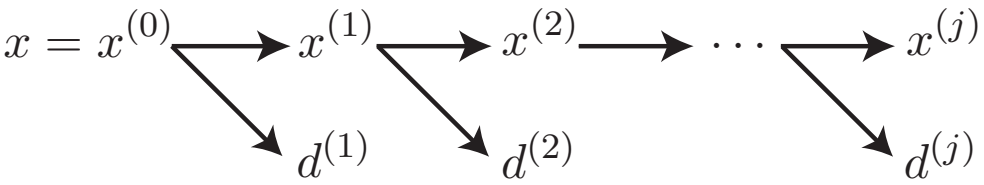


Figure 2.5

The Haar cascade: each stage separates scaling coefficients from details, which are stored from fine to coarse.

Book page 36; original figure 2.5.

Previous edition



Revised edition

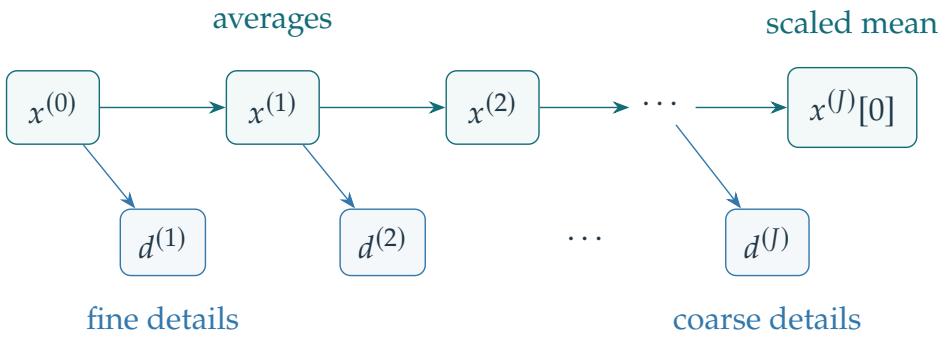
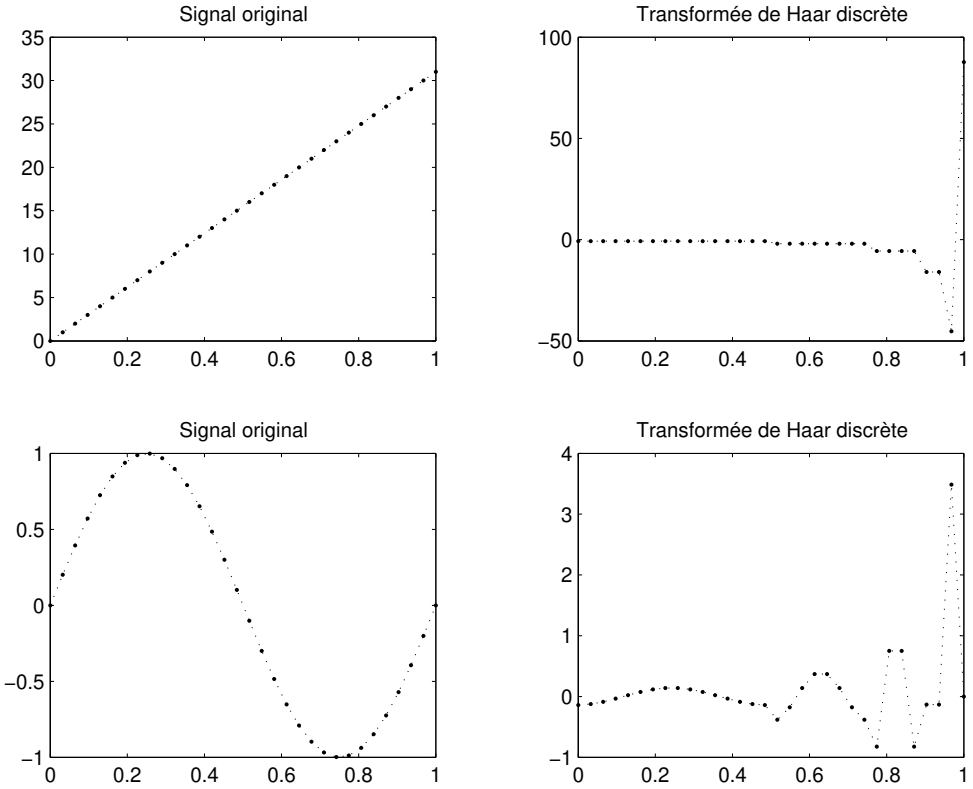


Figure 2.6

Two length-32 signals and their orthonormal Haar coefficients. Colored separators mark the detail scales; the final coefficient is the scaled mean.

Book page 37; original figure 2.6.

Previous edition



Revised edition

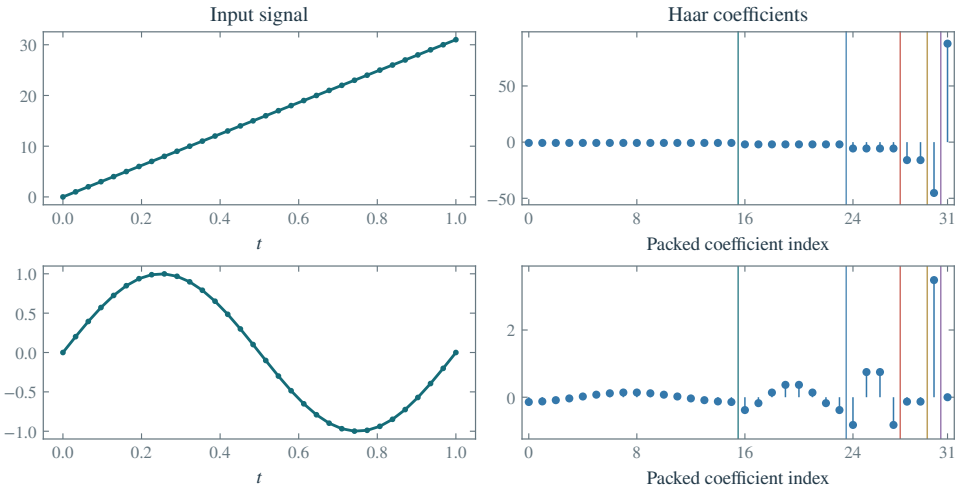
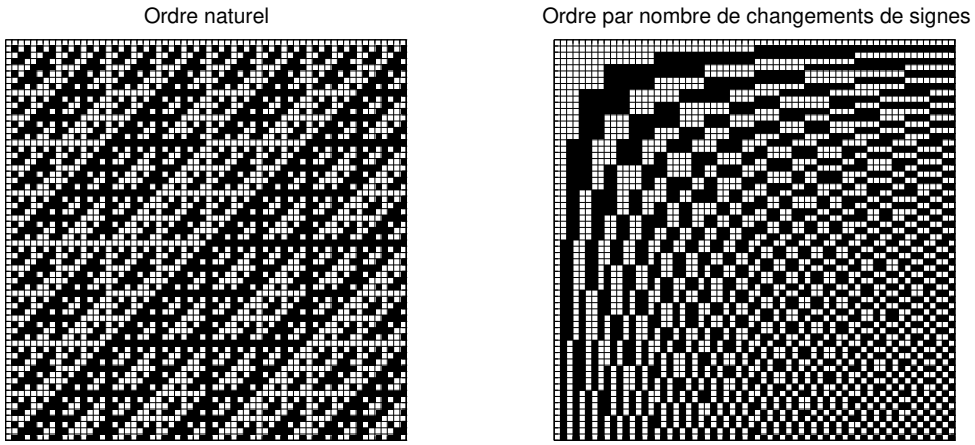


Figure 2.7

Walsh functions in natural order and in increasing sequency, with one basis function per row.

Book page 37; original figure 2.7.

Previous edition



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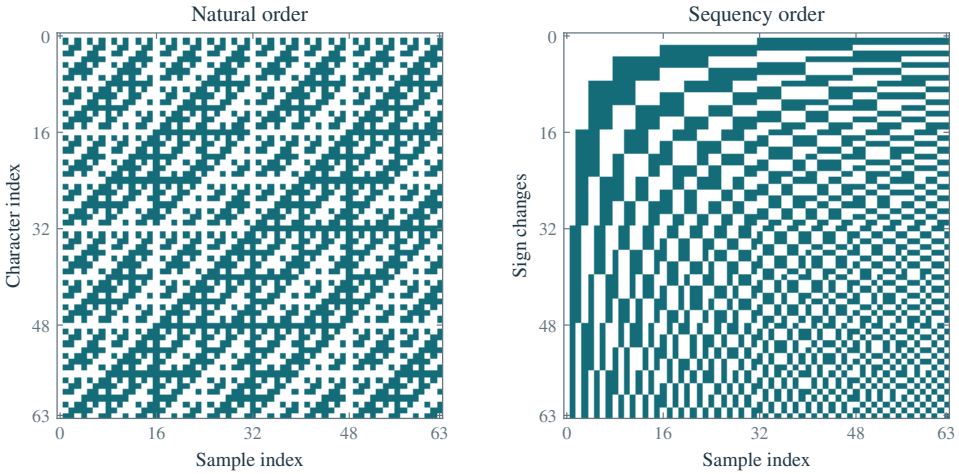
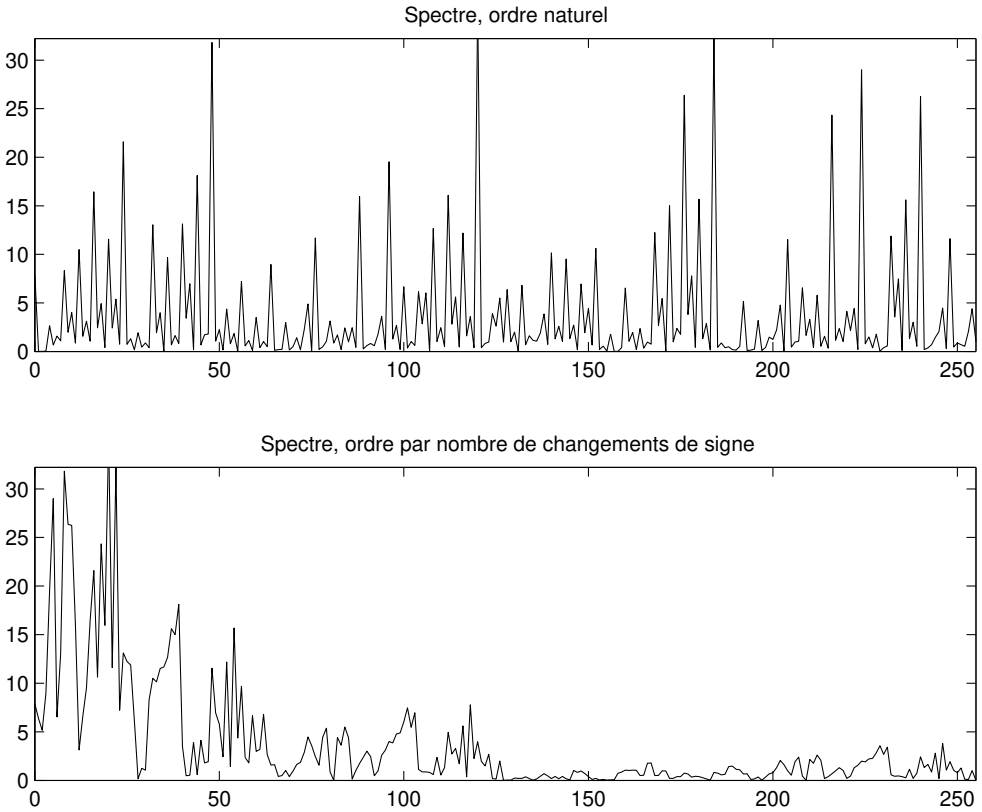


Figure 2.8

The Walsh spectrum of the oscillatory signal in Figure 2.9, in natural and sequency order.

Book page 38; original figure 2.8.

Previous edition



Revised edition

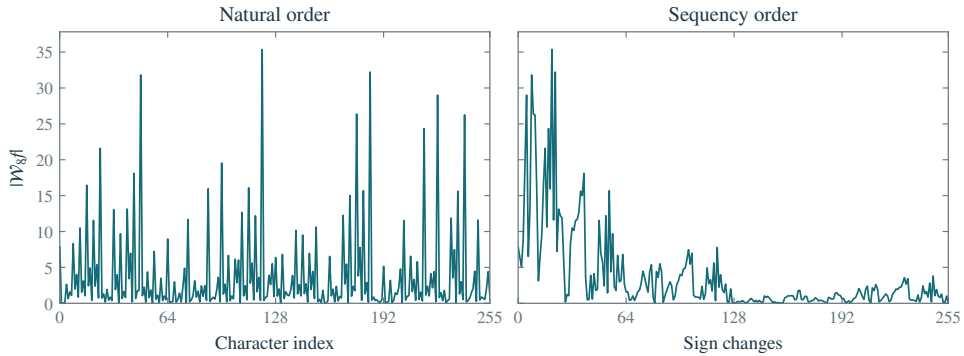
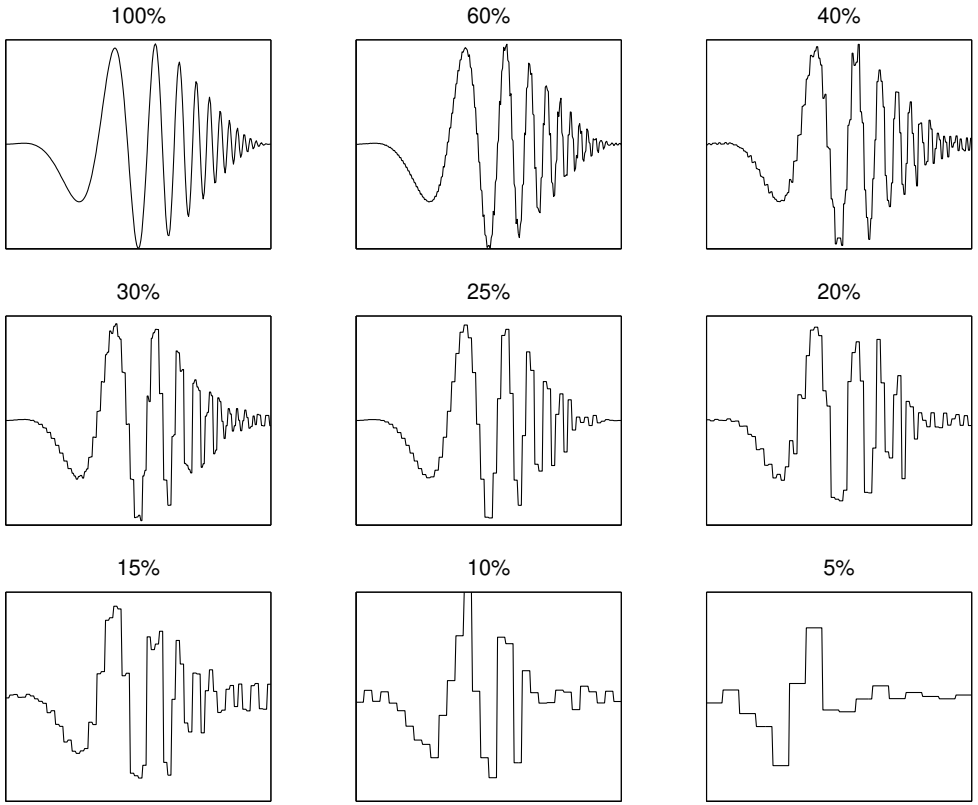


Figure 2.9

Progressive approximation of a signal by initial segments of the sequency-ordered Walsh basis. Each panel gives the exact retained coefficient count.

Book page 38; original figure 2.9.

Previous edition



Revised edition

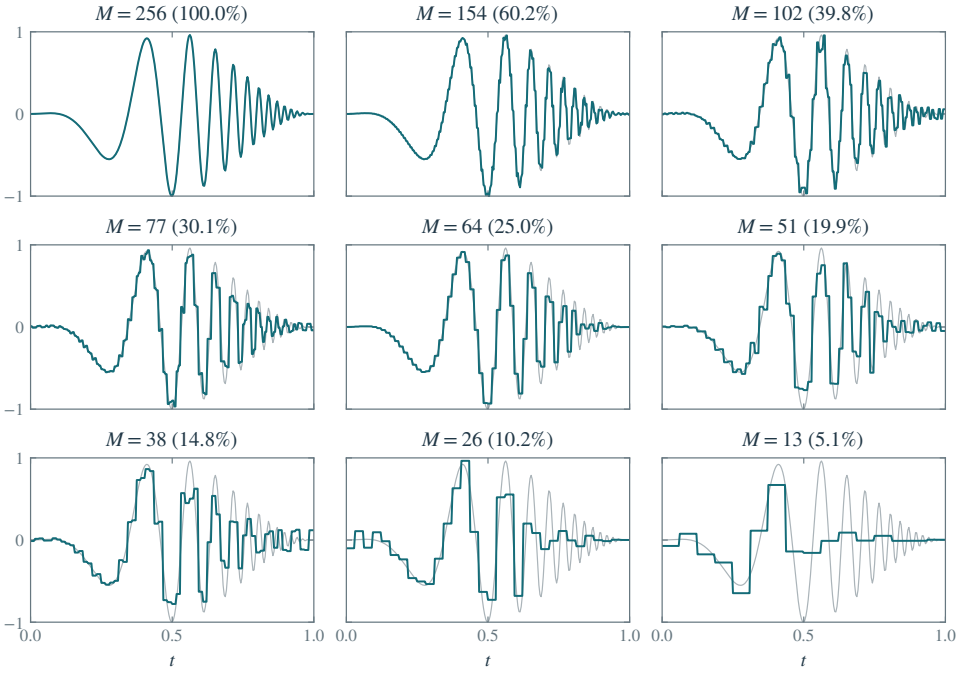
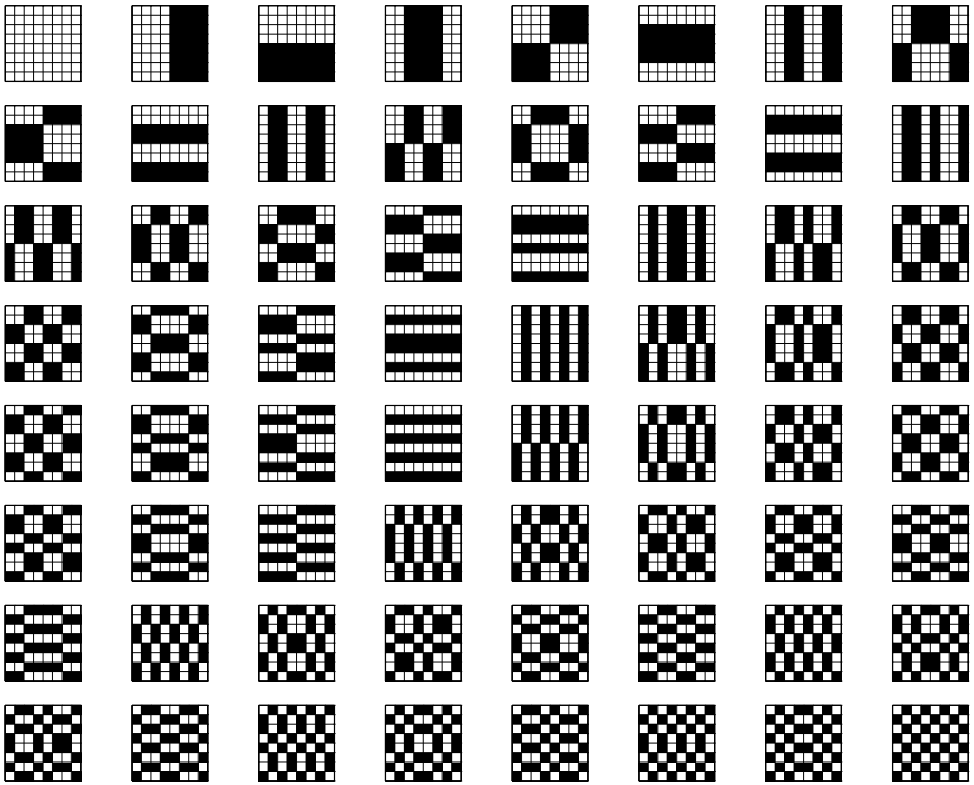


Figure 2.10

The 64 two-dimensional Walsh functions for 8 × 8 images, ordered by total sequency and then vertical sequency, from left to right and top to bottom.

Book page 39; original figure 2.10.

Previous edition



Revised edition

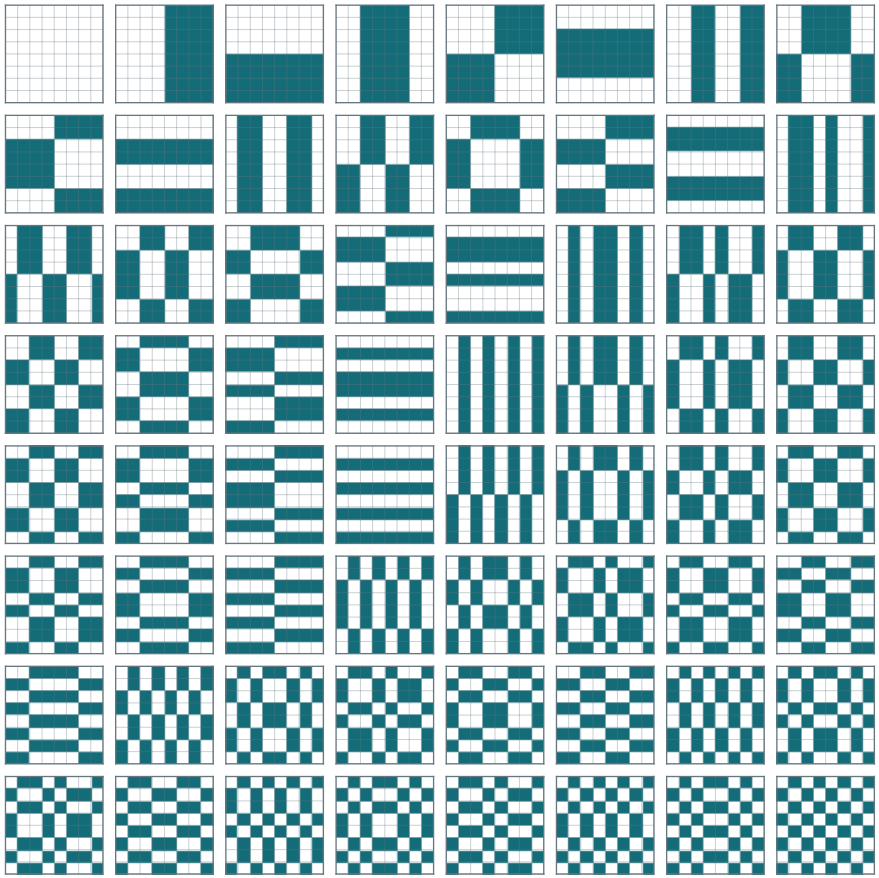


Figure 2.11

Walsh approximations of a synthetic letter A, with exact retained coefficient counts. Every panel uses the same grayscale range.

Book page 40; original figure 2.11.

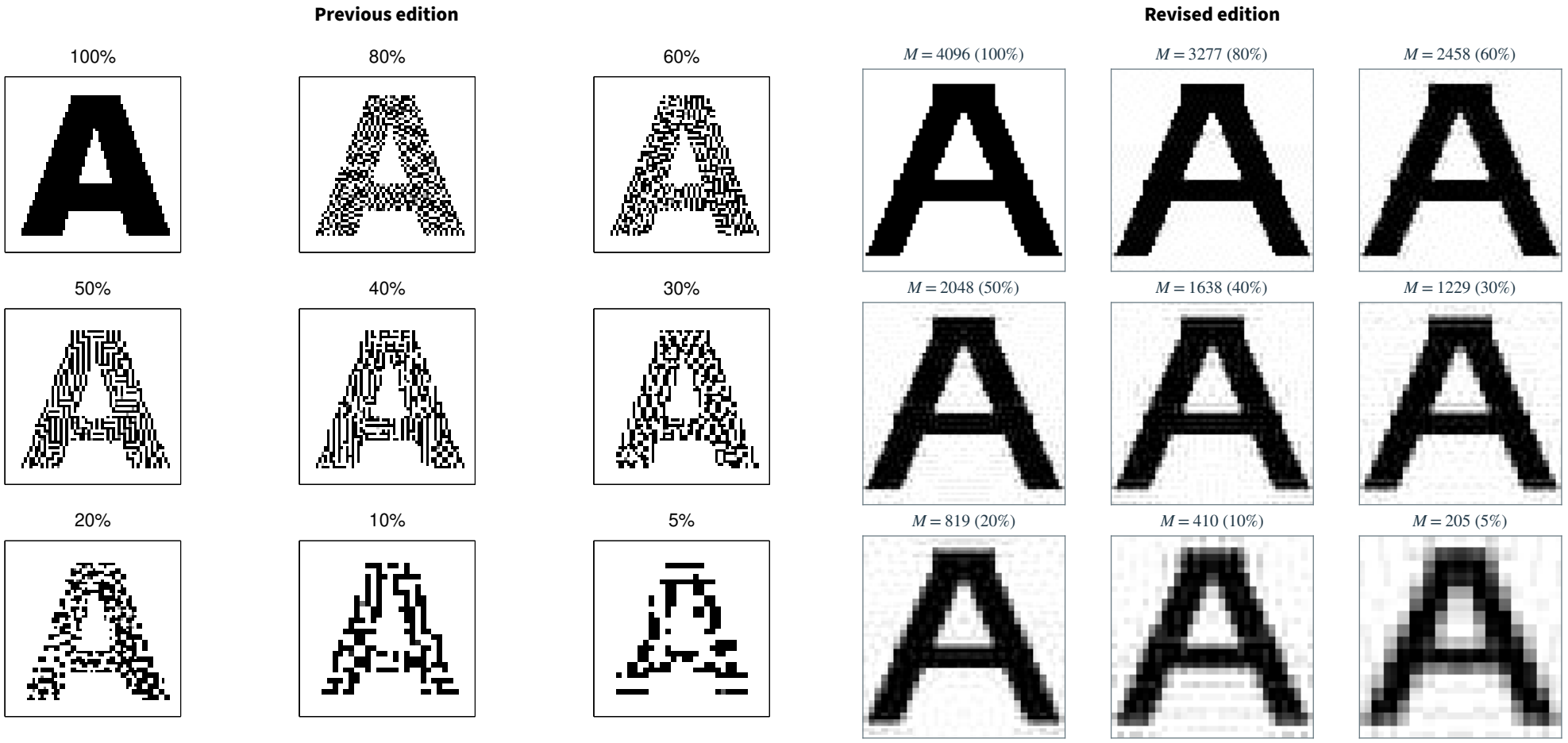
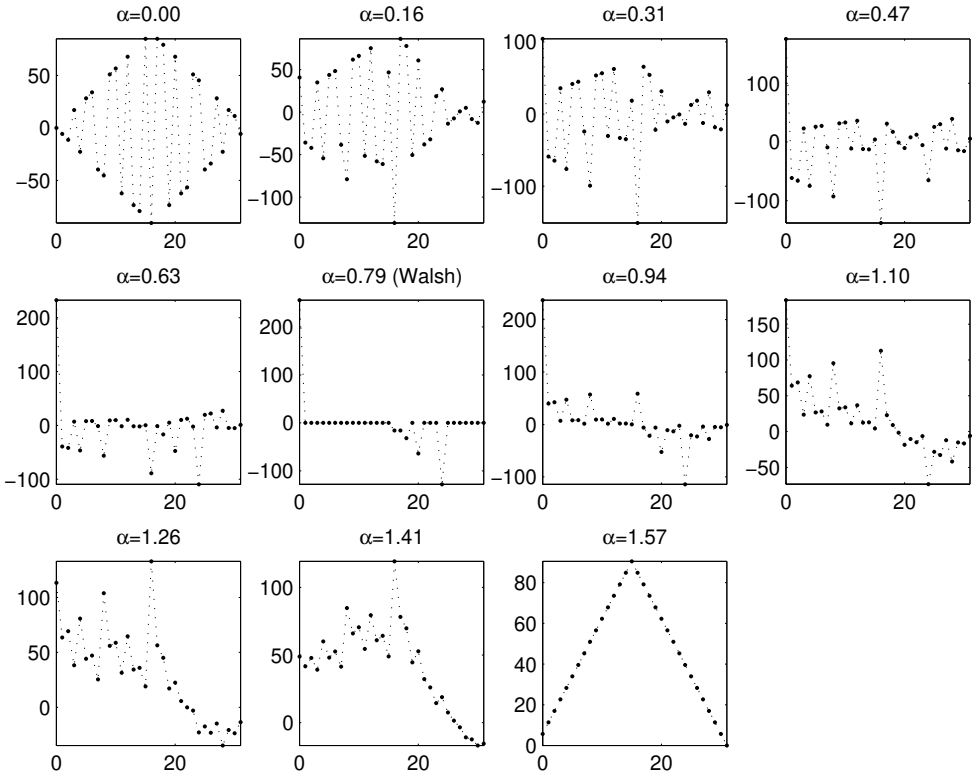


Figure 2.12

Normalized tensor transforms $2^{-n/2}A_\alpha^{\otimes n}$ of a triangular signal, with $n = 5$. The middle panel is the Walsh transform; the last reverses the input entries.

Book page 41; original figure 2.12.

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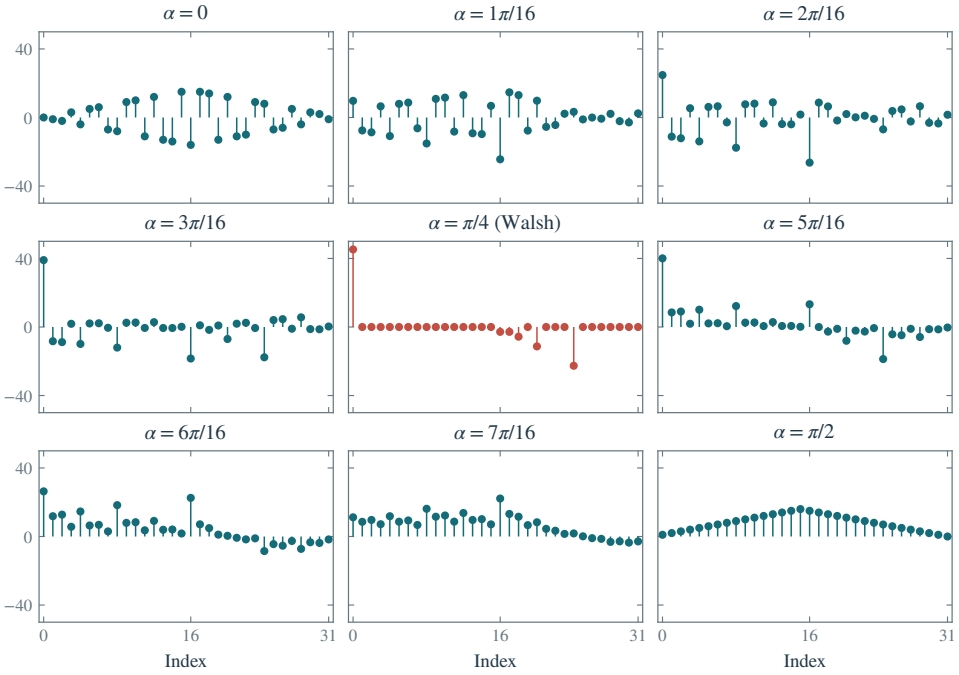
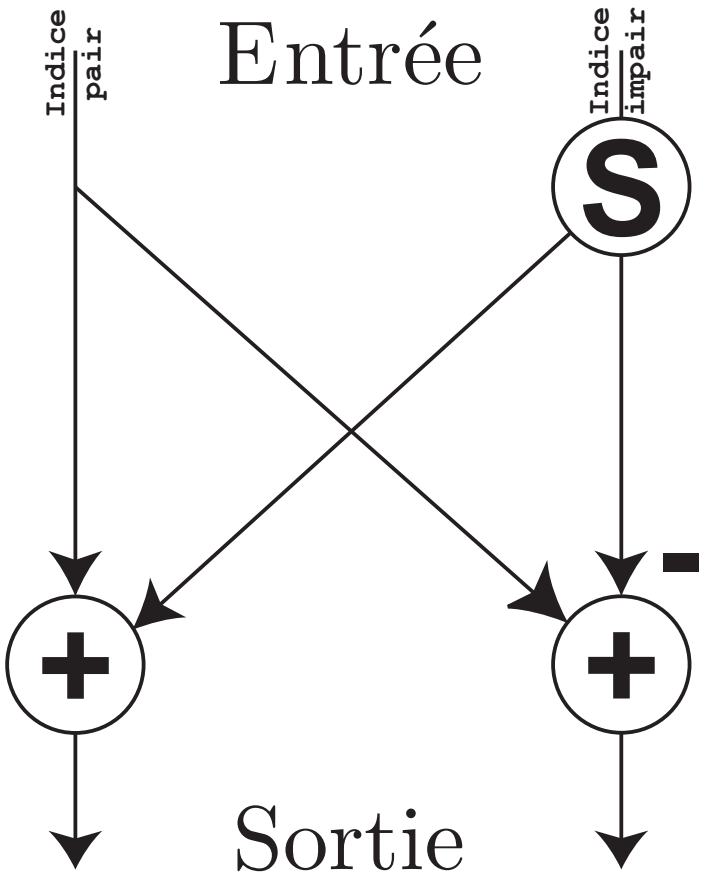


Figure 3.1

An FFT butterfly: the same weighted value wb enters the sum and the difference, with $w = \omega_N^{-k}$.

Book page 52; original figure 3.1.

Previous edition



Revised edition

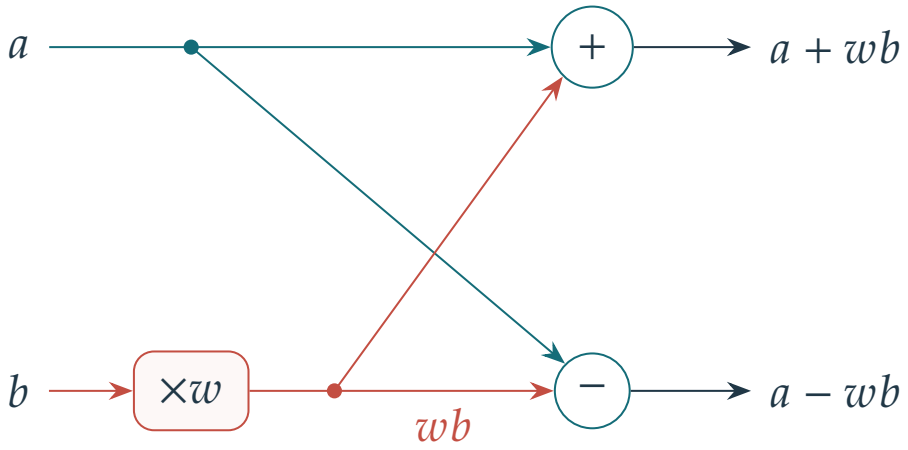
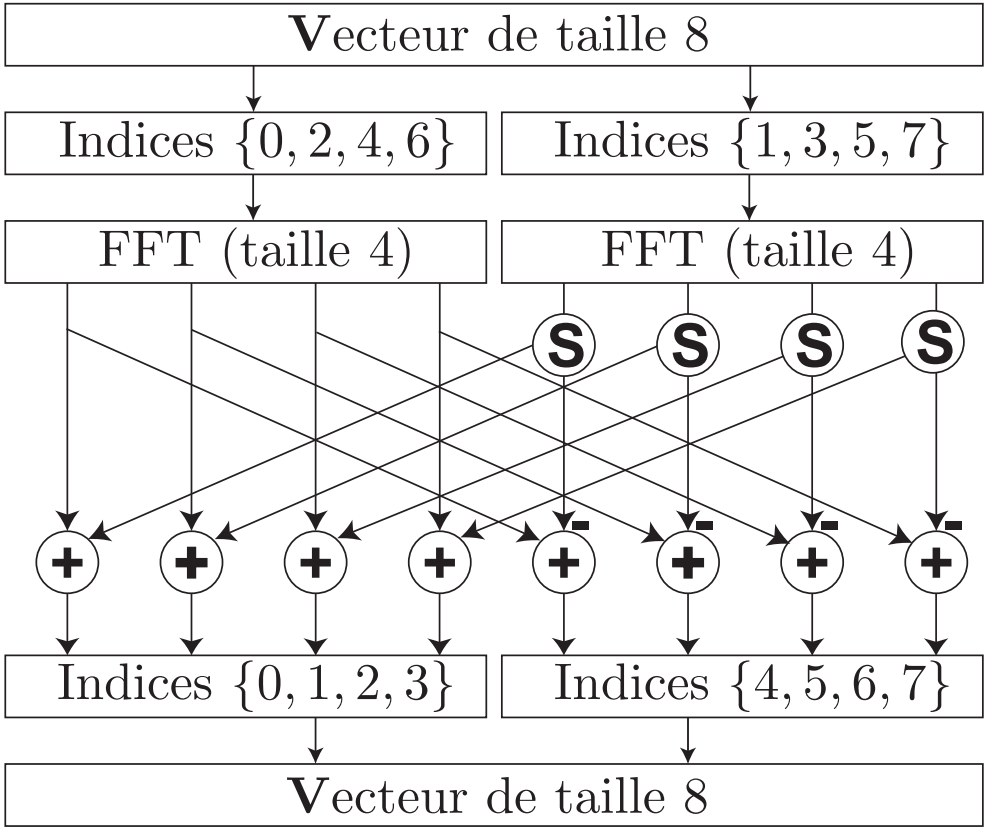


Figure 3.2

The combination stage of an eight-point DIT FFT. Two four-point transforms produce E_k and O_k; each butterfly combines E_k with $\omega_8^{-k} O_k$.

Book page 53; original figure 3.2.

Previous edition



Revised edition

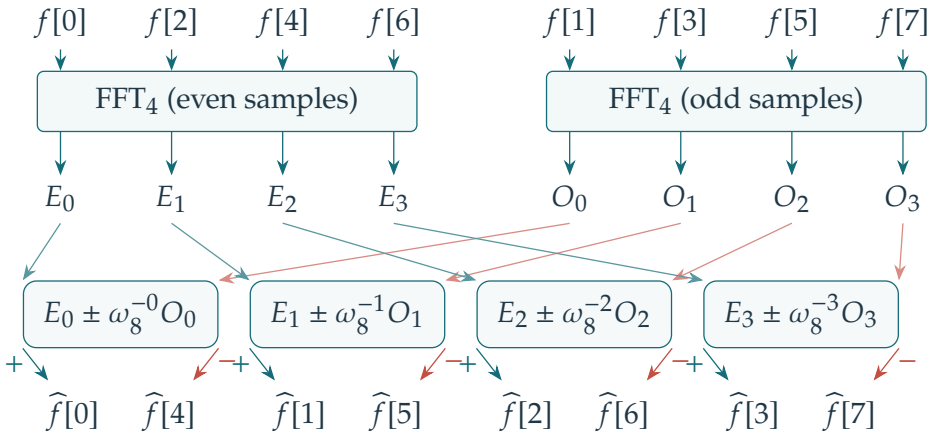
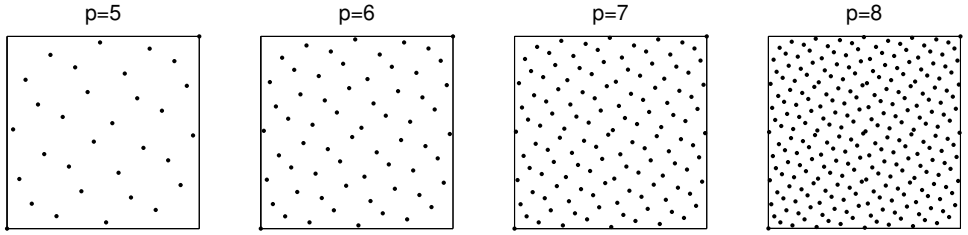


Figure 3.3

Bit-reversal permutation matrices for $N = 32, 64, 128, 256$. A dot in row i and column j means that the bits of j are those of i in reverse order.

Book page 54; original figure 3.3.

Previous edition



Revised edition

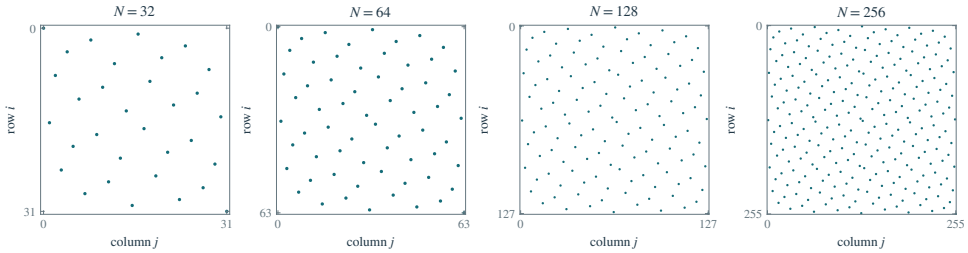
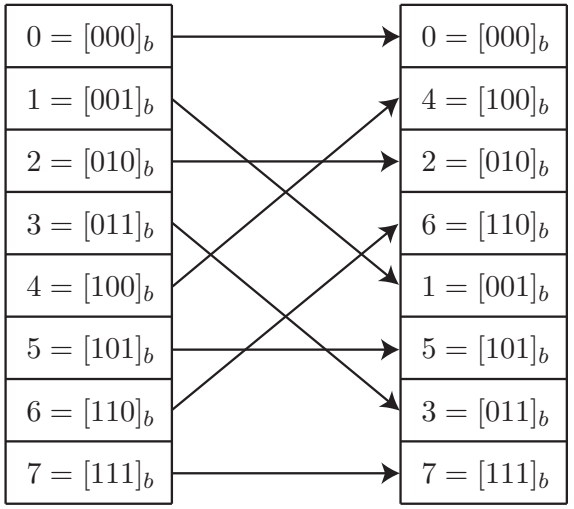


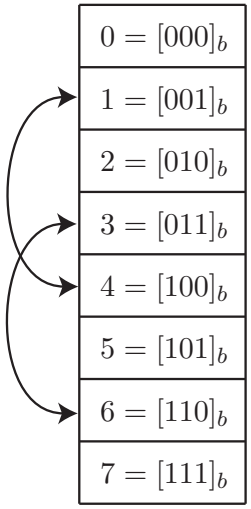
Figure 3.4

Bit reversal on eight entries. The binary labels make the swaps $1 \leftrightarrow 4$ and $3 \leftrightarrow 6$ explicit; the other four indices are fixed.

Book page 55; original figure 3.4.

Previous edition





Revised edition

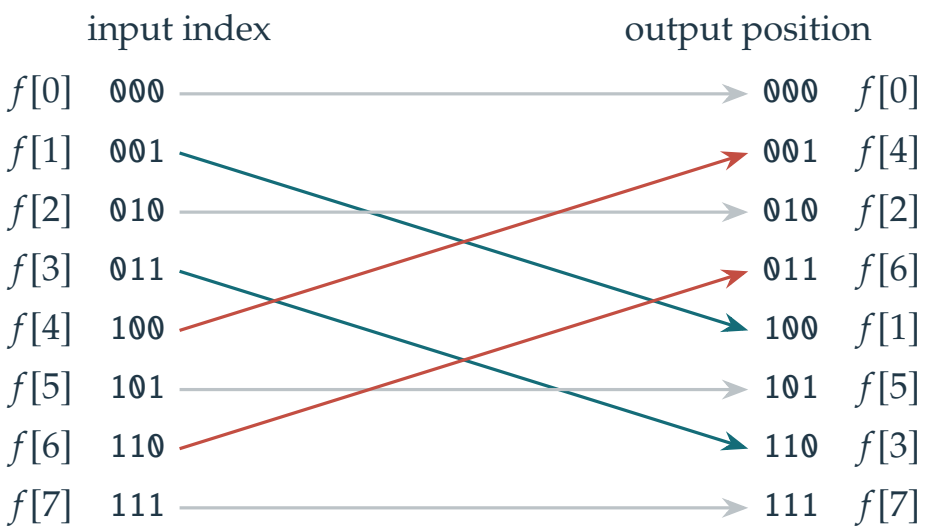
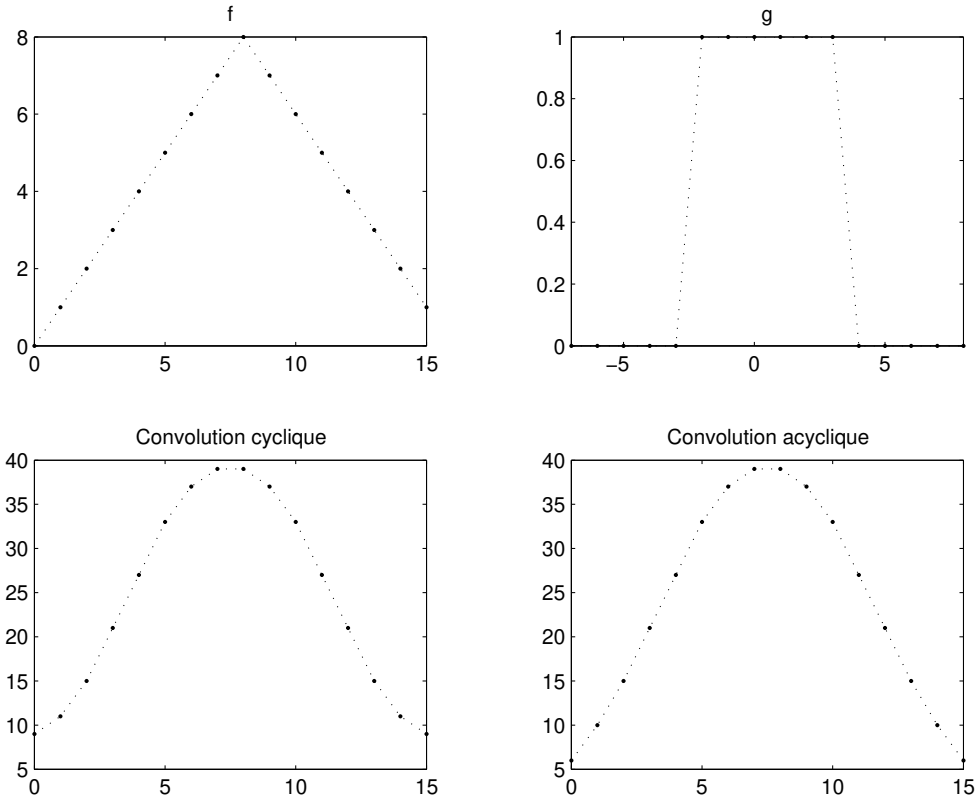


Figure 3.5

Circular and zero-extended linear convolution of a triangular signal with a unit-sum five-tap box filter. The outputs agree away from the endpoints.

Book page 57; original figure 3.5.

Previous edition



Revised edition

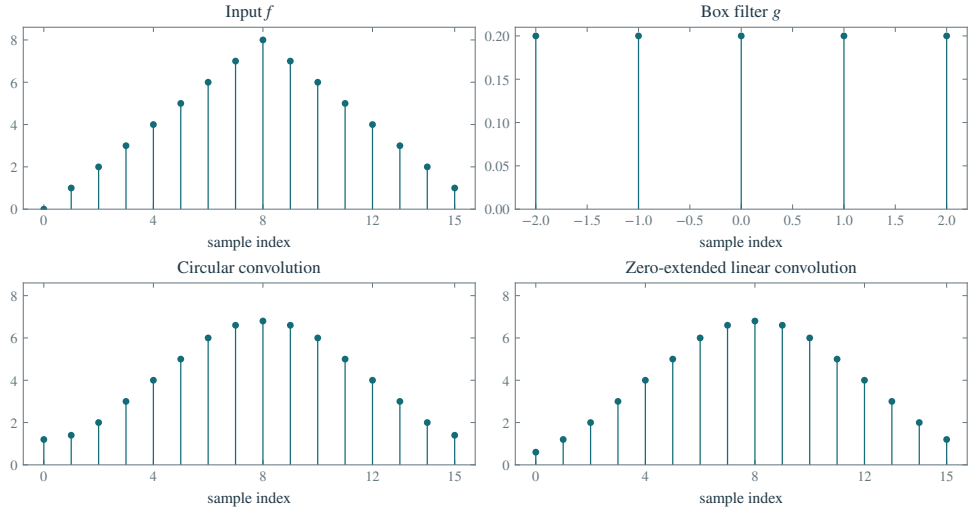
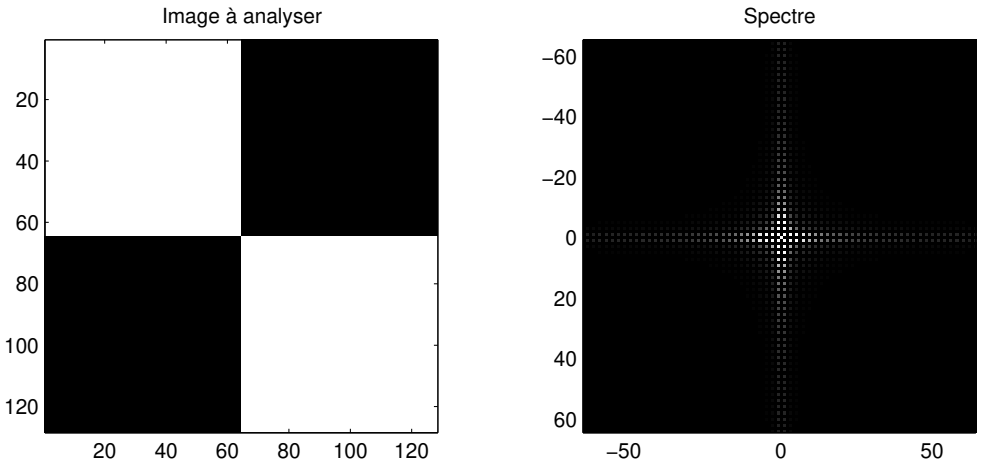


Figure 3.6

A two-dimensional checker pattern and its centered Fourier magnitude. The logarithmic color scale reveals the weaker frequencies.

Book page 59; original figure 3.6.

Previous edition



Revised edition

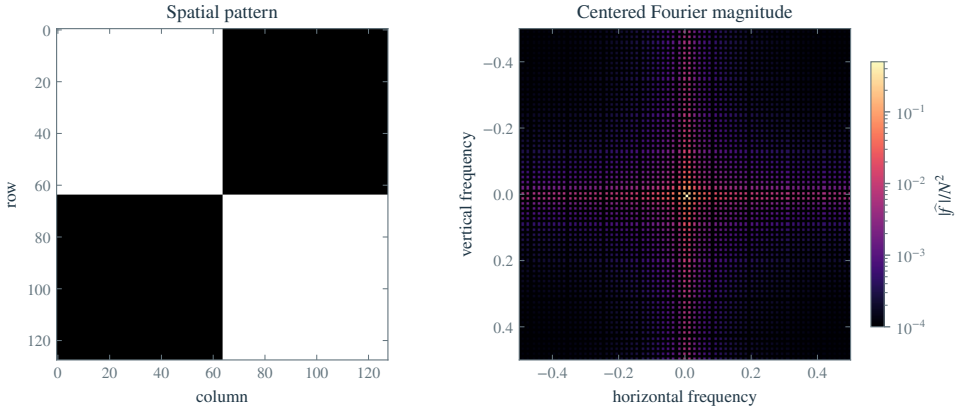
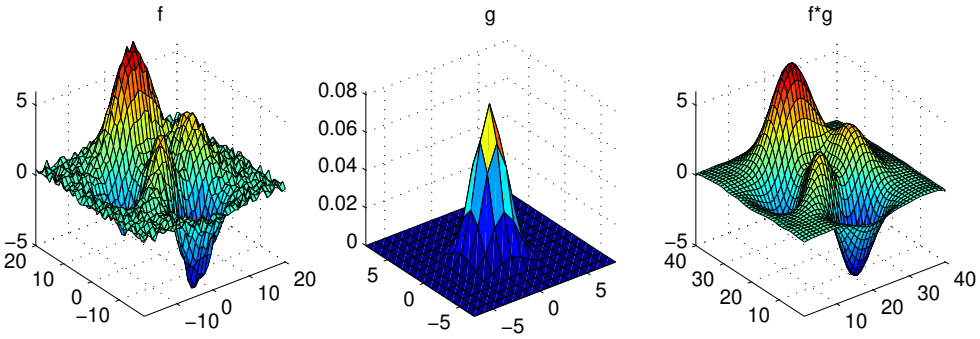


Figure 3.7

Circular Gaussian smoothing of a noisy surface. The discrete kernel has unit sum; its center is shifted to index $(0, 0)$ before the FFT.

Book page 59; original figure 3.7.

Previous edition



Revised edition

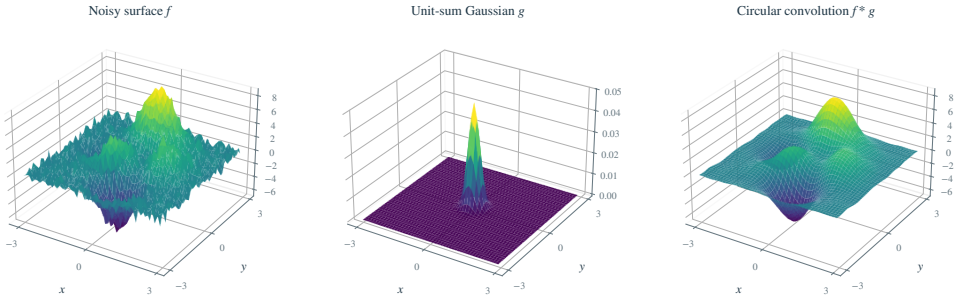
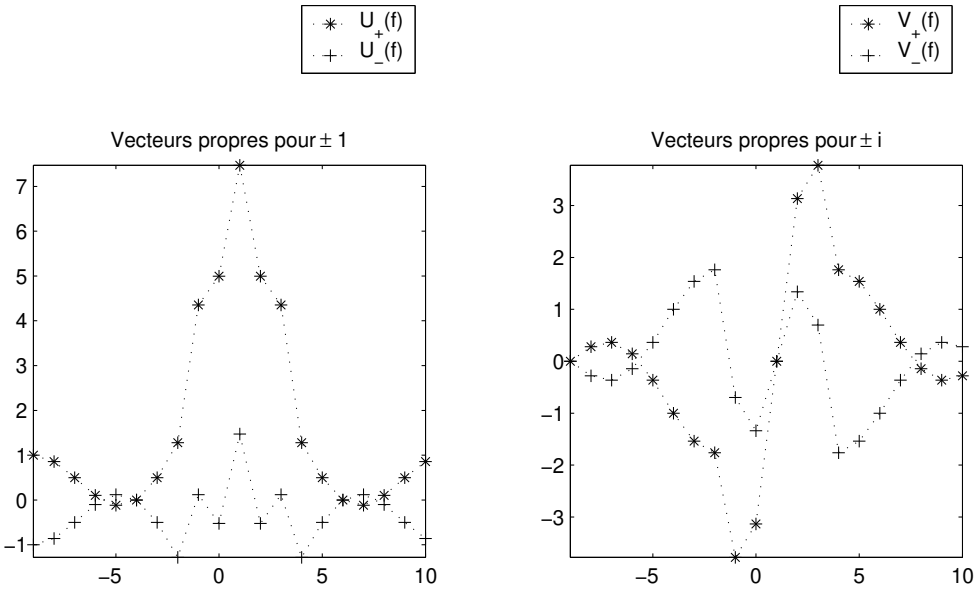


Figure 3.8

The four orthogonal components $P_\lambda f$ of a real signal for the normalized DFT U. Their sum is f; even and odd components appear on separate rows.

Book page 61; original figure 3.8.

Previous edition



Revised edition

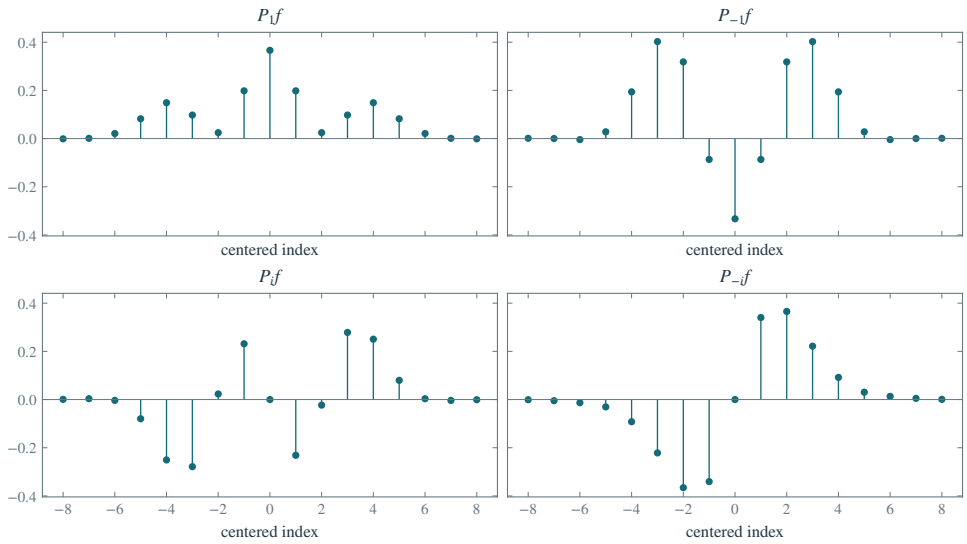
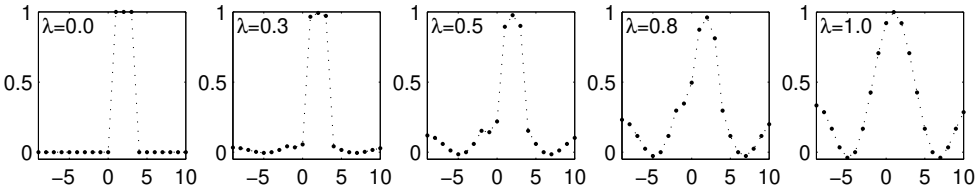


Figure 3.9

The unitary transforms $U^{\alpha} f$ for $0 \leq \alpha \leq 1$, using the fixed phases in the text. Real and imaginary parts are shown separately.

Book page 61; original figure 3.9.

Previous edition



Revised edition

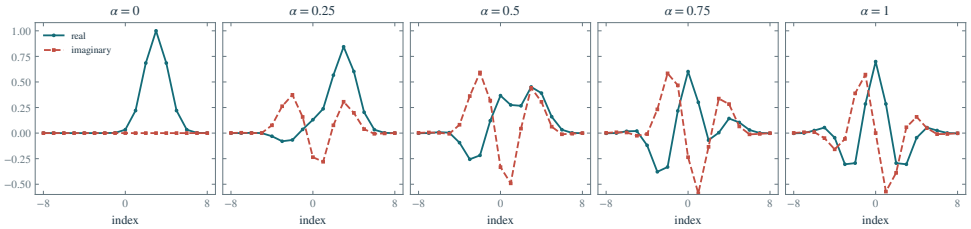
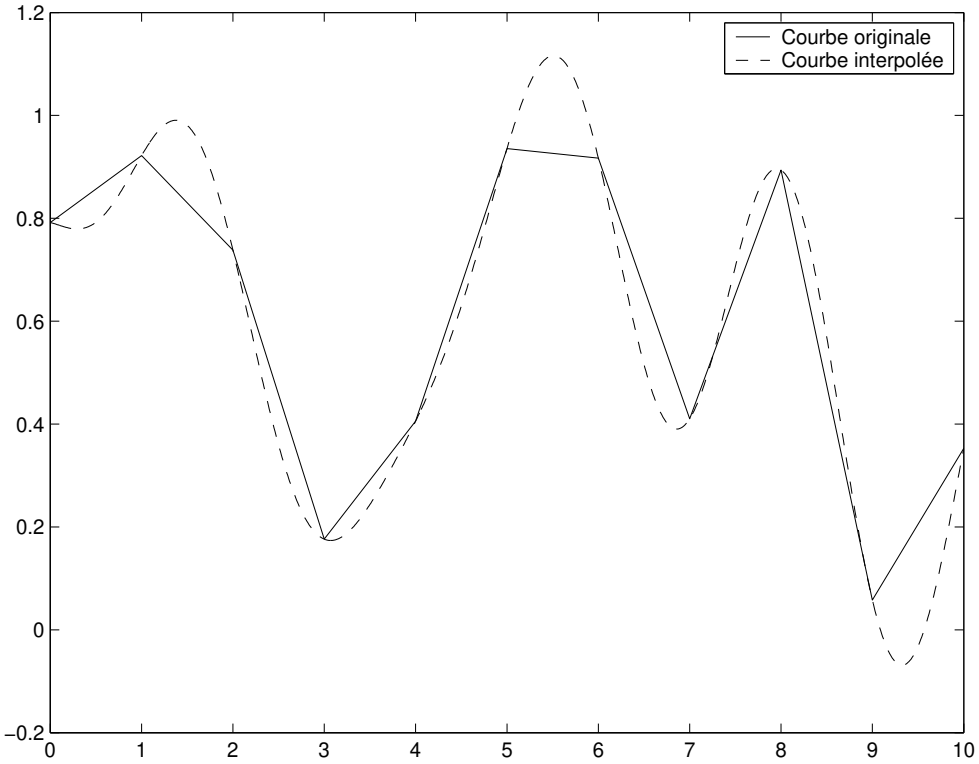


Figure 3.10

Periodic trigonometric interpolation of eleven samples. The faint polygon joins the samples for reference; it is not an underlying continuous signal.

Book page 63; original figure 3.10.

Previous edition



Revised edition

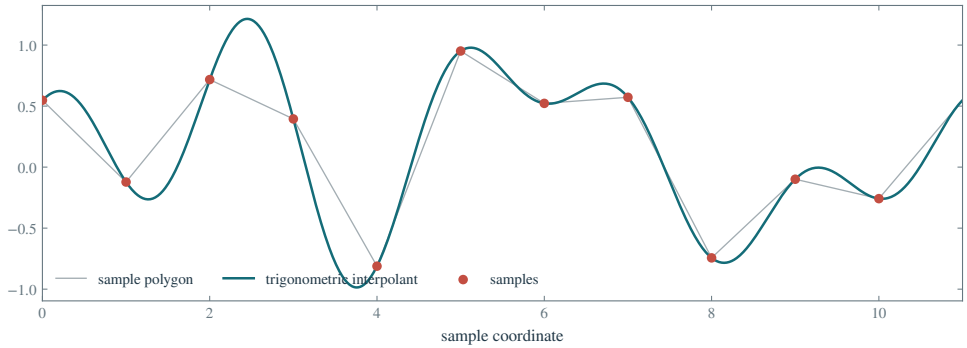
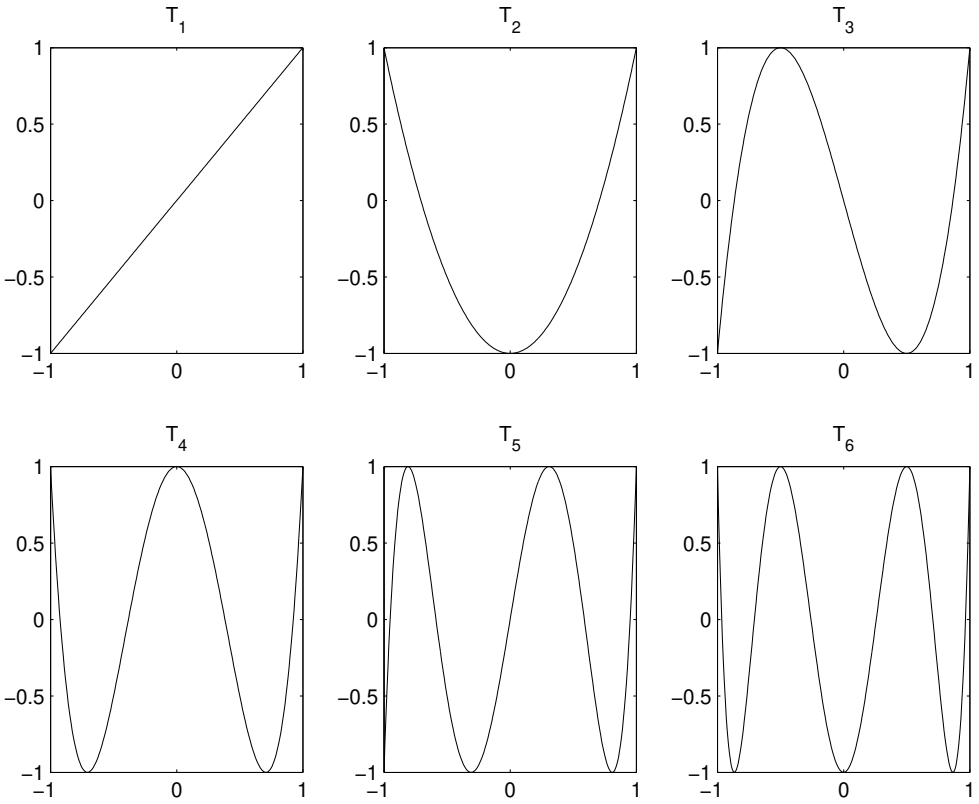


Figure 3.11

The Chebyshev polynomials $T_k(x) = \cos(k \arccos x)$ for $k = 1, \ldots, 6$.

Book page 64; original figure 3.11.

Previous edition



Revised edition

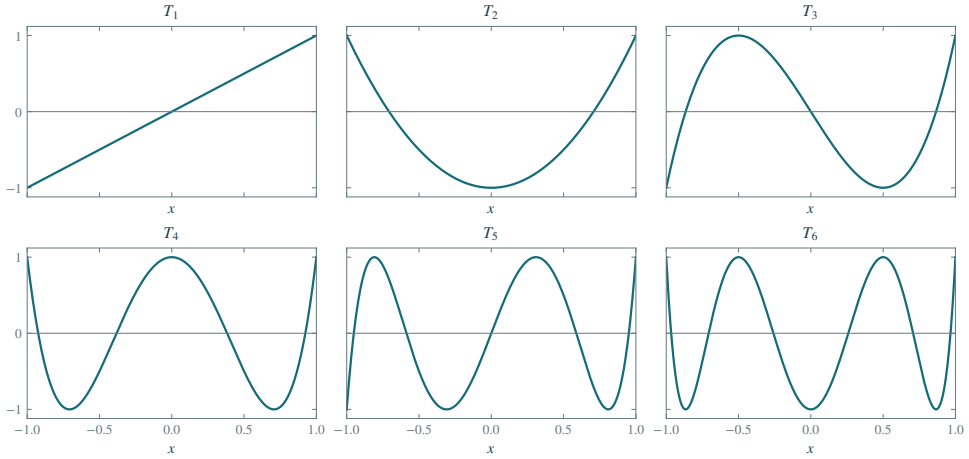
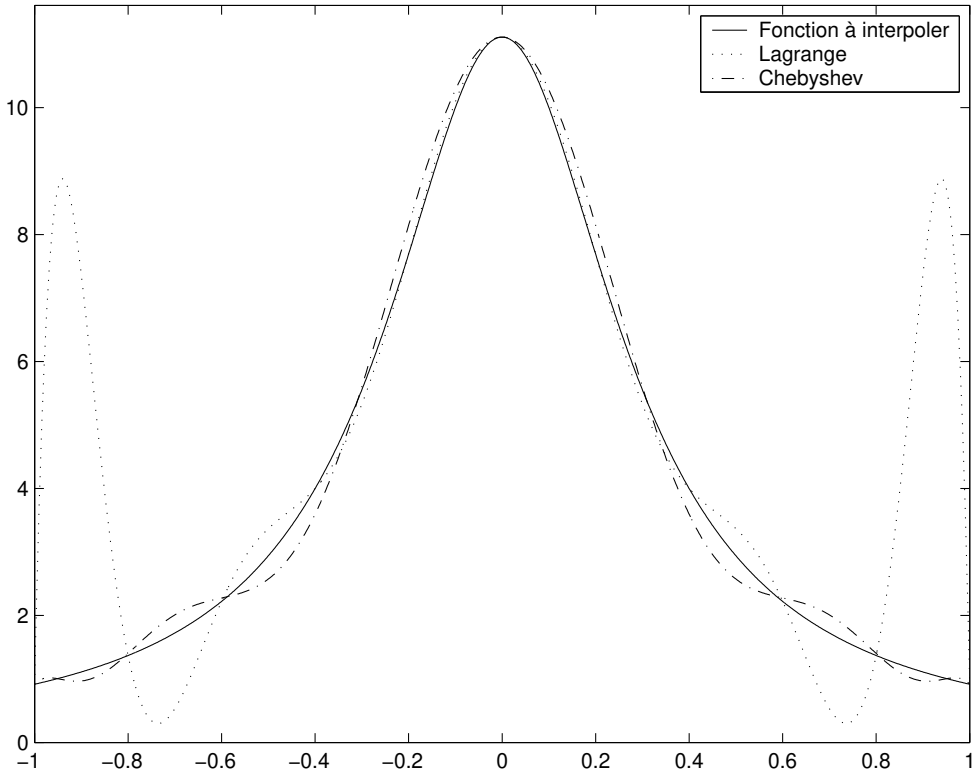


Figure 3.12

Degree-ten polynomial interpolation of $f(x) = 1/(0.3^2 + x^2)$ at eleven equally spaced points and at the roots of T_{11}. Both are Lagrange interpolants; the choice of nodes controls the endpoint oscillations.

Book page 65; original figure 3.12.

Previous edition



Revised edition

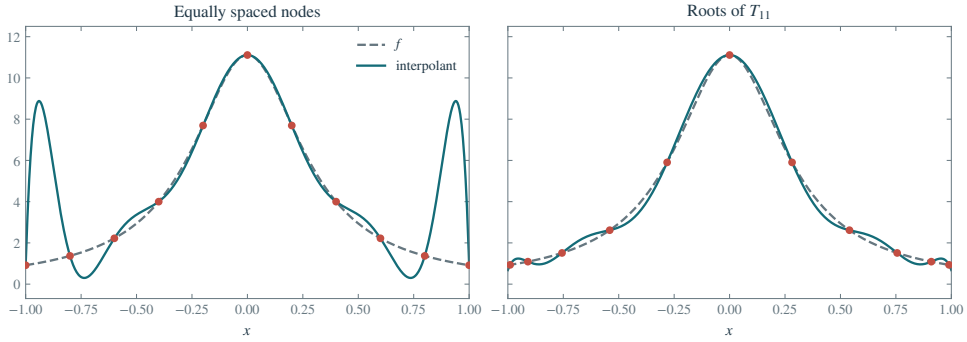
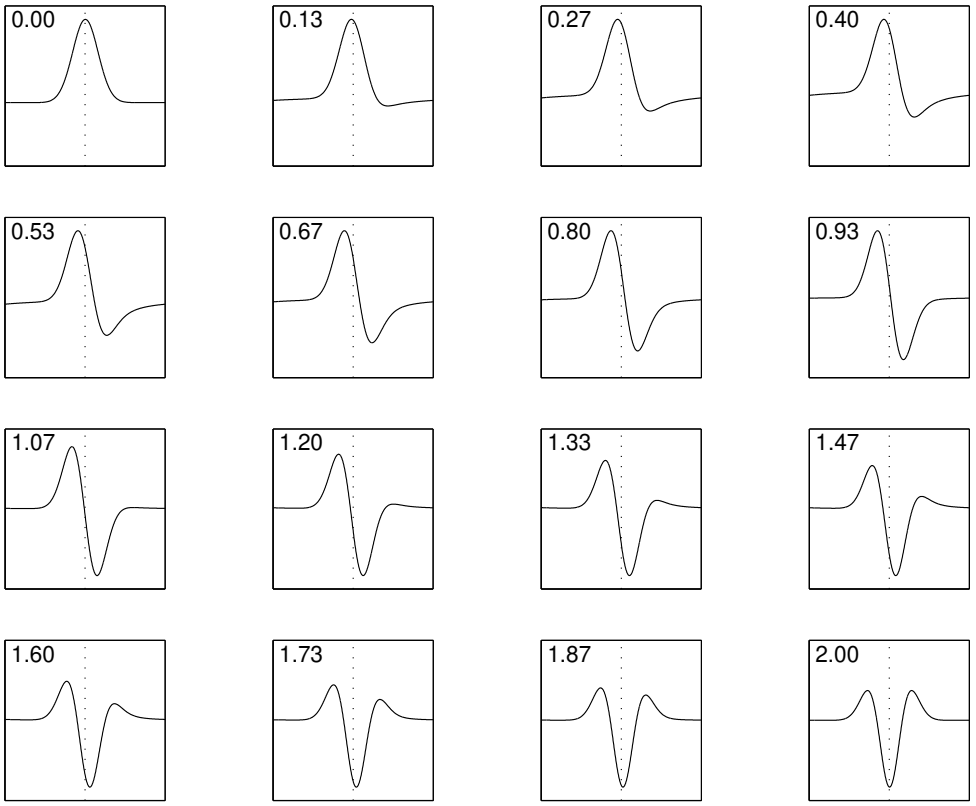


Figure 3.13

Fractional derivatives of $e^{-x^2/2}$ for orders $0 \le \alpha \le 2$, computed from the multiplier $(\iota\xi)^\alpha$ with the phase convention in the text.

Book page 66; original figure 3.13.

Previous edition



Revised edition

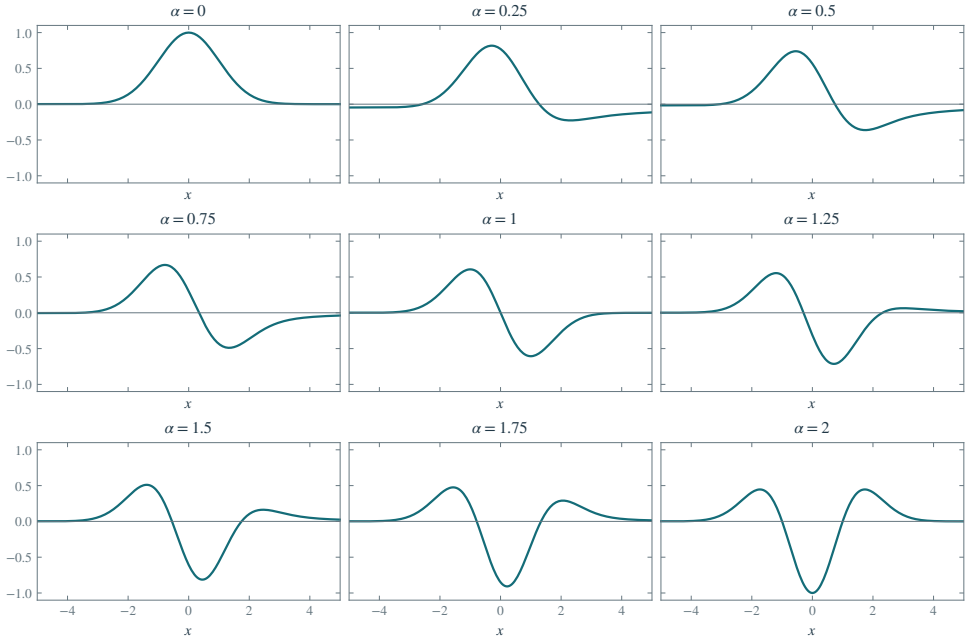
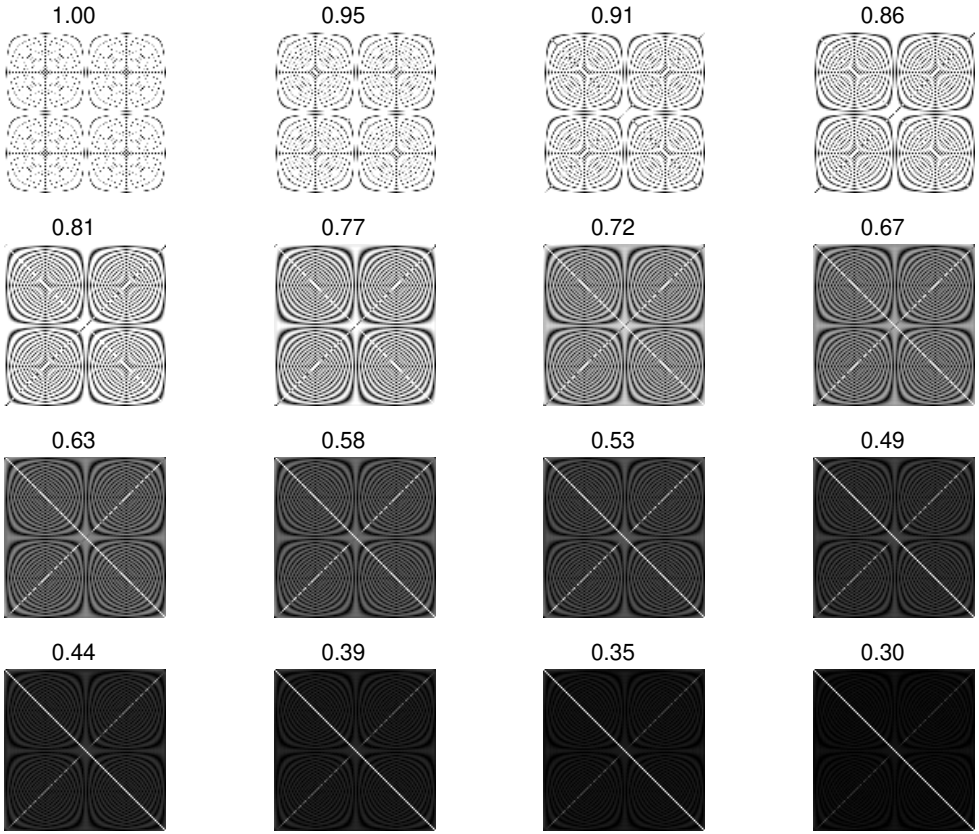


Figure 3.14

Entrywise magnitudes of the unitary matrices U^{α}, on one common nonlinear color scale. At $\alpha = 1$ every magnitude equals $1/\sqrt{N}$; at $\alpha = 2$ the matrix is cyclic reversal.

Book page 66; original figure 3.14.

Previous edition



Revised edition

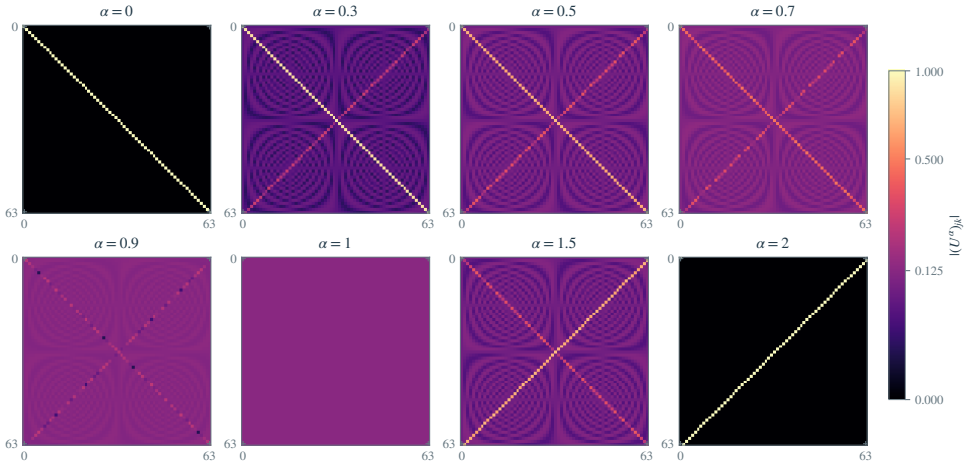
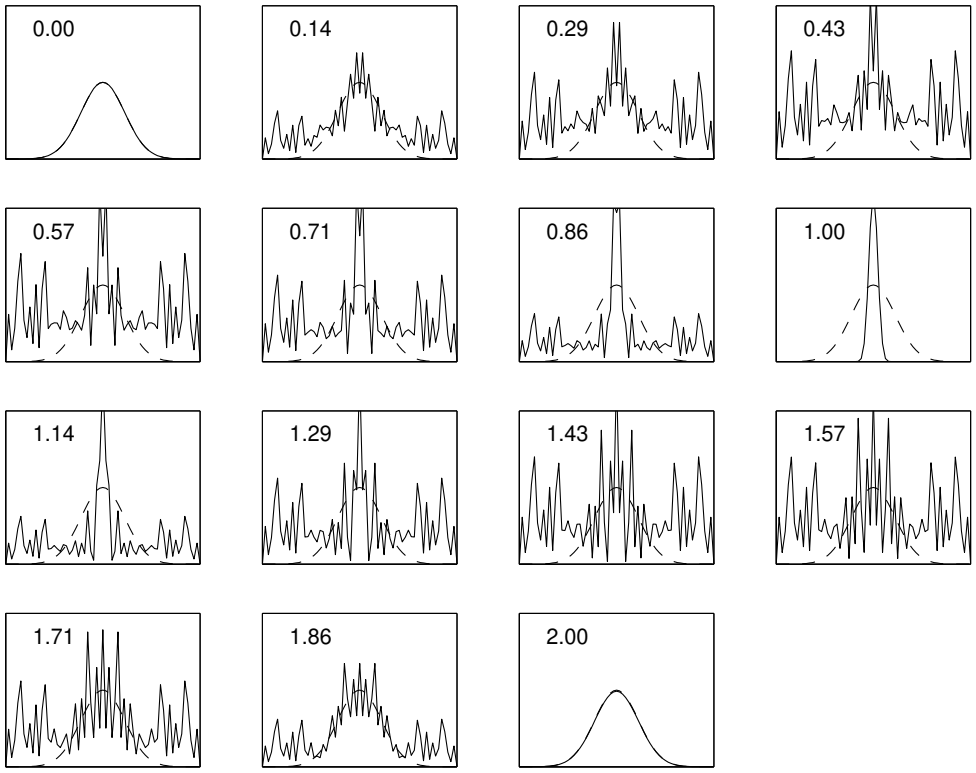


Figure 3.15

Unitary fractional Fourier transforms of a discrete Gaussian. The fixed eigenspace phases give a continuous family; at $\alpha = 2$ this even input is recovered.

Book page 67; original figure 3.15.

Previous edition



Revised edition

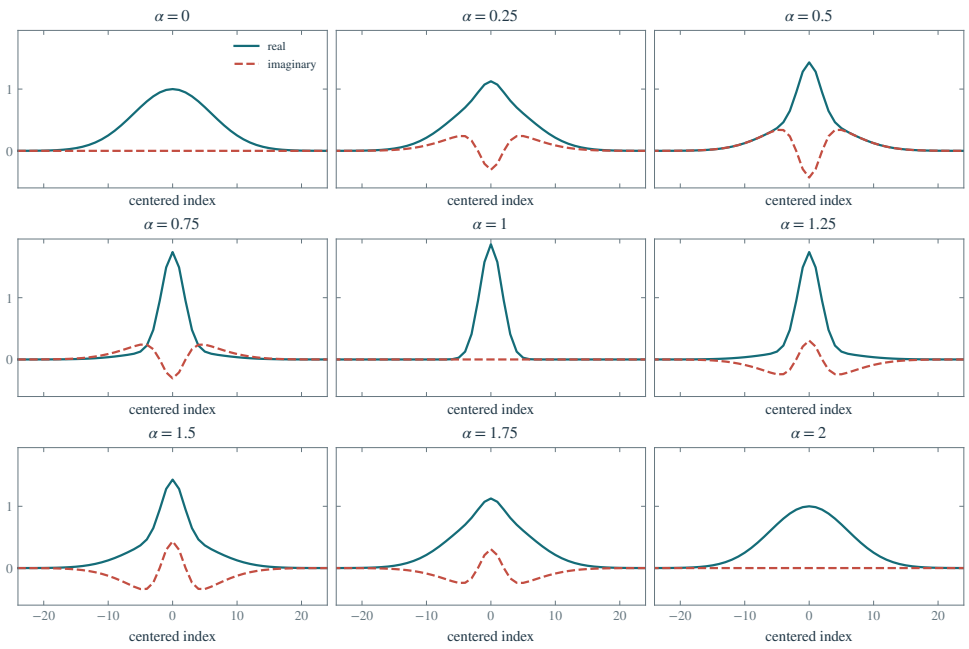
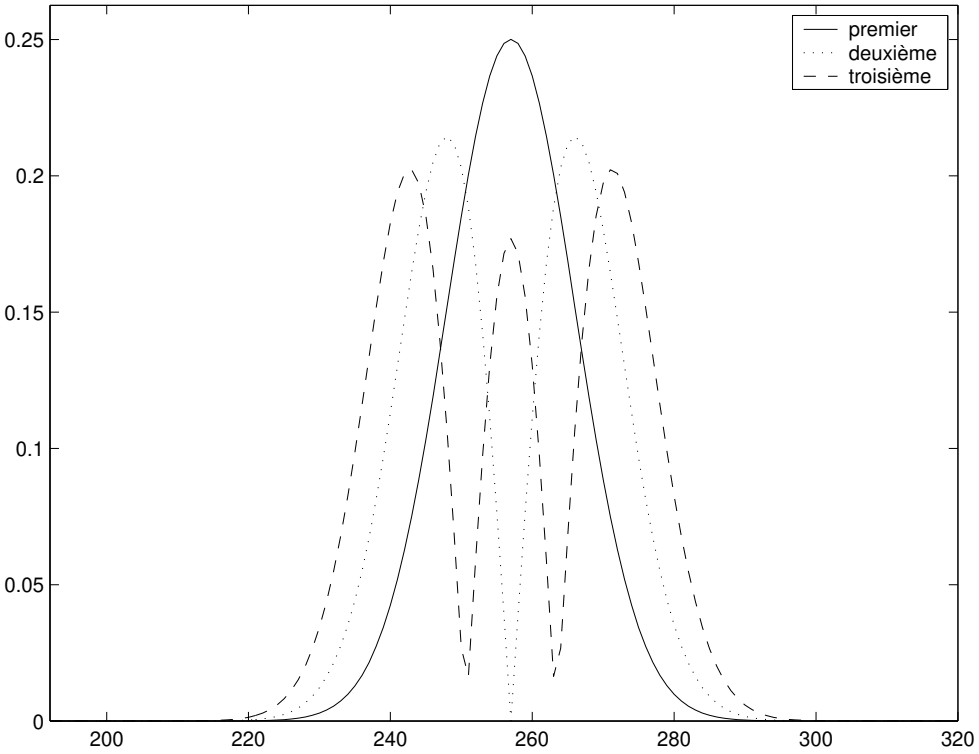


Figure 3.16

Magnitudes of the first three vectors in the structured DFT eigenbasis for $N = 128$, ordered by decreasing eigenvalue of S. The index is centered for display.

Book page 68; original figure 3.16.

Previous edition



Revised edition

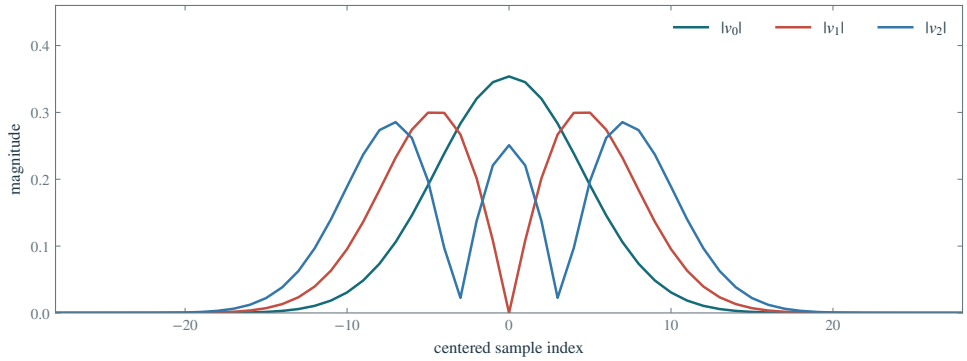
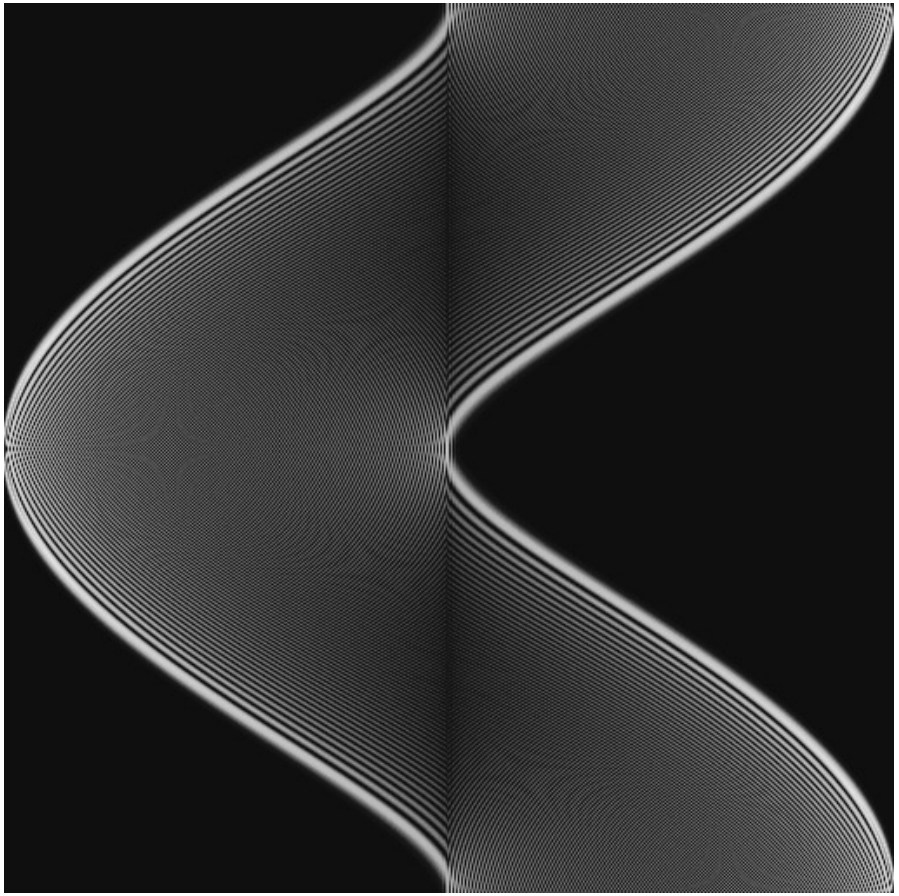


Figure 3.17

Magnitudes of the structured orthonormal eigenbasis for $N = 128$. Columns follow decreasing eigenvalues of S, with even parity first in a tie; rows show centered sample indices.

Book page 68; original figure 3.17.

Previous edition



Revised edition

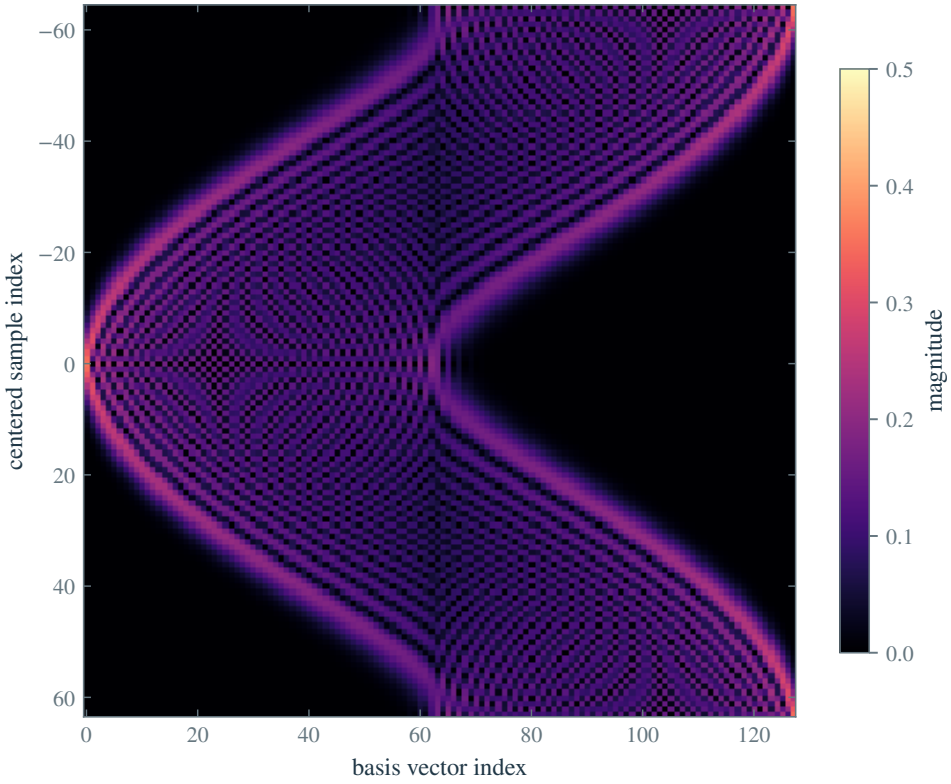
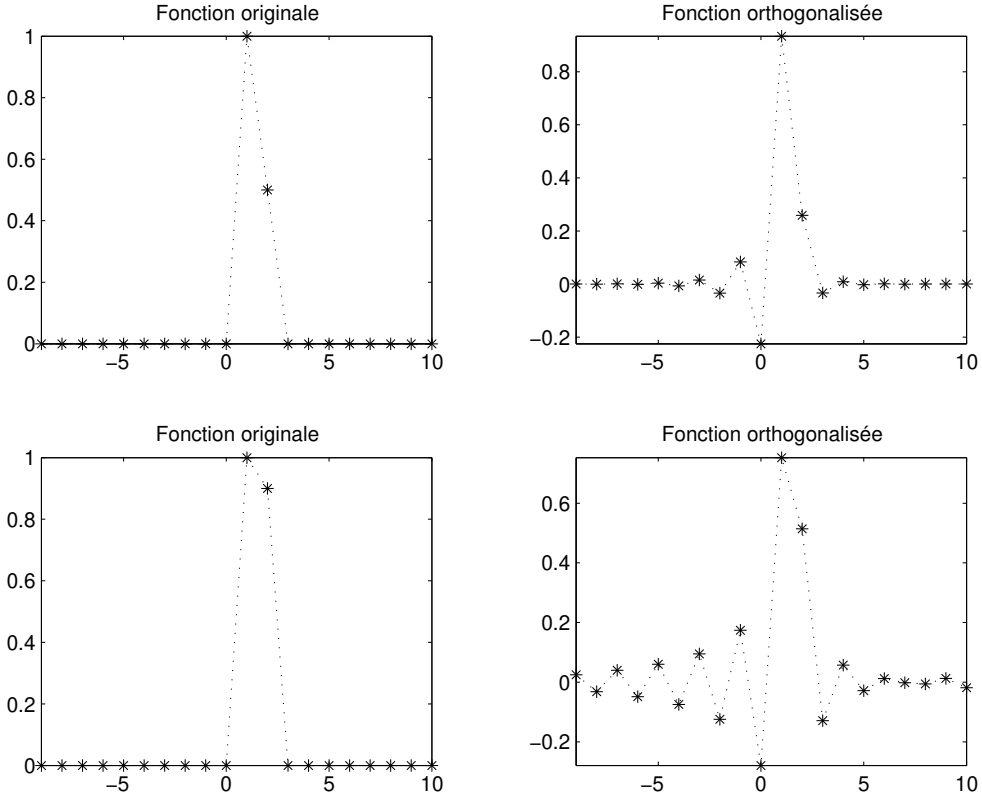


Figure 3.18

Orthogonalizing the cyclic translates of $f = \delta_0 + a\delta_1$ on $\mathbb{Z}/20\mathbb{Z}$, for $a = 0.5$ and $a = 0.9$. The output g satisfies $|\widehat{g}| = \sqrt{20}$, as required by the normalized inner product.

Book page 69; original figure 3.18.

Previous edition



Revised edition

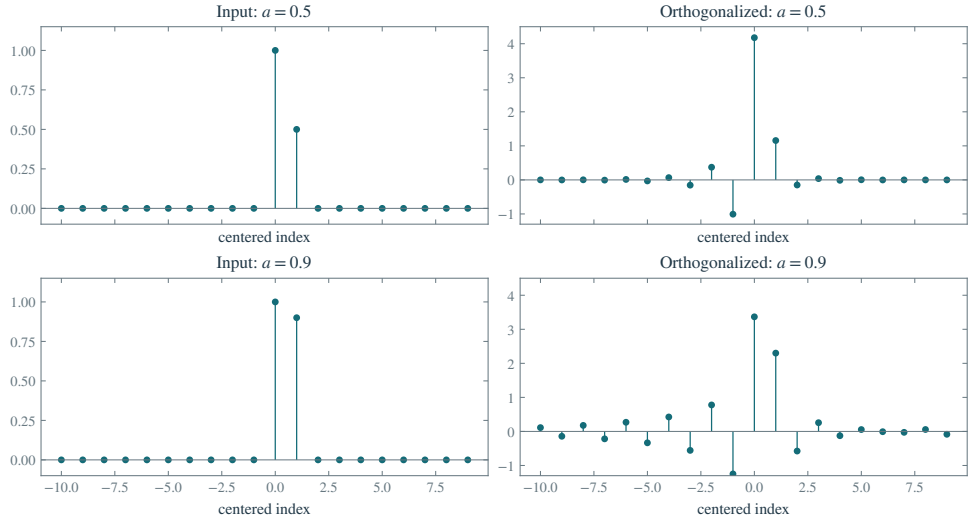
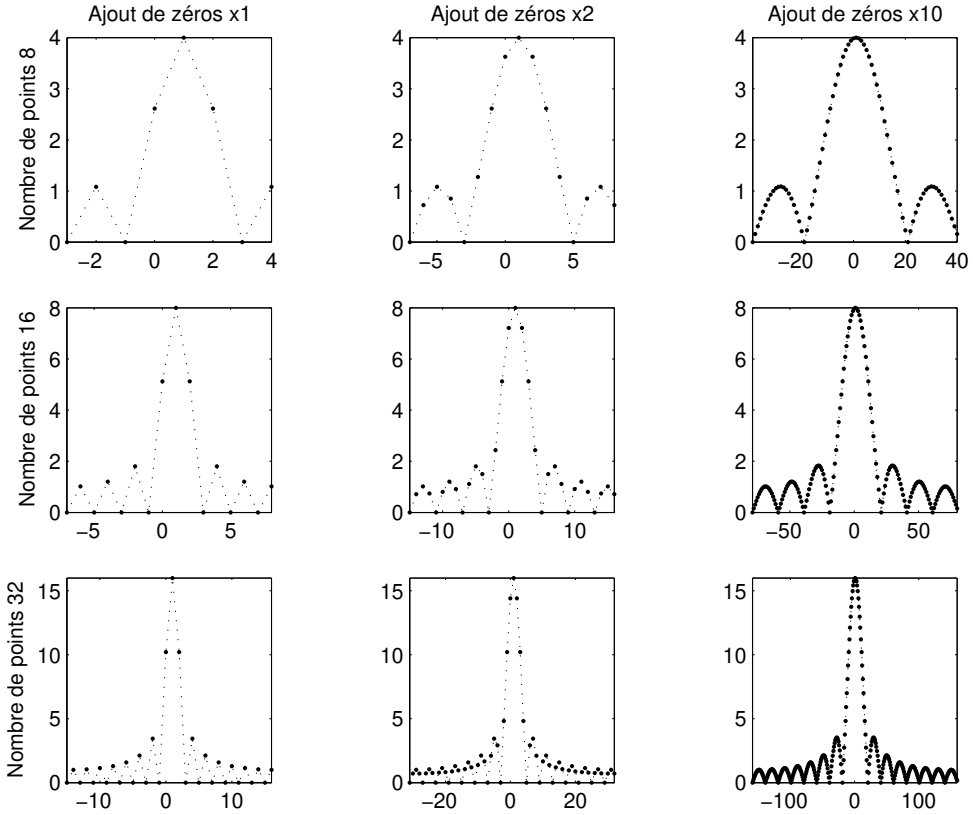


Figure 4.1

Sampling and zero padding for $f = \mathbf{1}_{[1/2,1)}$. Rows change the number N of measured samples; columns change only the padded length M. The DFT magnitudes include the quadrature factor $1/N$; the faint curve is the continuous-transform magnitude.

Book page 80; original figure 4.1.

Previous edition



Revised edition

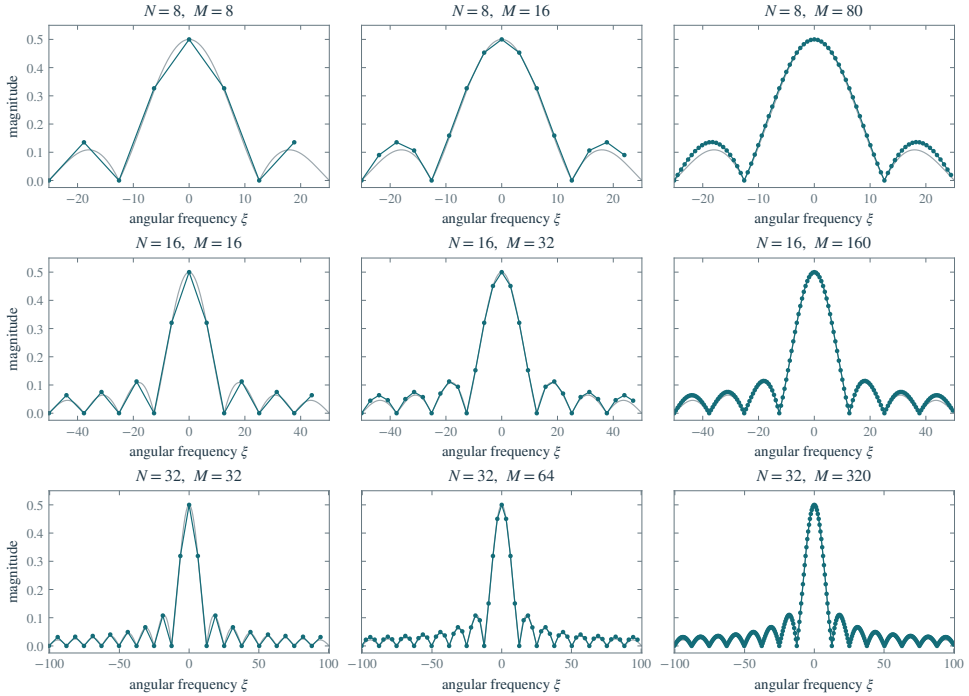
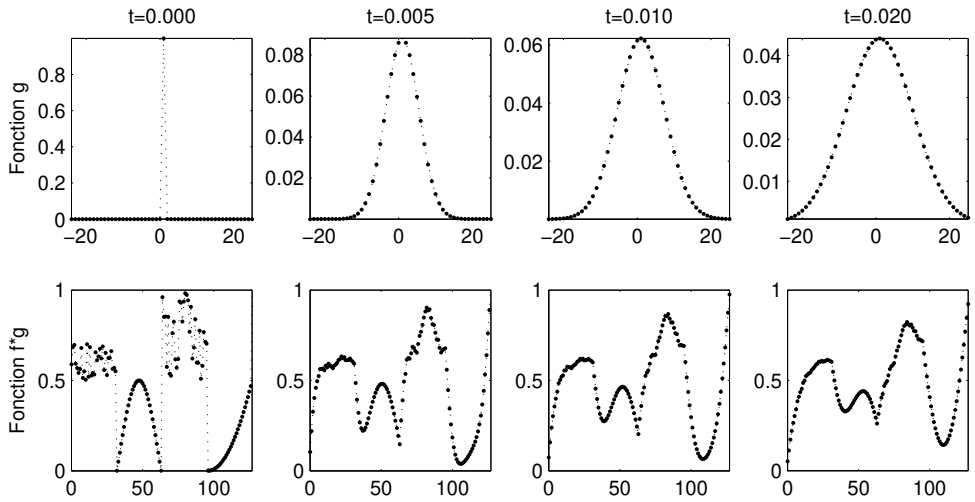


Figure 4.2

Gaussian kernels and their action on one fixed irregular signal. Widths are in sample units, and the signal is reflected at its endpoints. The first column is the exact identity kernel.

Book page 81; original figure 4.2.

Previous edition



Revised edition

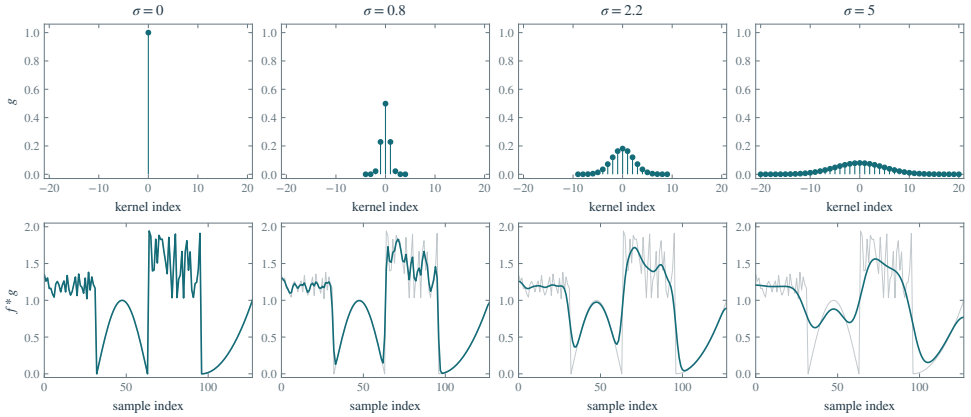
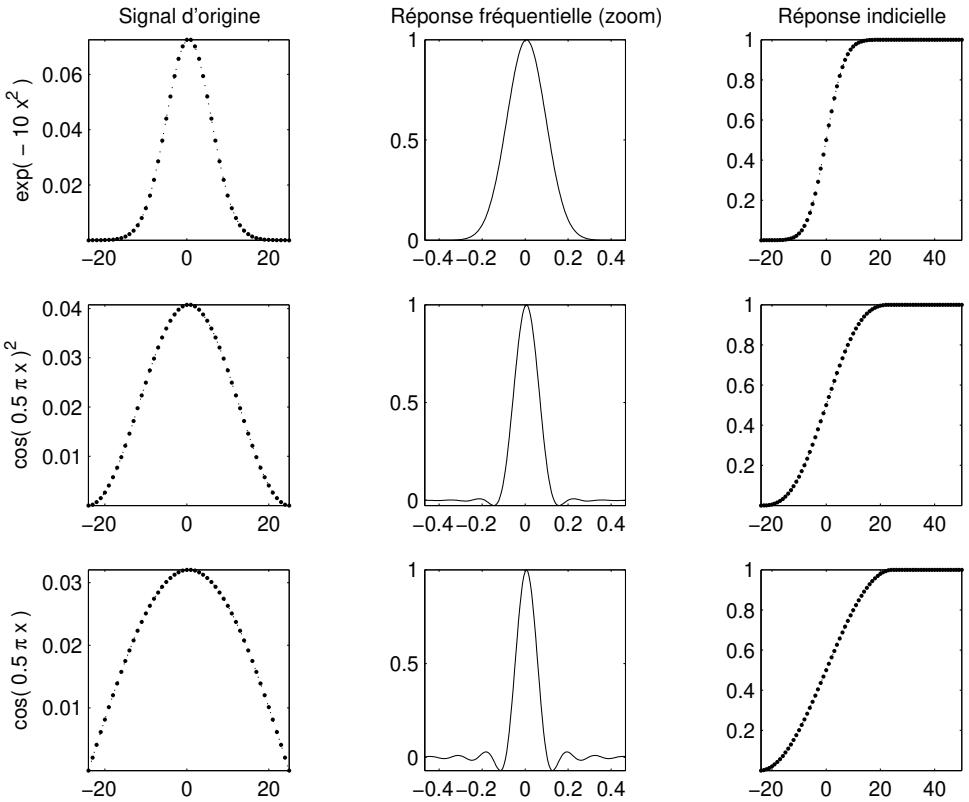


Figure 4.3

Impulse, frequency, and step responses of three unit-sum symmetric FIR kernels. The frequency plots show a neighborhood of zero; the step responses are cumulative sums of the impulse responses.

Book page 82; original figure 4.3.

Previous edition



Revised edition

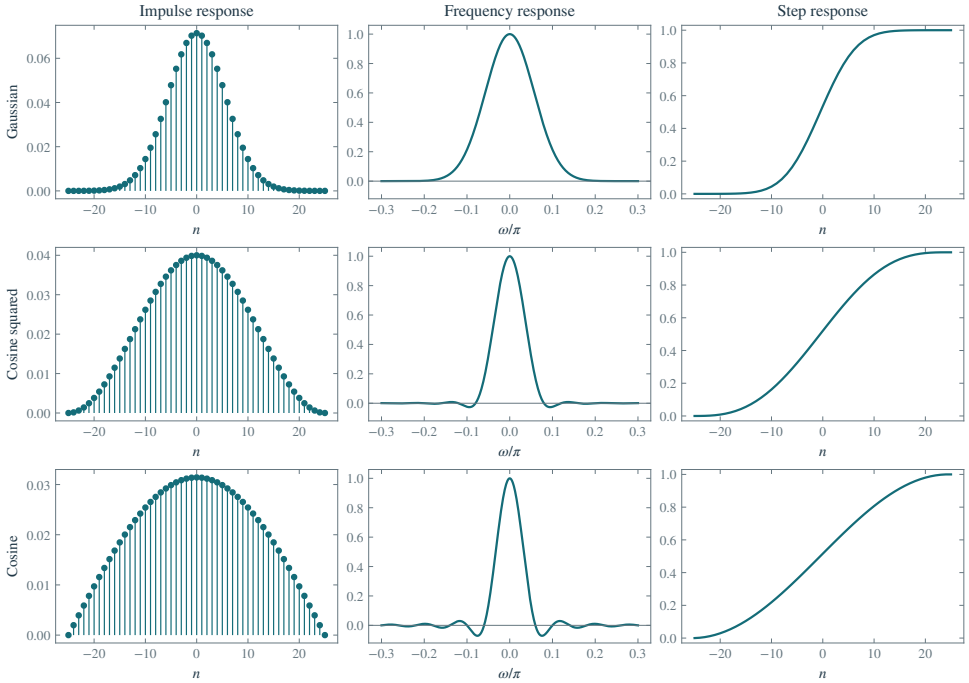
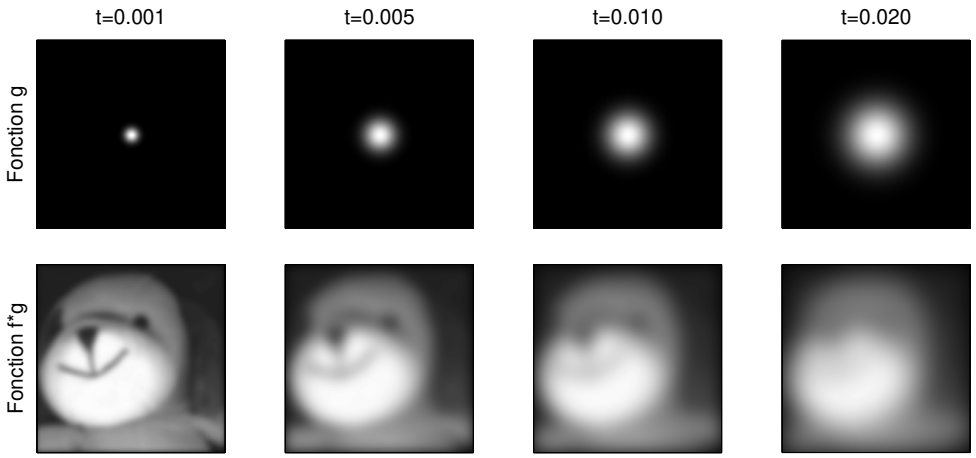


Figure 4.4

Discrete Gaussian kernels (top) and smoothing of the flower image (bottom), with reflected boundaries. Each kernel is displayed relative to its own peak; all image panels share the grayscale range $[0, 1]$.

Book page 83; original figure 4.4.

Previous edition



Revised edition

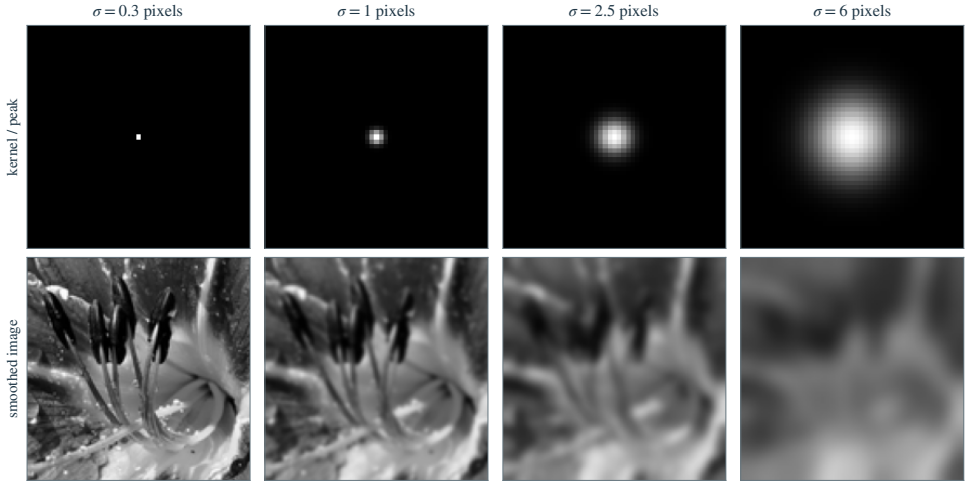


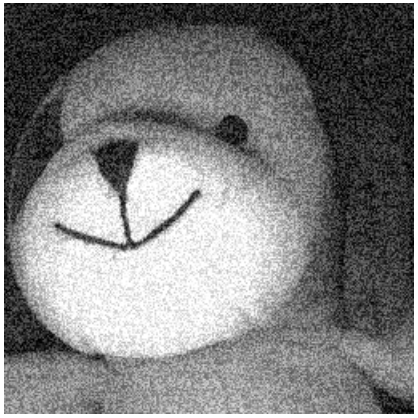
Figure 4.5

A flower image corrupted by additive Gaussian noise of standard deviation 0.12, and Gaussian smoothing with width 0.9 pixels. The displayed PSNR values use the same clean reference; intensity clipping is for display only.

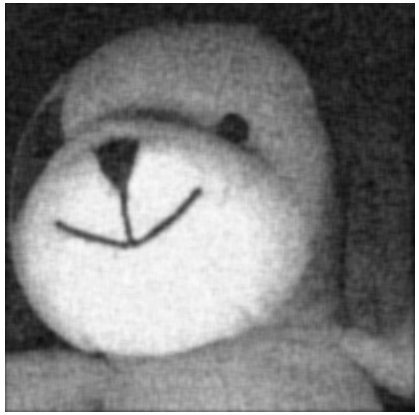
Book page 83; original figure 4.5.

Previous edition

Image bruitée



Après filtrage par une gaussienne



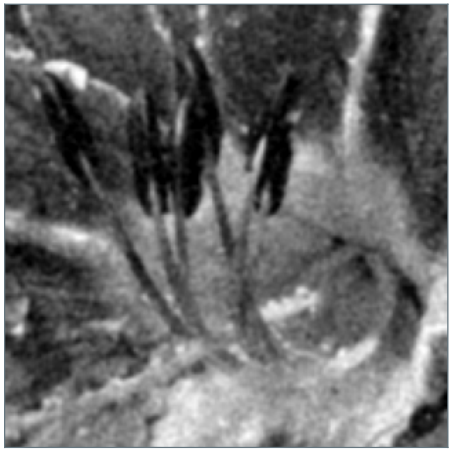
Revised edition

Noisy image



PSNR = 18.5 dB

Gaussian smoothing



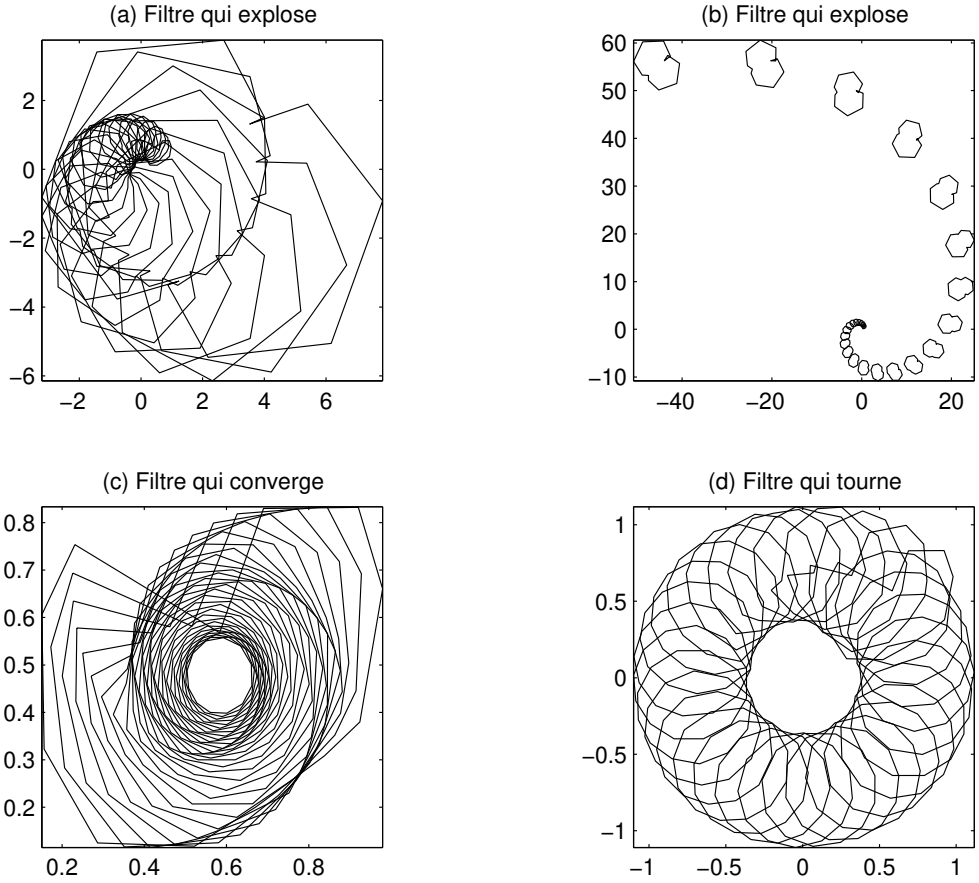
PSNR = 24.9 dB

Figure 4.6

Four regimes of polygon filtering. Curves progress from the initial polygon to later iterates. Expanding active multipliers cause divergence; unit-modulus phases can keep the trajectory bounded without convergence.

Book page 84; original figure 4.6.

Previous edition



Revised edition

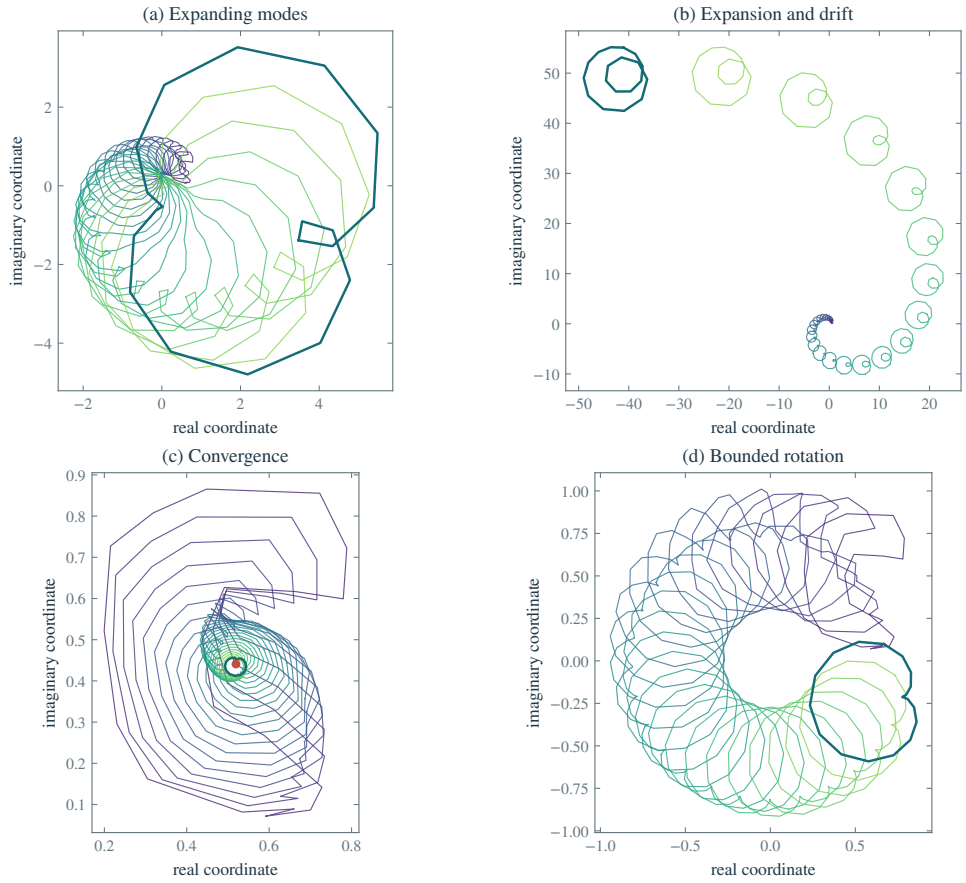
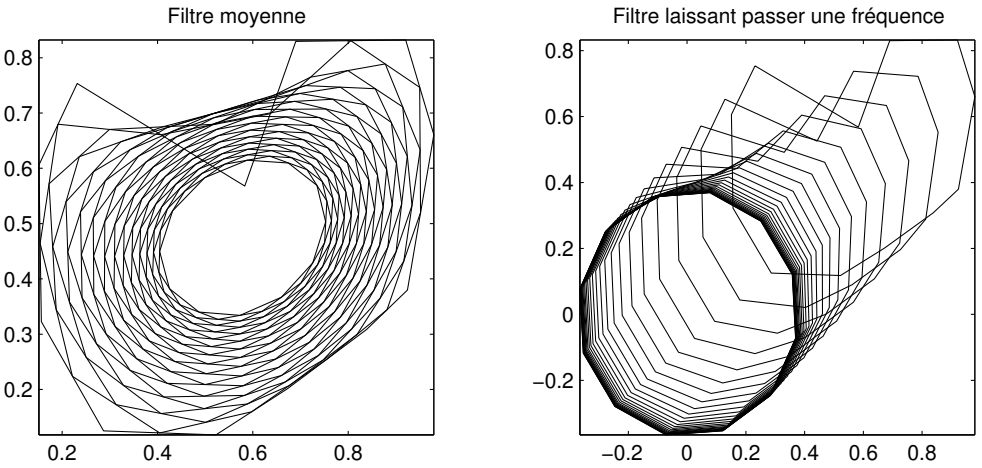


Figure 4.7

Successive midpoint averaging converges to the vertex centroid (left). Keeping only frequency one produces a regular polygon of radius $|\widehat{\Pi}[1]|/N$ (right). Red marks indicate the limits.

Book page 85; original figure 4.7.

Previous edition



Revised edition

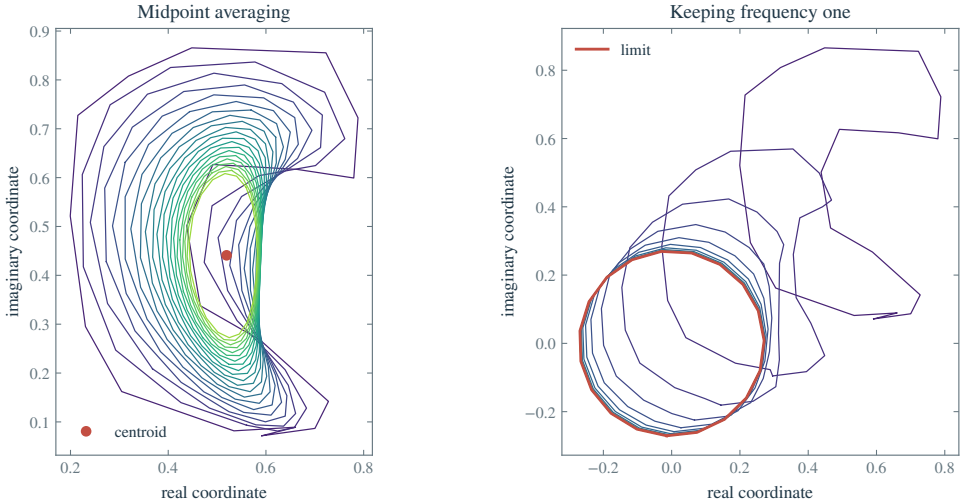
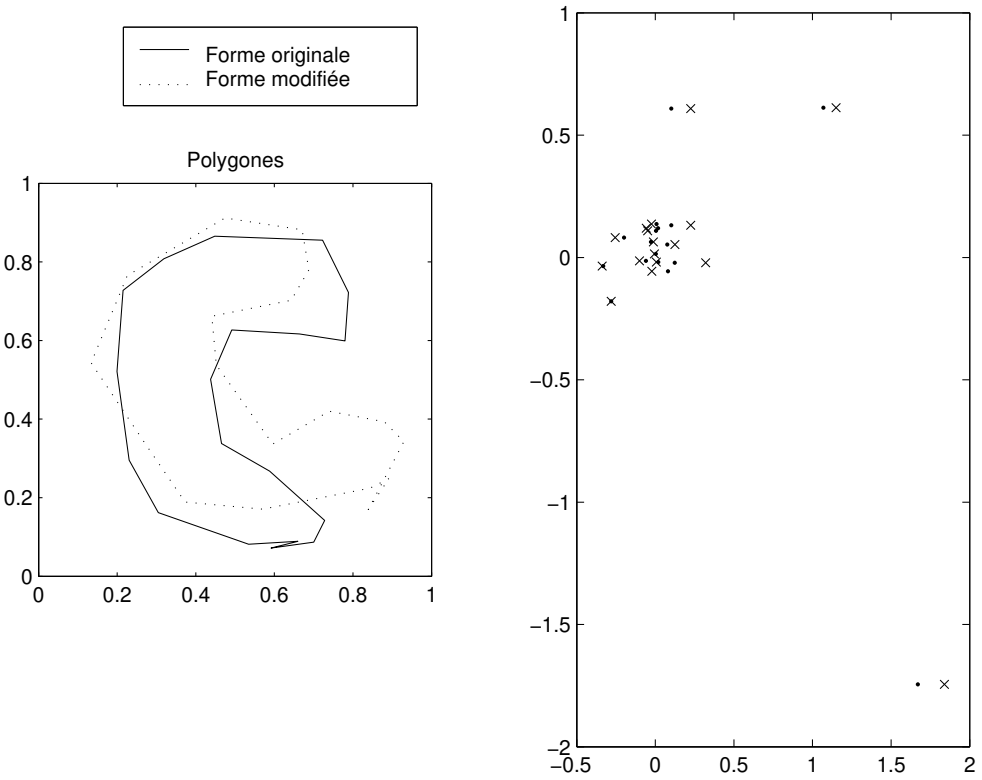


Figure 4.8

A polygon and a perturbed, rotated version, with their magnitude-based Fourier descriptors. The perturbation changes the descriptor; the rotation alone would leave it unchanged.

Book page 86; original figure 4.8.

Previous edition



Revised edition

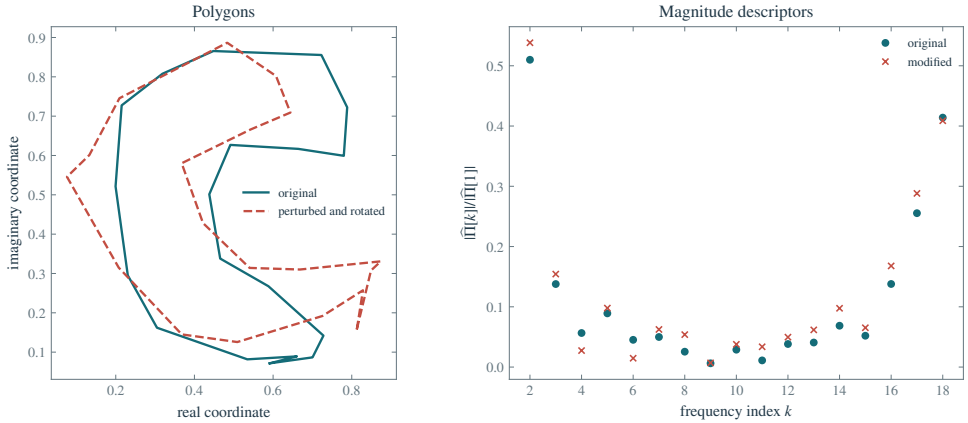
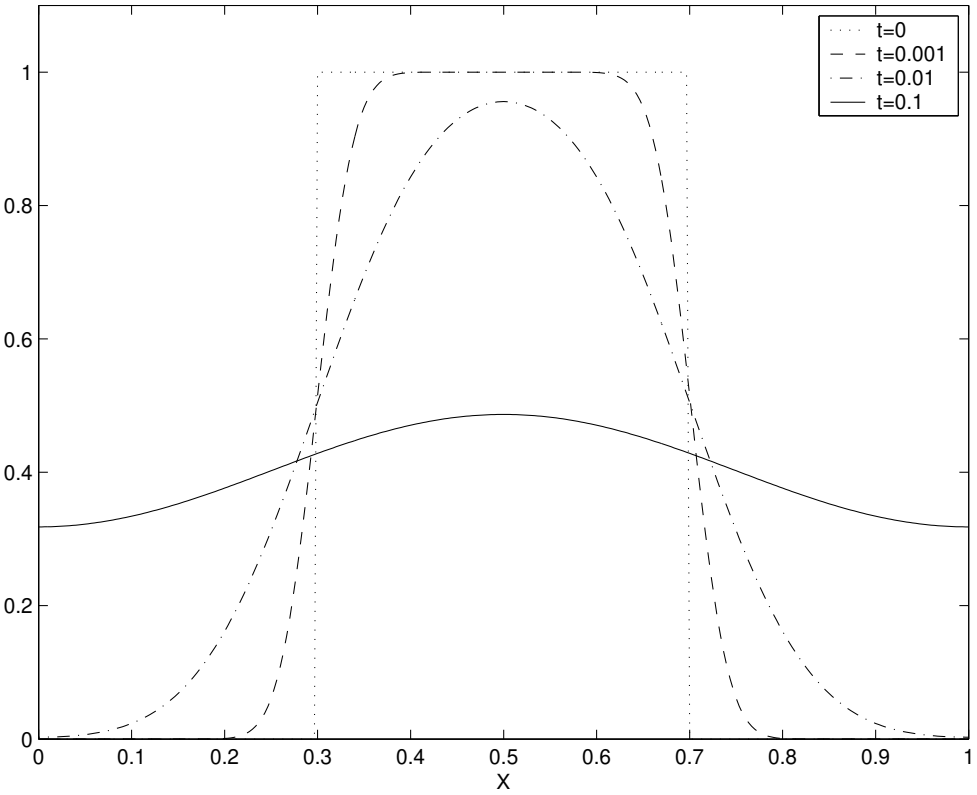


Figure 4.9

Spectral heat evolution of an interval indicator on the unit circle, with conductivity $\kappa = 1/2$. The mean is preserved and the nonconstant Fourier modes decay.

Book page 88; original figure 4.9.

Previous edition



Revised edition

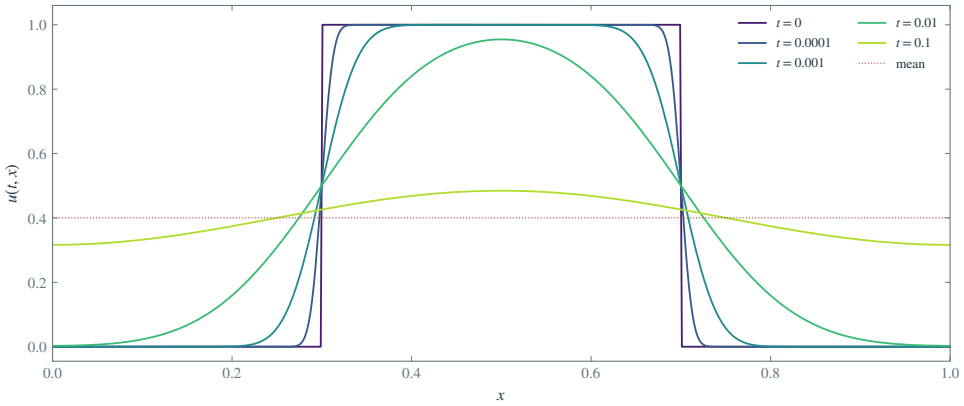
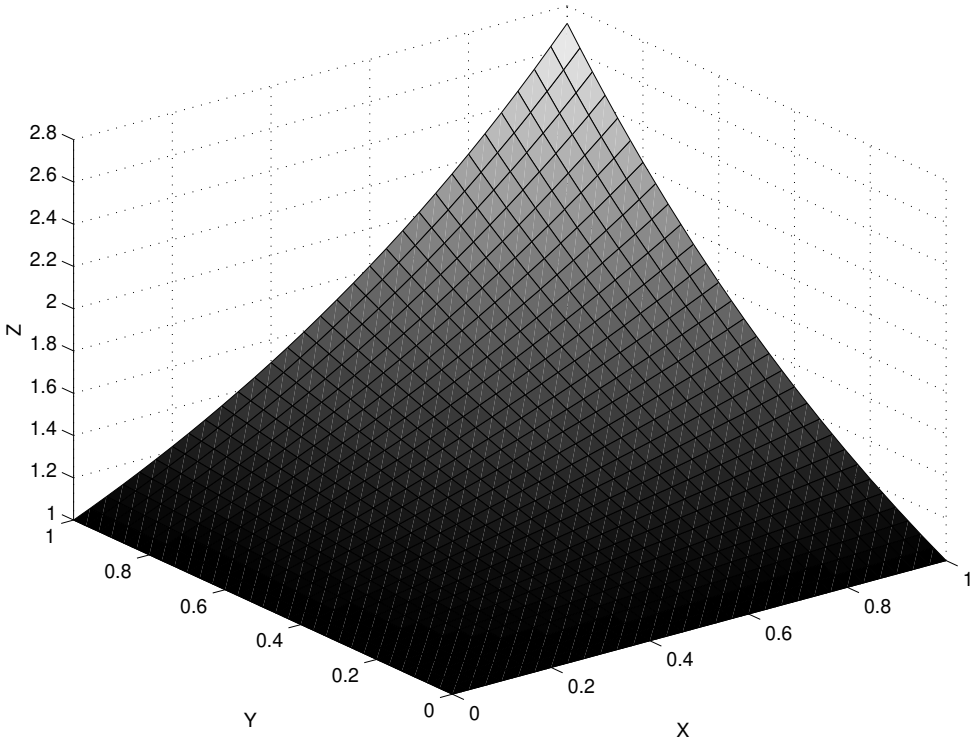


Figure 4.10

Dirichlet Poisson solution with reference $u(x, y) = e^{xy}$, on a grid with $N = 32$ intervals per axis.

Book page 89; original figure 4.10.

Previous edition



Revised edition

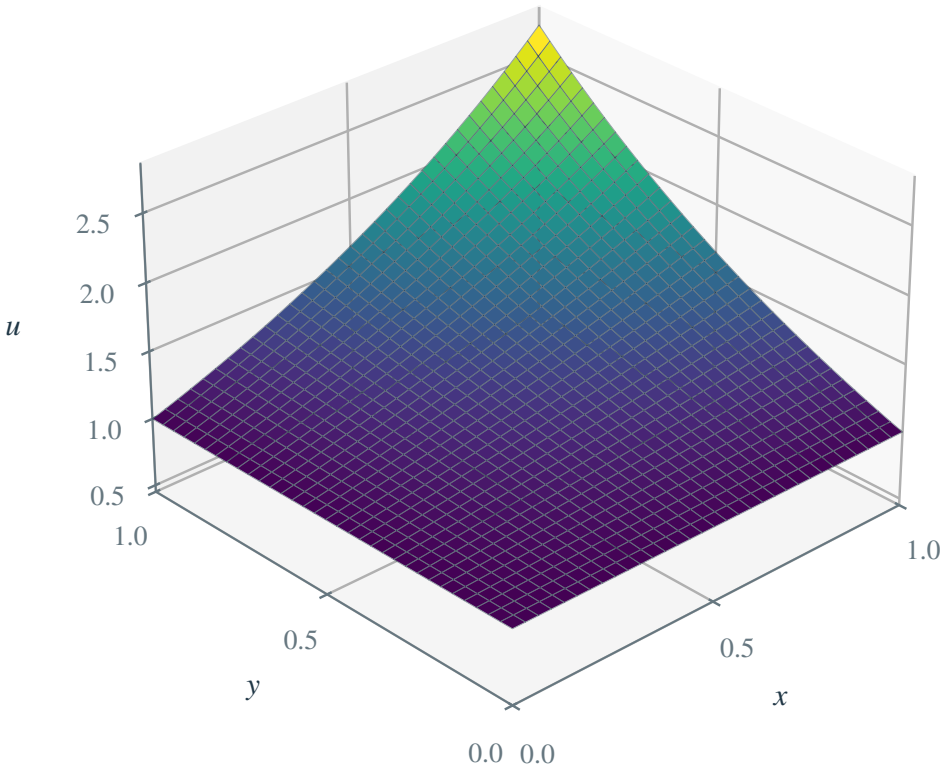
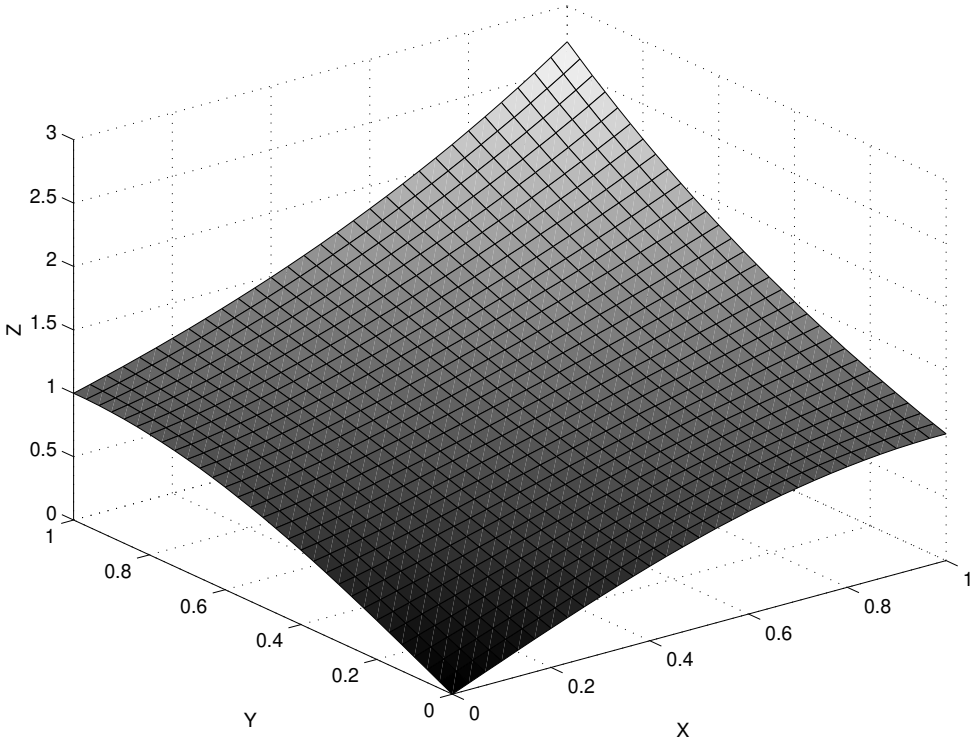


Figure 4.11

The same source with boundary values lowered by $0.45\sin(\pi s)$ on $x = 0$ and $y = 0$. All four corner values remain compatible.

Book page 89; original figure 4.11.

Previous edition



Revised edition

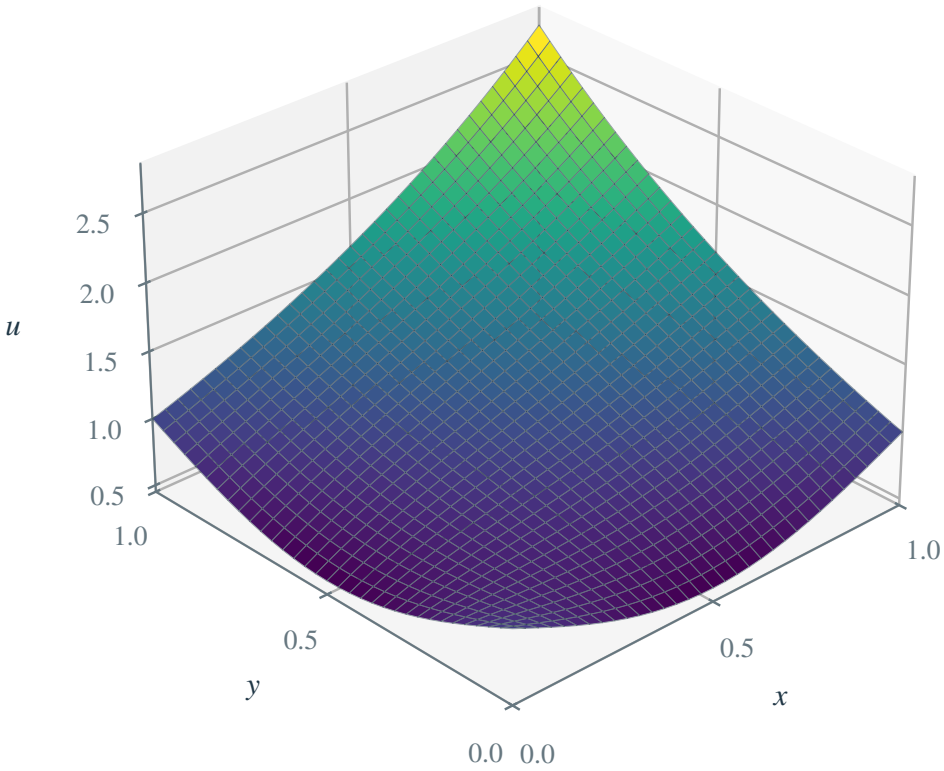
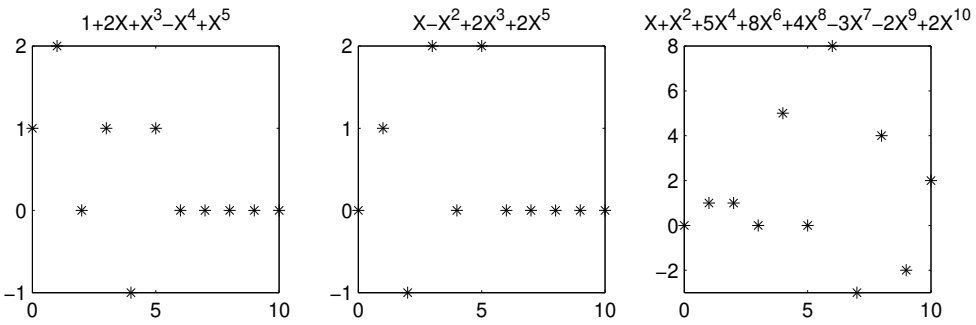


Figure 4.12

Coefficient vectors of P, Q, and their product. Length-eleven FFT convolution agrees with the ordinary degree-ten polynomial product.

Book page 90; original figure 4.12.

Previous edition



Revised edition

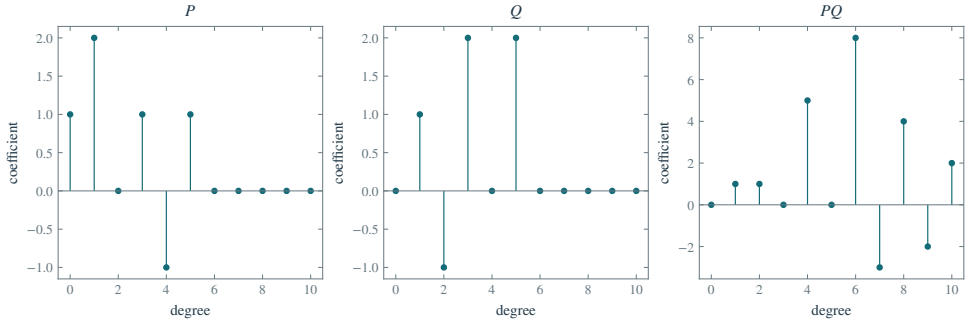
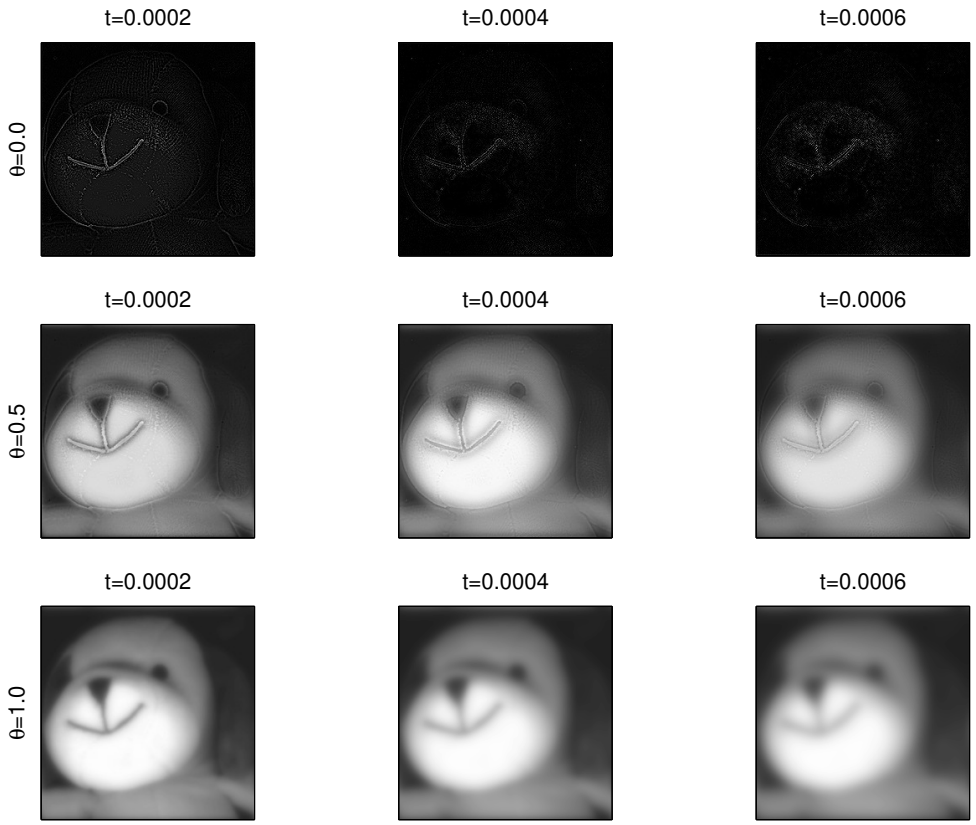


Figure 4.13

Three time steps for the same flower image, with $N = 128$ and $h = 0.0002$. Explicit Euler is unstable; Crank–Nicolson is stable but weakly damps high frequencies; backward Euler damps them strongly. All images use $[0, 1]$, and each panel reports its actual value range.

Book page 92; original figure 4.13.

Previous edition



Revised edition

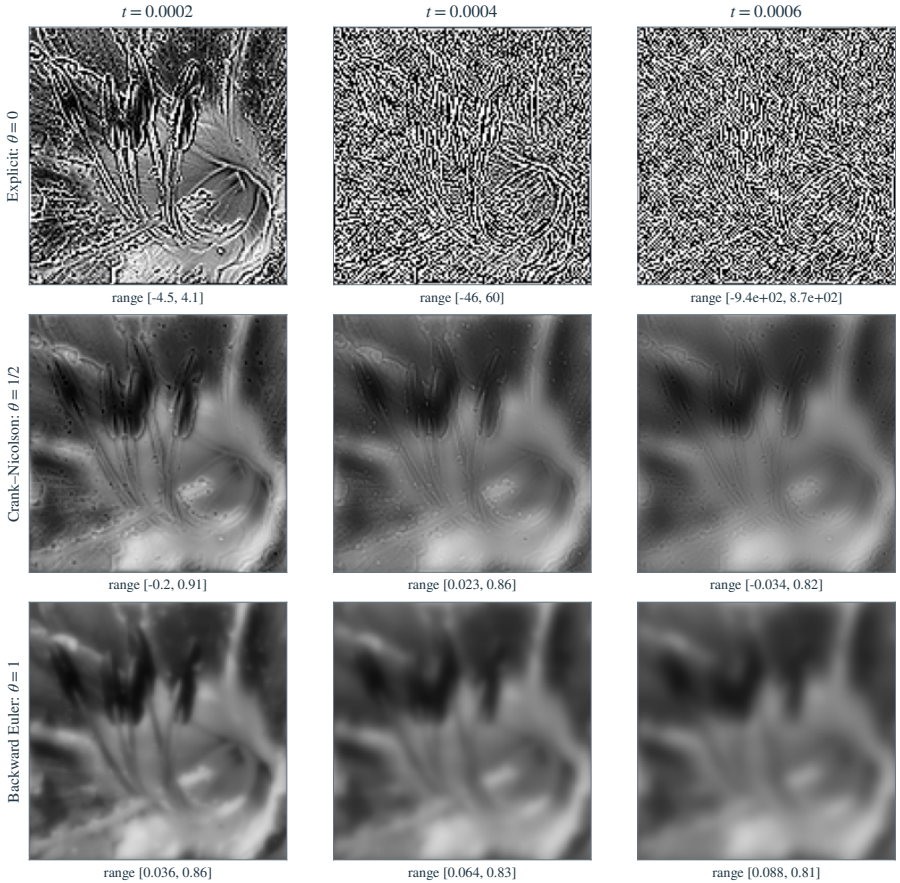


Figure 4.14

Template matching in the flower image. The red box identifies the extracted patch; crosses mark its location in the raw and normalized correlation maps. Only complete-template positions are included.

Book page 94; original figure 4.14.

Previous edition

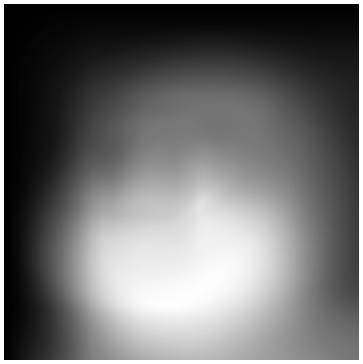
(a) Image d'origine



(b) Image extraite



(c) Correlation



(d) Correlation normalisée



Revised edition

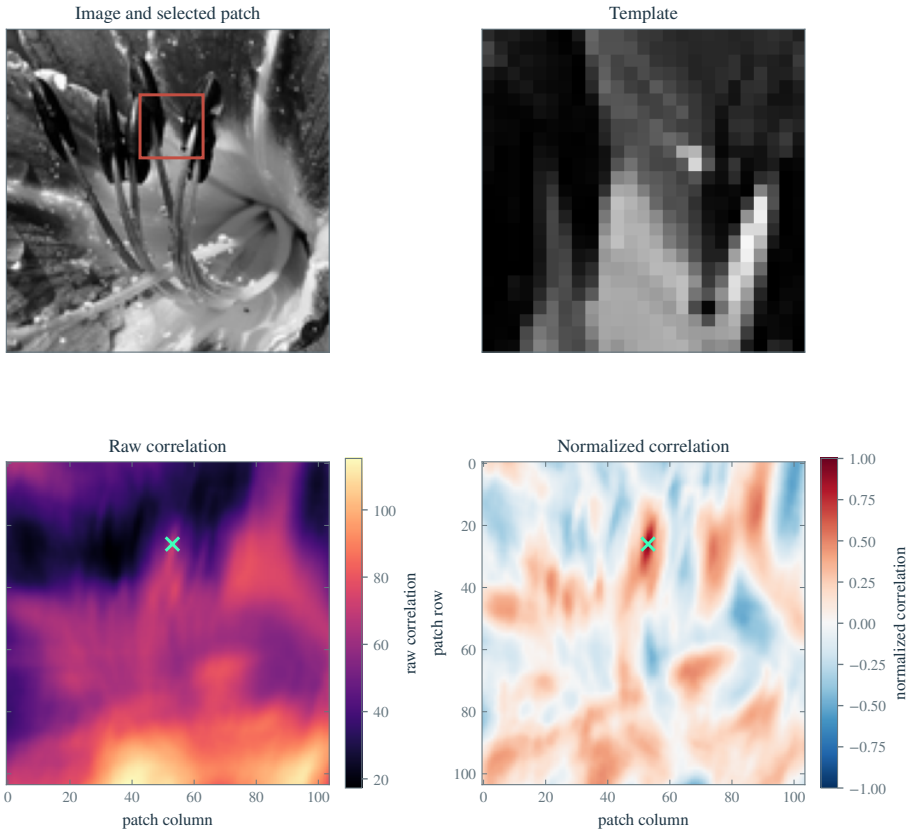


Figure 4.15

Rotations of the flower image by three Fourier shears. The odd grid avoids a Nyquist ambiguity; a white margin limits wraparound. Cropping and intensity clipping are applied only to the displayed results.

Book page 95; original figure 4.15.

Previous edition



Revised edition

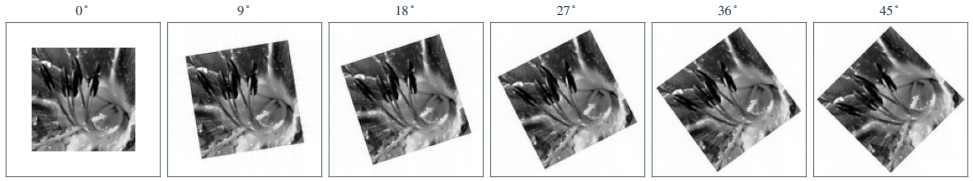
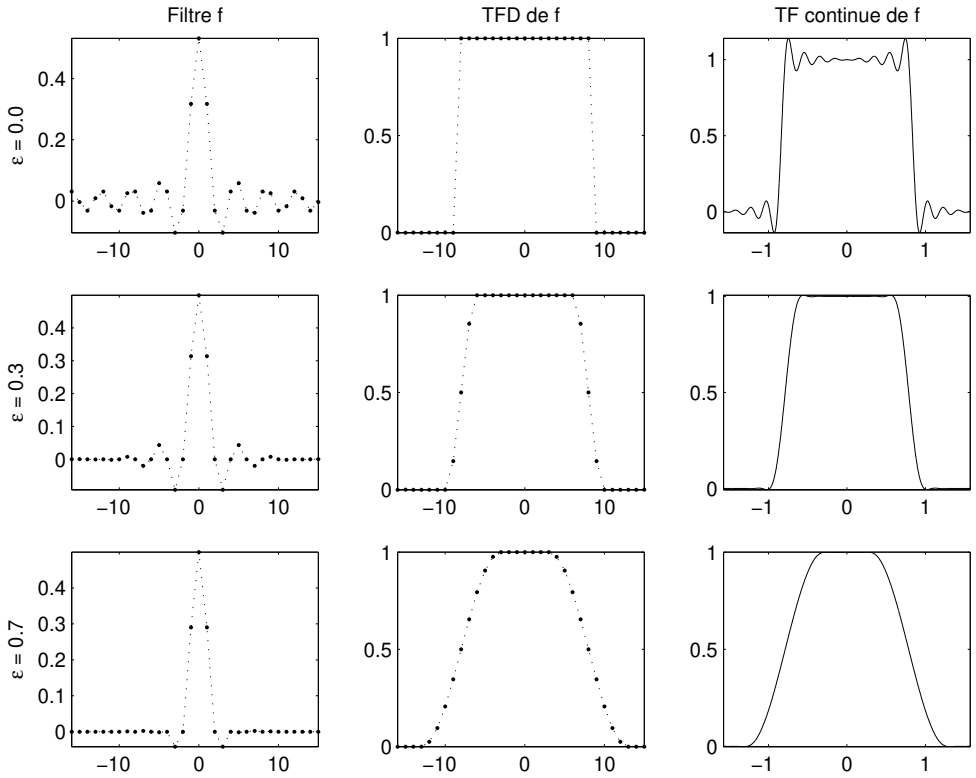


Figure 4.16

Low-pass filters with increasingly wide raised-cosine transitions. The columns show the centered impulse response, the prescribed DFT samples, and the densely sampled FIR frequency response.

Book page 96; original figure 4.16.

Previous edition



Revised edition

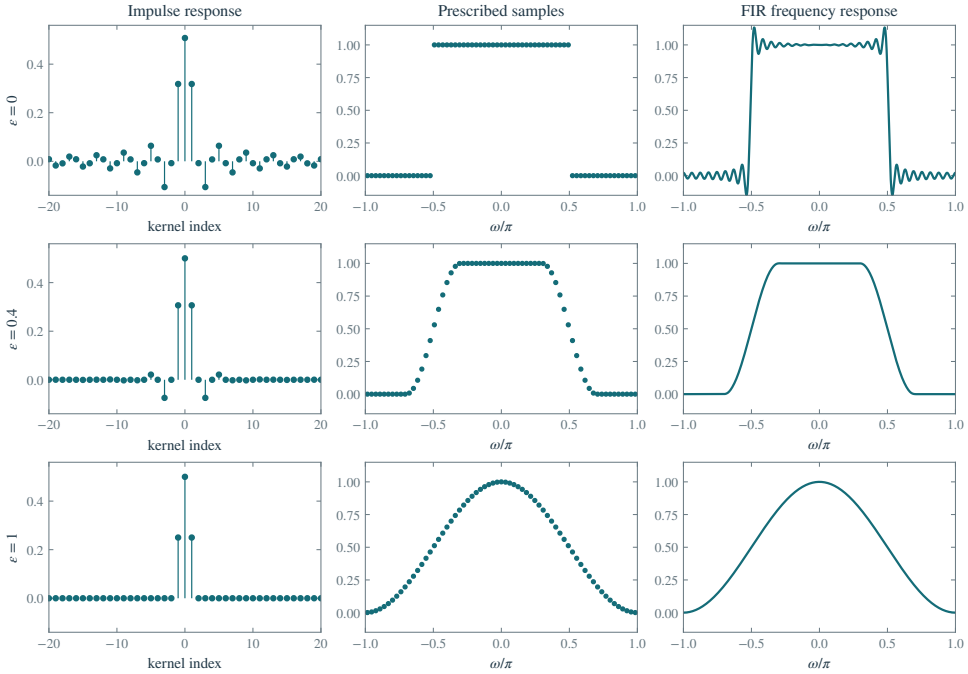
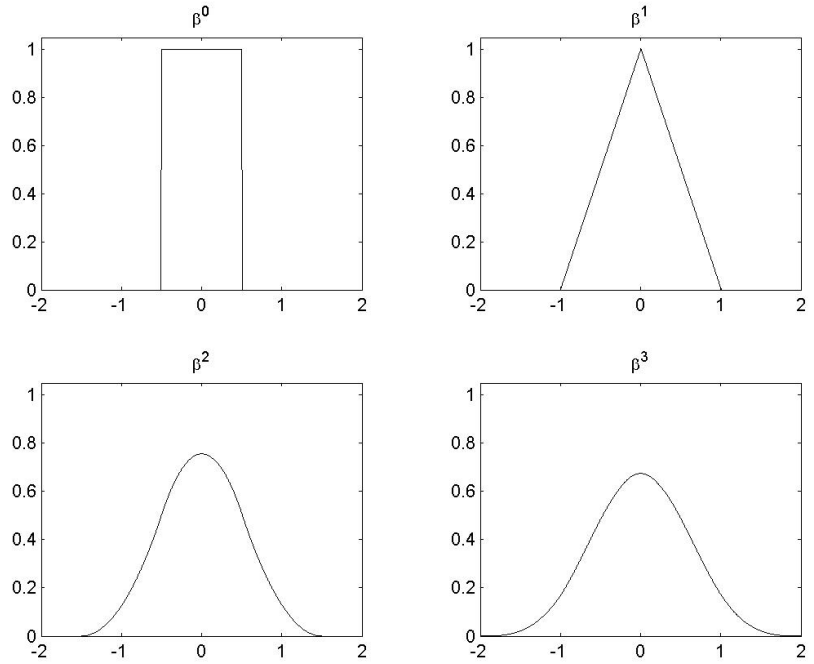


Figure 4.17

Centered B-splines of degrees zero through three. Their support is $[-(n+1)/2, (n+1)/2]$.

Book page 97; original figure 4.17.

Previous edition



Revised edition

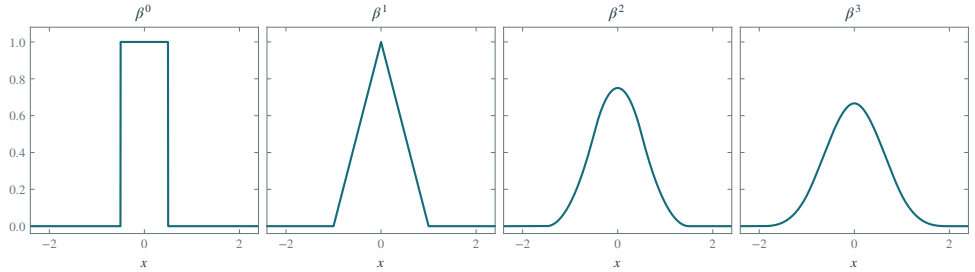
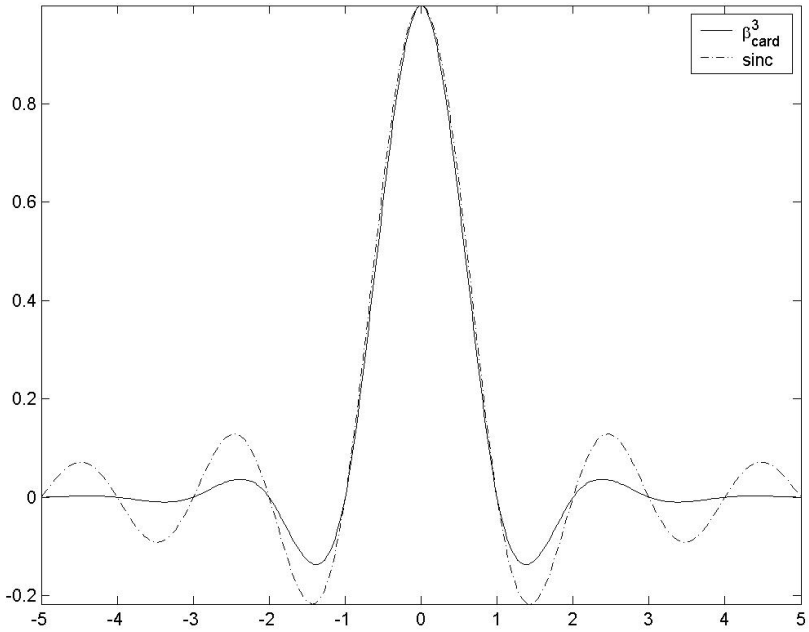


Figure 4.18

The cubic cardinal kernel, the cubic B-spline, and sinc. The cardinal kernel and sinc equal one at zero and vanish at every other integer; the cubic B-spline requires a prefilter.

Book page 97; original figure 4.18.

Previous edition



Revised edition

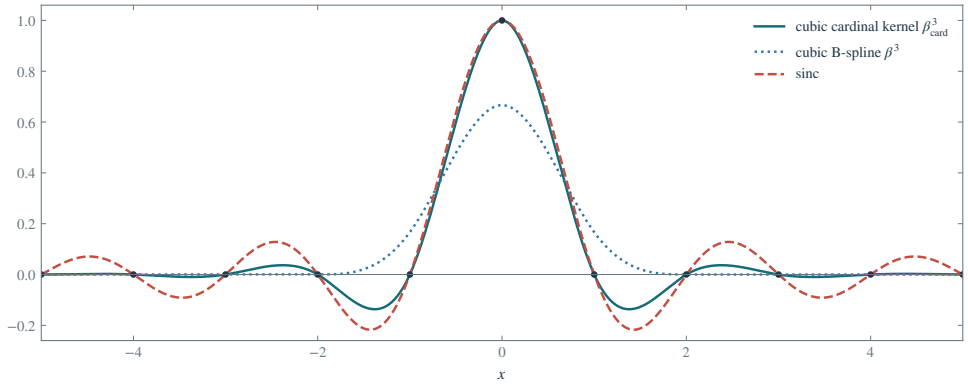
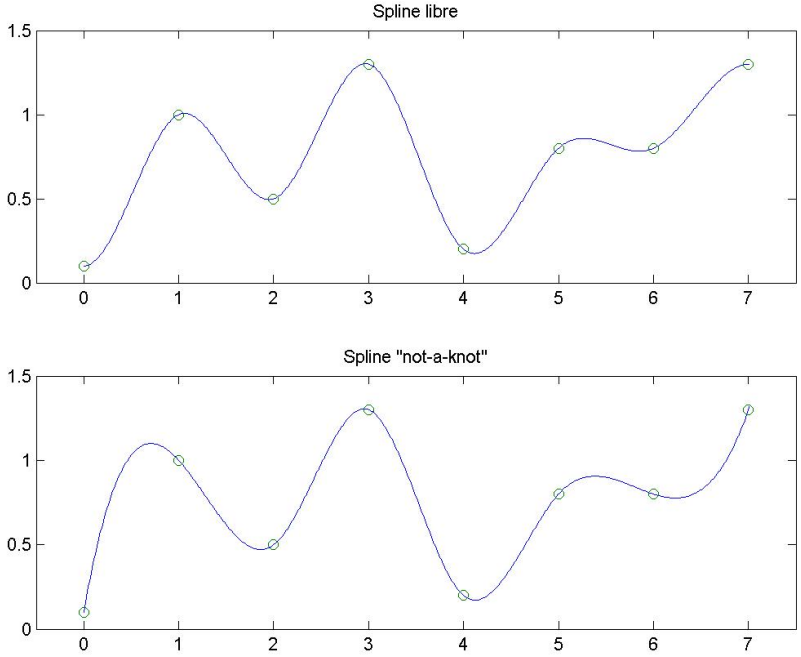


Figure 4.19

Cubic interpolation of the same eight samples with natural and not-a-knot boundary conditions. Natural conditions set the endpoint second derivatives to zero.

Book page 98; original figure 4.19.

Previous edition



Revised edition

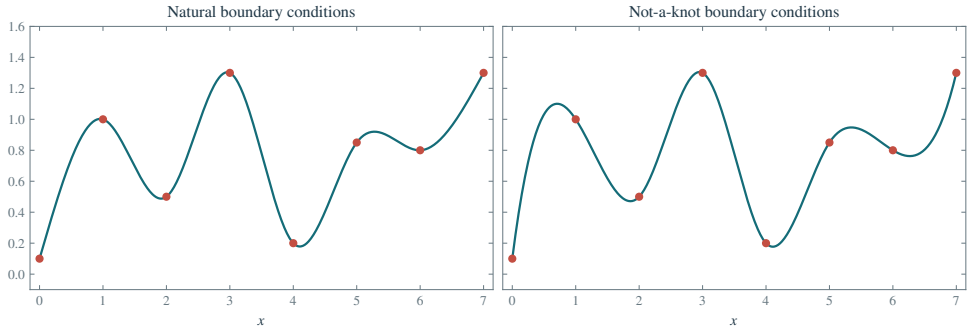
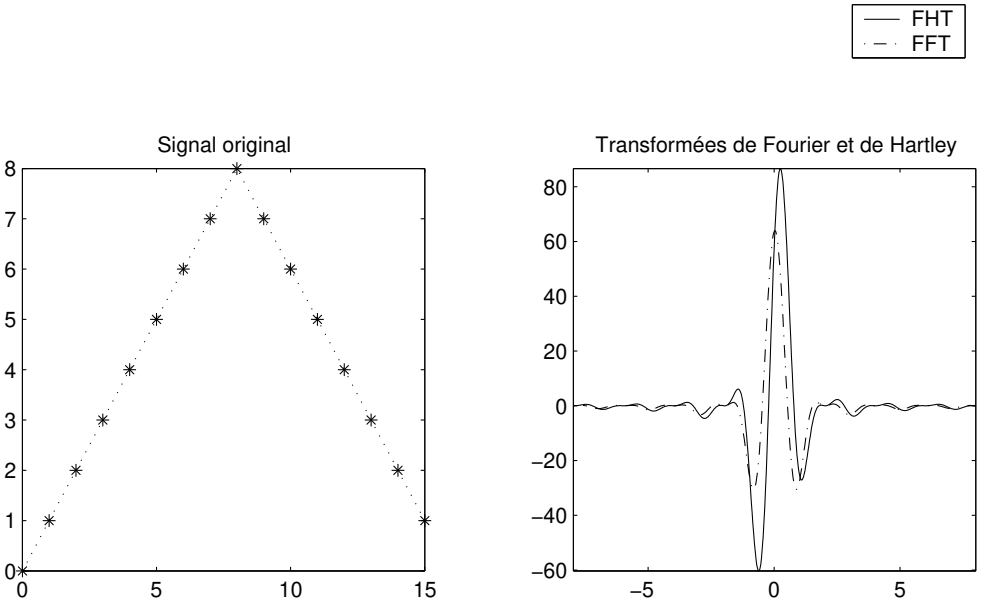


Figure 5.1

A triangular signal and its densely evaluated Fourier and Hartley spectra. The Hartley spectrum equals the real part minus the imaginary part of the DFT. Zero padding changes only the frequency grid.

Book page 111; original figure 5.1.

Previous edition



Revised edition

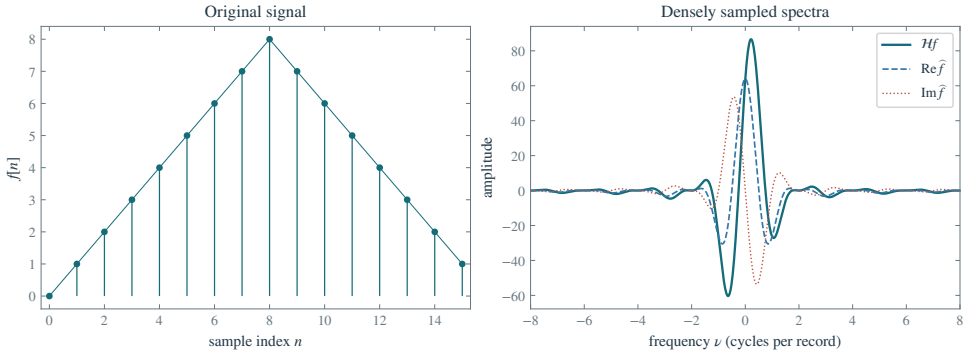
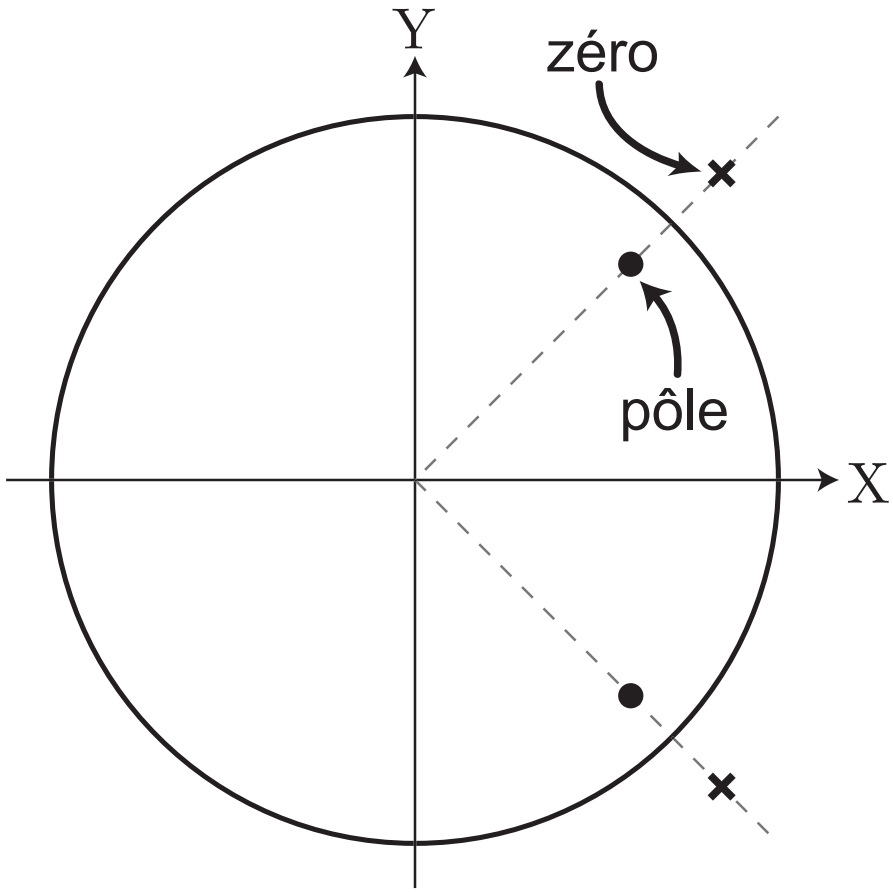


Figure 5.2

Zeros on the unit circle and poles at radius 0.9 for a notch filter at angular frequencies $\pm\pi/4$. Circles mark zeros; crosses mark poles.

Book page 113; original figure 5.2.

Previous edition



Revised edition

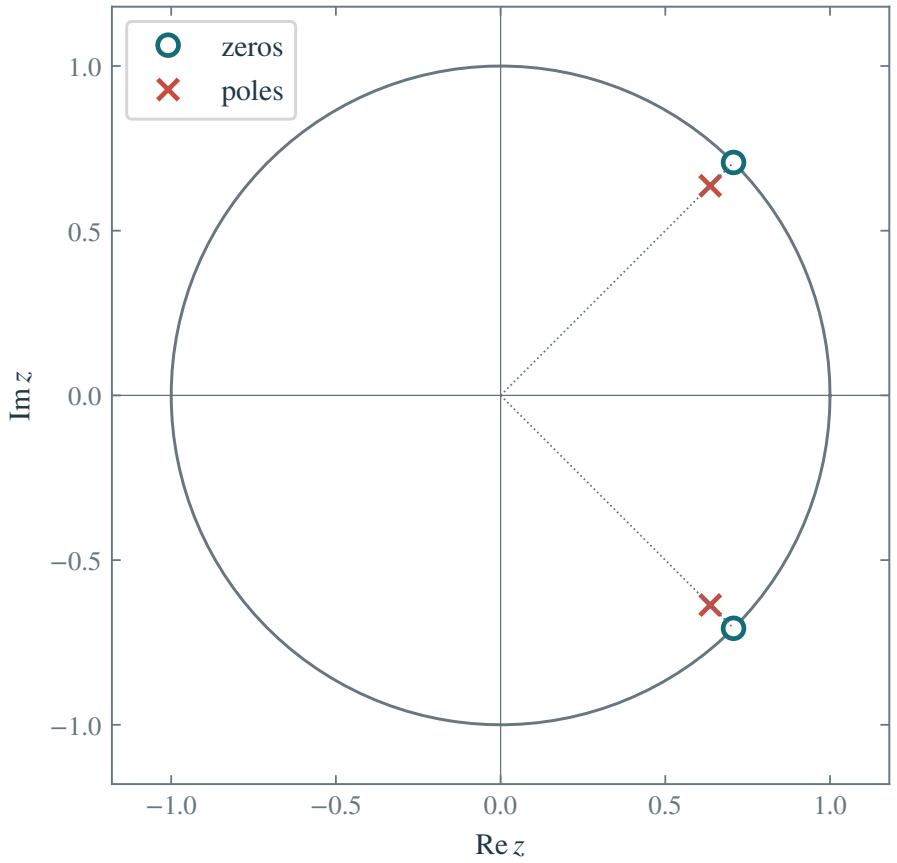
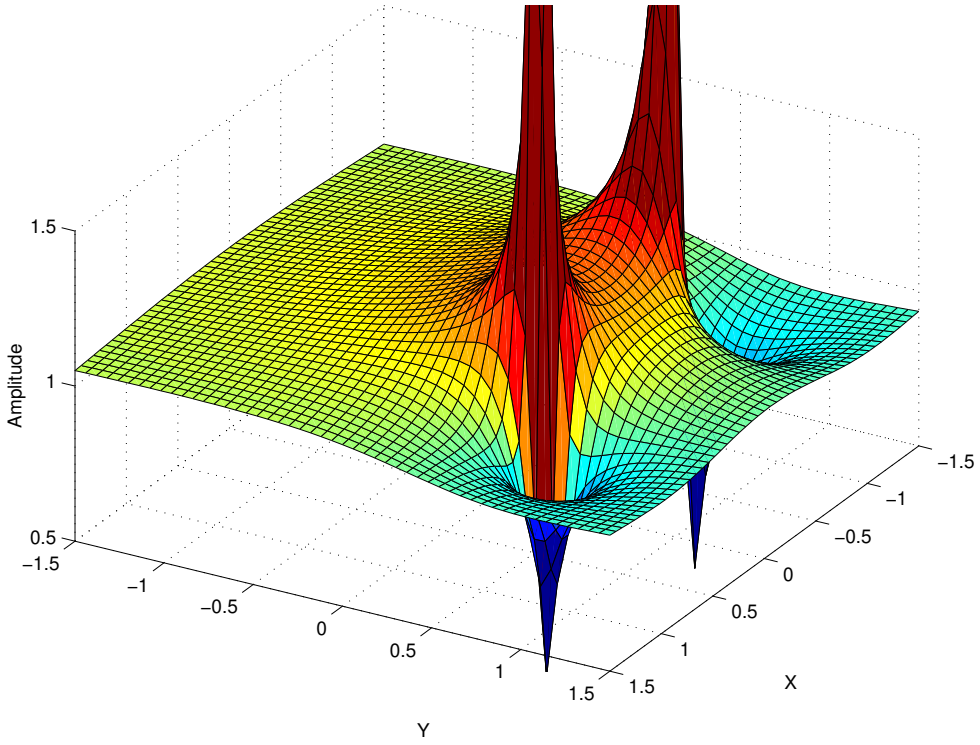


Figure 5.3

Magnitude of the rational notch-filter transfer function in the complex plane. Values above two are clipped to reveal the zeros and pole neighborhoods. The causal Laurent series converges for $|z| > 0.9$.

Book page 114; original figure 5.3.

Previous edition



Revised edition

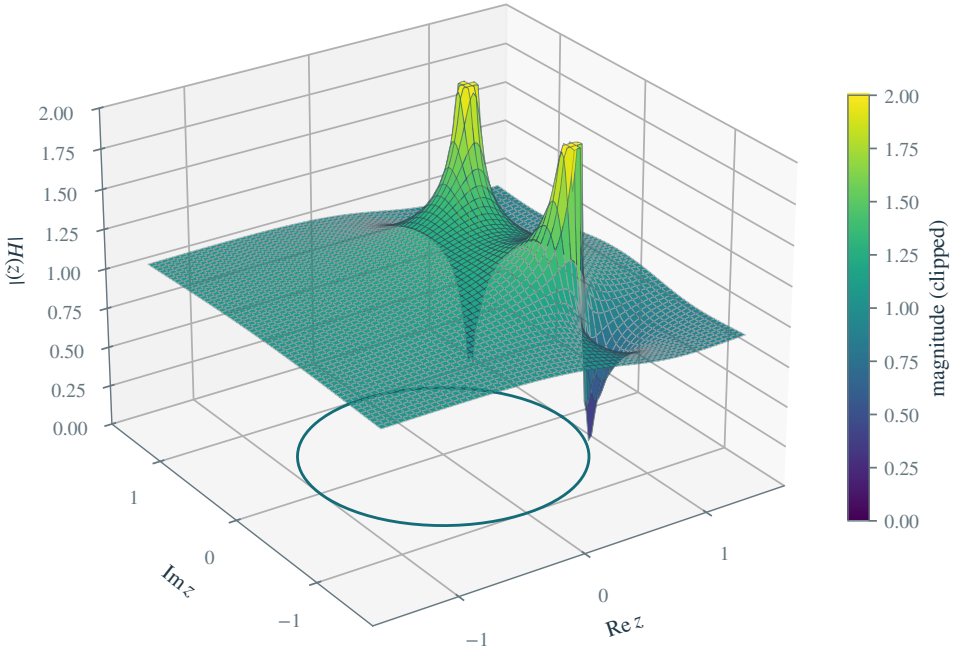
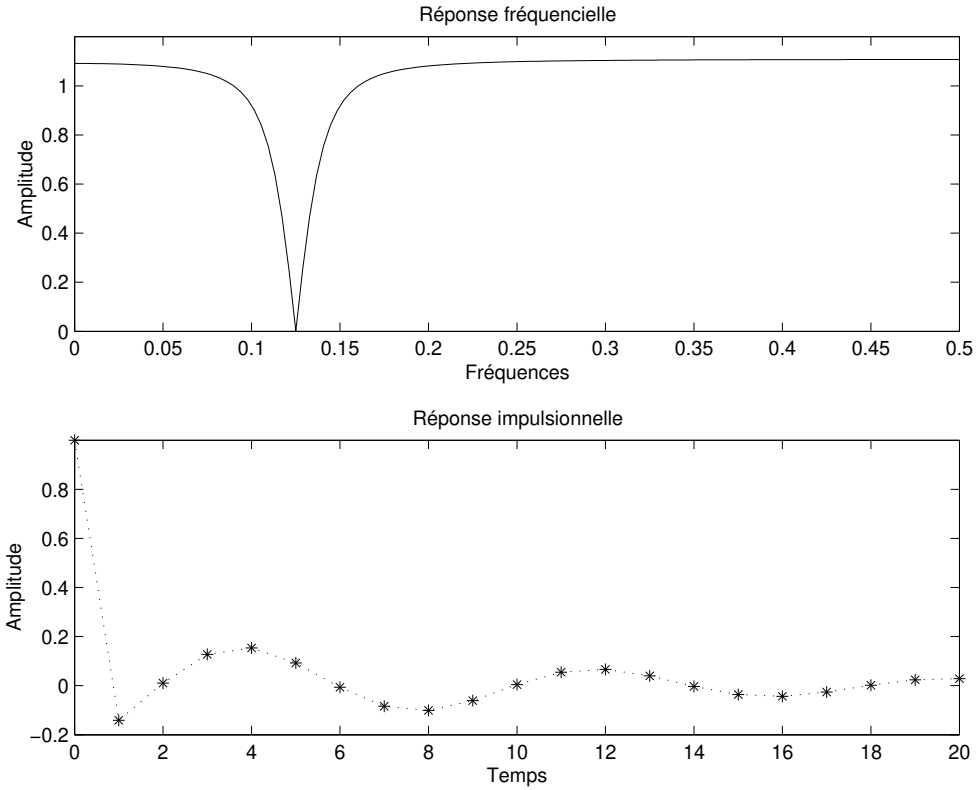


Figure 5.4

Frequency-response magnitude and causal impulse response of the notch filter. The response vanishes at angular frequency $\pi/4$.

Book page 114; original figure 5.4.

Previous edition



Revised edition

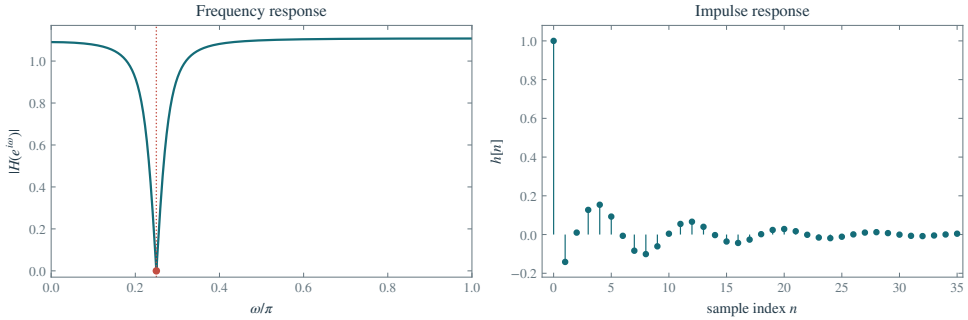
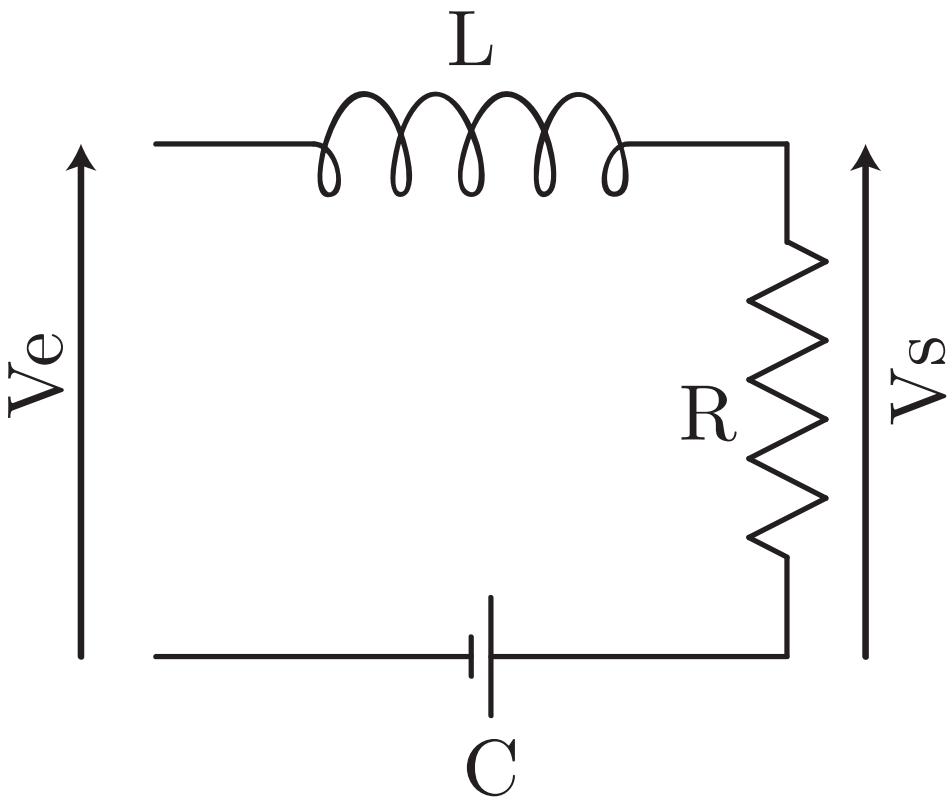


Figure 5.5

A series RLC circuit, with the output voltage V_s measured across the resistor and the input V_e across the two left terminals.

Book page 115; original figure 5.5.

Previous edition



Revised edition

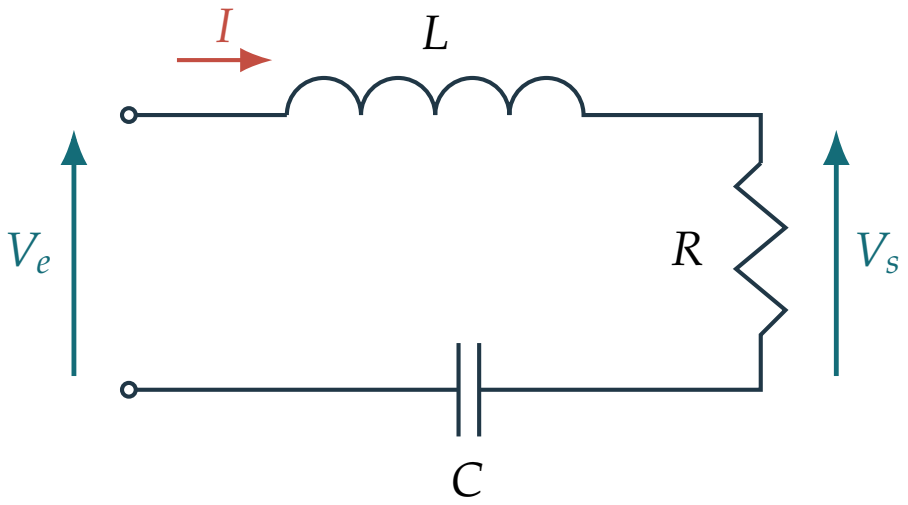
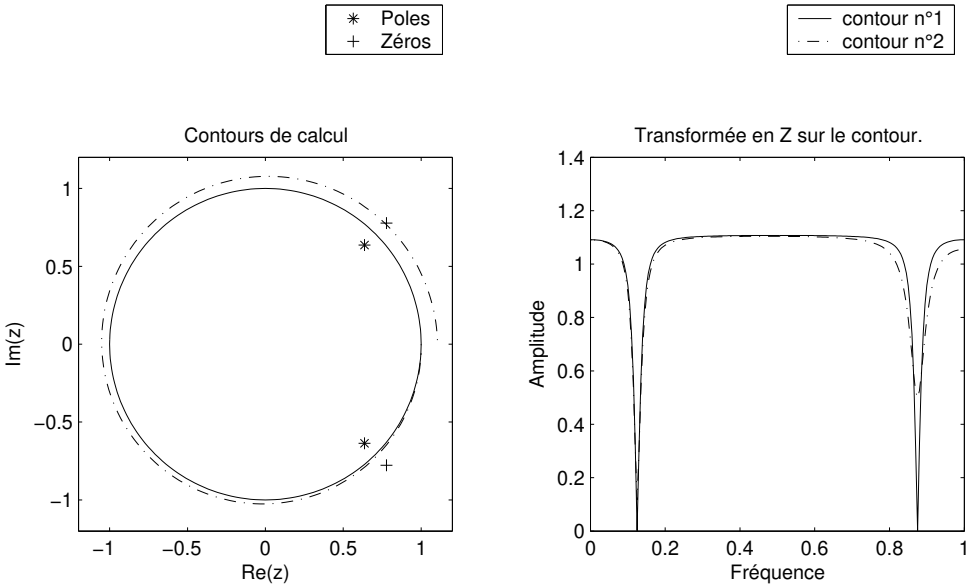


Figure 5.6

The notch-filter transfer function evaluated on a unit circle and an outward logarithmic spiral. Both contours lie in the causal region of convergence; leaving the unit circle changes the depth of the notches.

Book page 116; original figure 5.6.

Previous edition



Revised edition

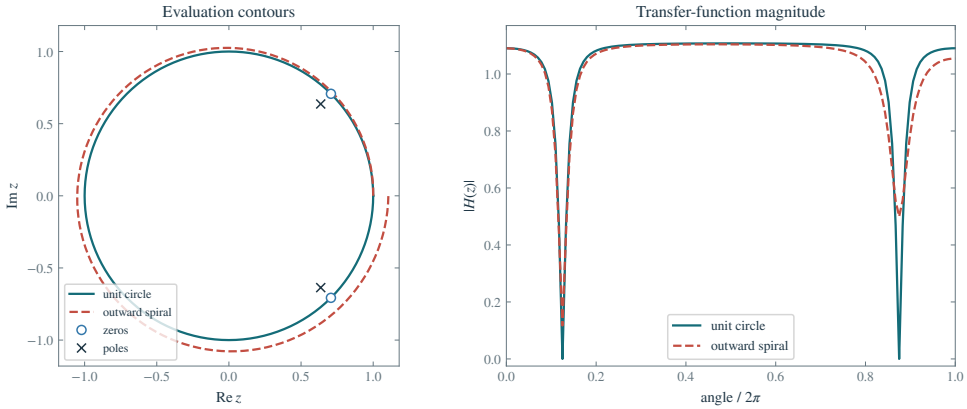
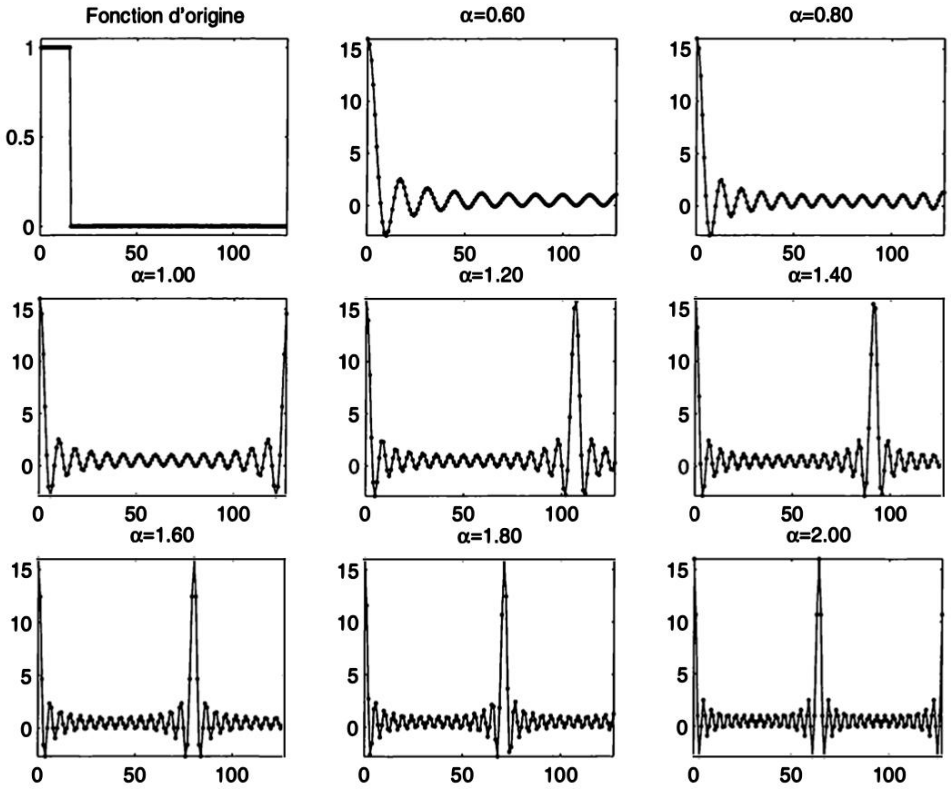


Figure 5.7

Scaled-frequency Fourier transforms of an interval indicator. The real and imaginary parts share the same scale. Changing α changes the sampled frequency grid.

Book page 118; original figure 5.7.

Previous edition



Revised edition

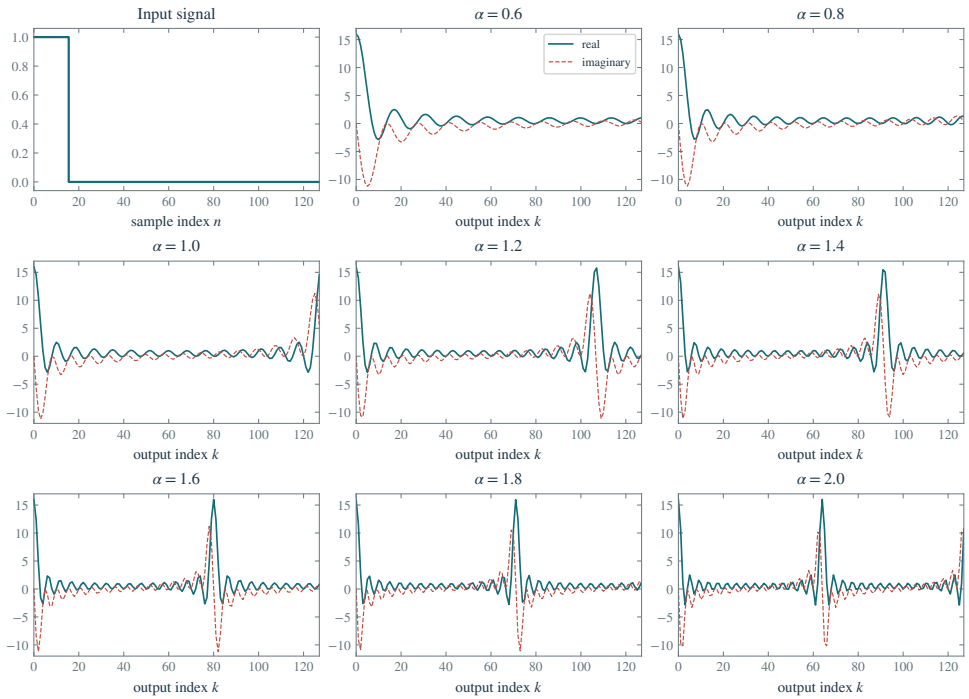
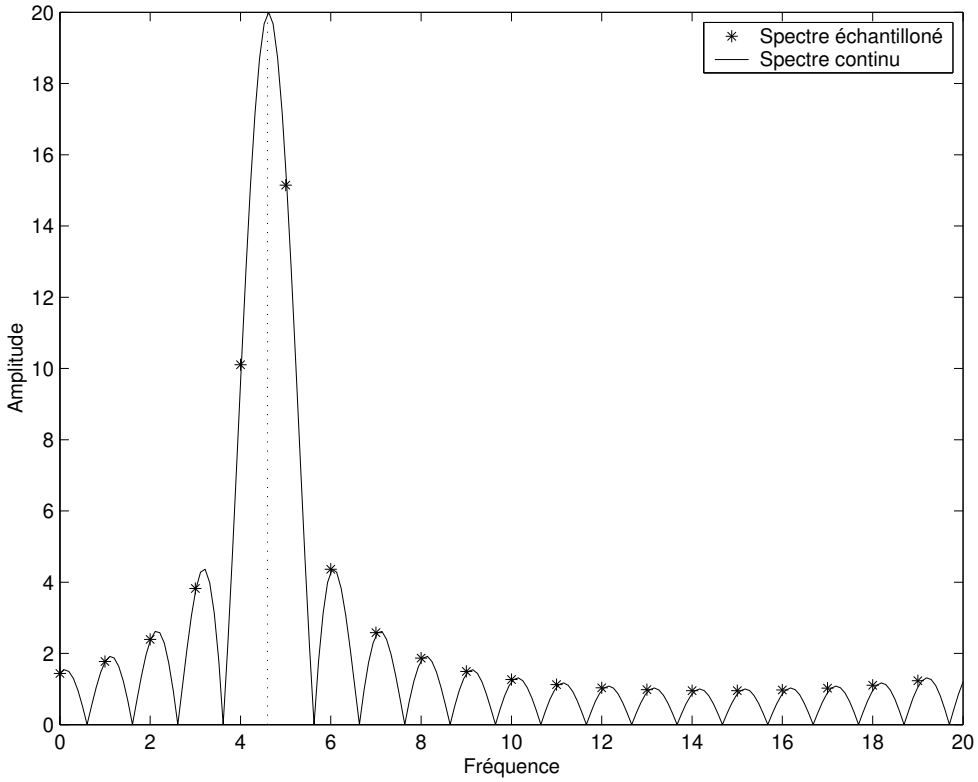


Figure 5.8

Finite-record spectra of sinusoids with integer and noninteger numbers of cycles. The continuous curves evaluate the finite sum; the dots are DFT samples. The vertical line marks the generating frequency.

Book page 118; original figure 5.8.

Previous edition



Revised edition

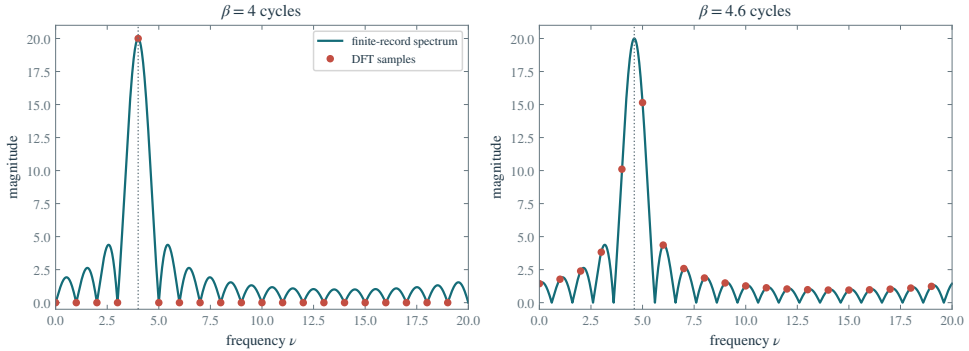
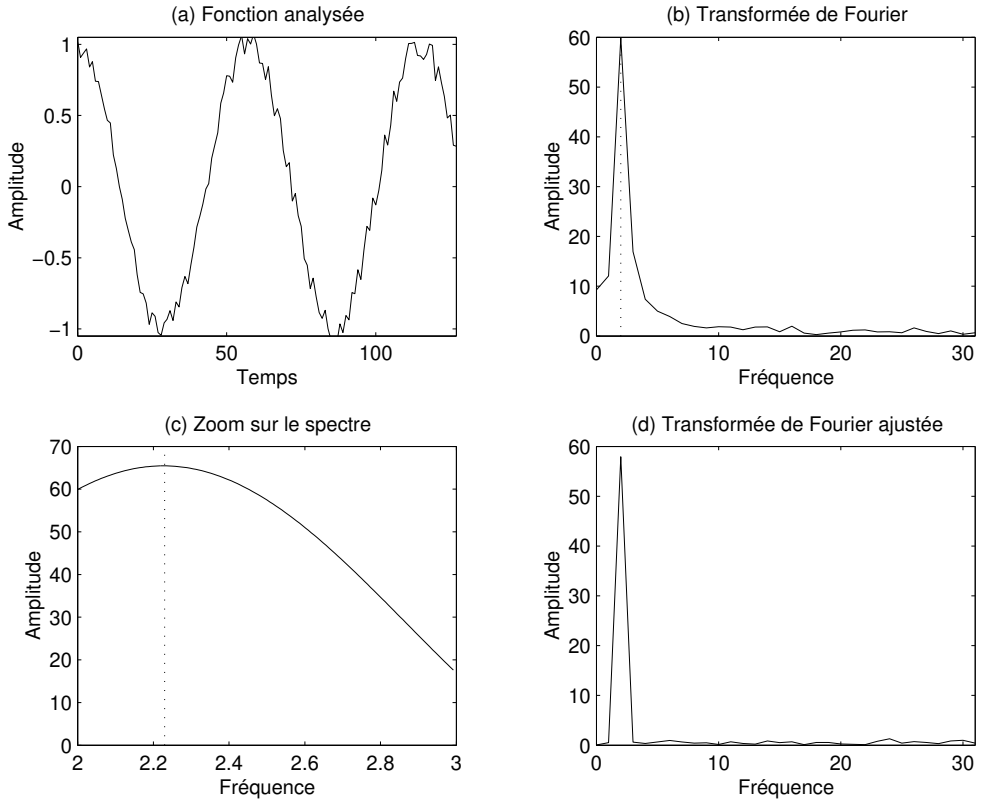


Figure 5.9

A noisy cosine, its coarse spectrum, a local spectral zoom, and an adjusted spectrum after shortening the record using the estimated peak. Spectral magnitudes are divided by the respective record lengths. The generating frequency is distinguished from its estimate.

Book page 119; original figure 5.9.

Previous edition



Revised edition

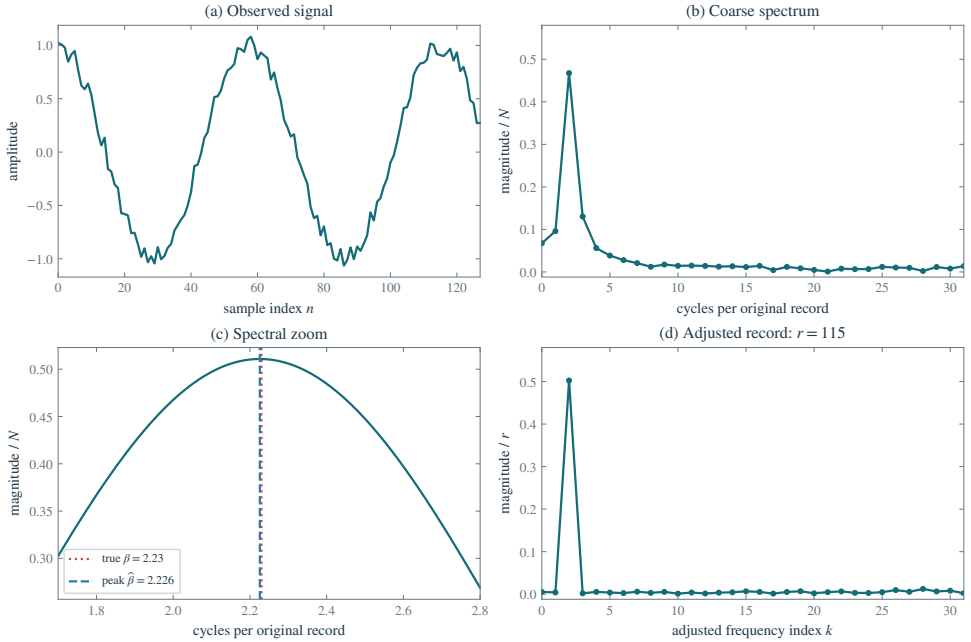
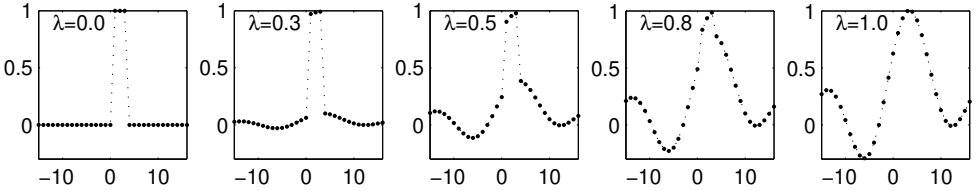


Figure 5.10

Real and imaginary parts of fixed-branch unitary Hartley powers $V^t f$, with $V = \mathcal{H}/\sqrt{N}$. All panels use the same amplitude scale and preserve the norm.

Book page 120; original figure 5.10.

Previous edition



Revised edition

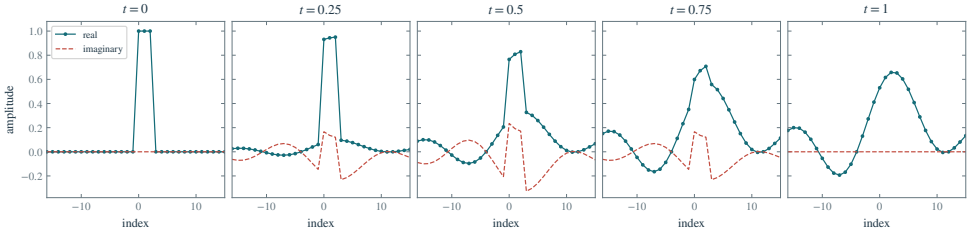
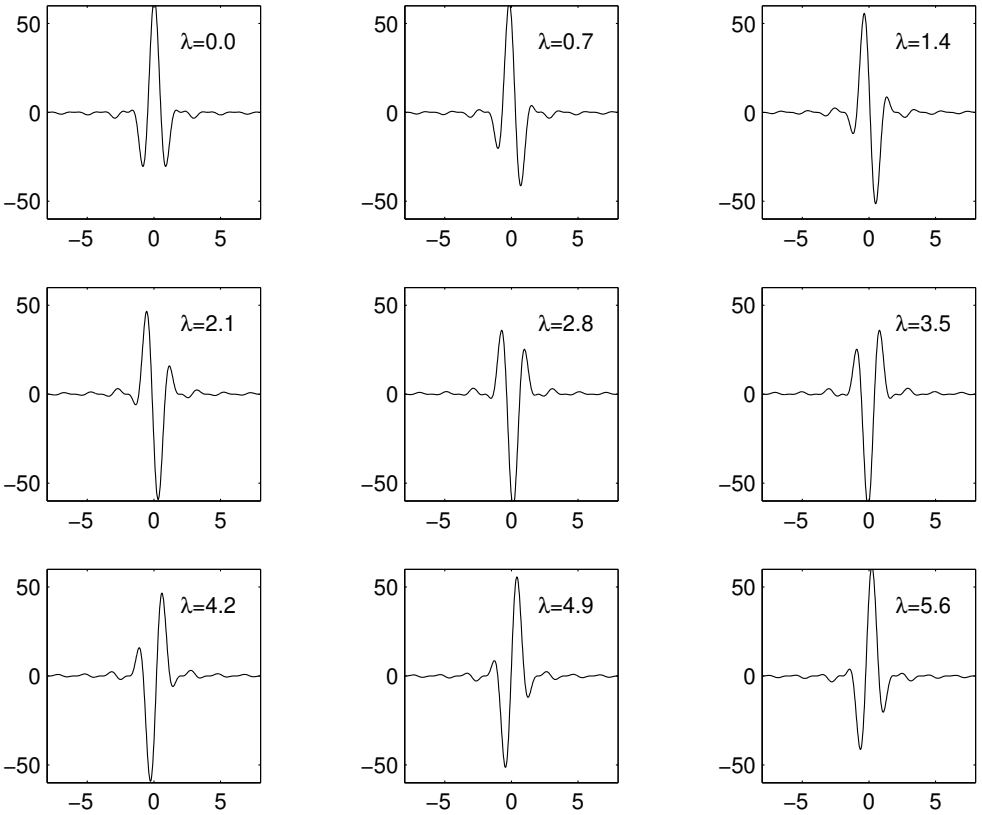


Figure 5.11

Generalized Hartley transforms of the same triangular signal. The phase λ changes the relative cosine and sine contributions; the two endpoint panels illustrate the 2π periodicity.

Book page 121; original figure 5.11.

Previous edition



Revised edition

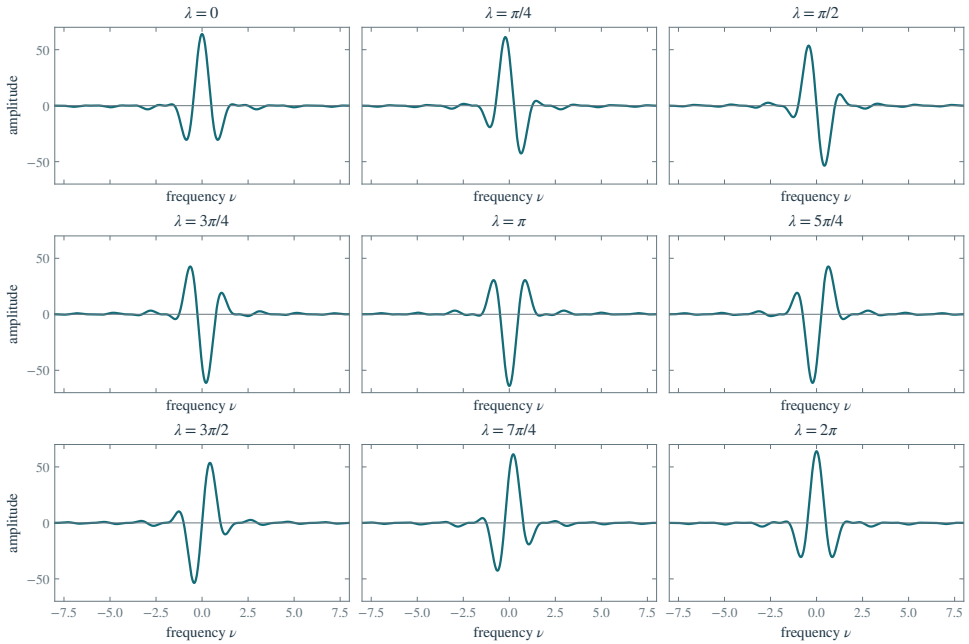
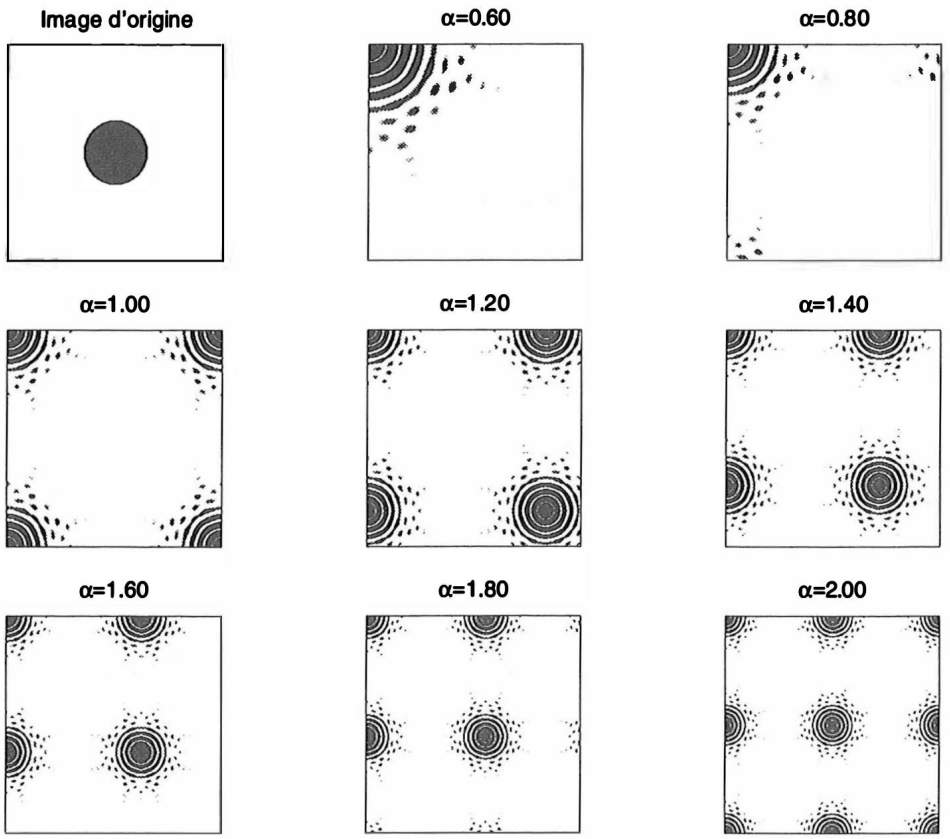


Figure 5.12

Magnitudes of scaled-frequency two-dimensional Fourier transforms of a disk. Spectral panels share the contrast mapping $\log(1 + 100|G_2|/\sum f)/\log(101)$, shown dark on white; the input panel uses ordinary grayscale.

Book page 123; original figure 5.12.

Previous edition



Revised edition

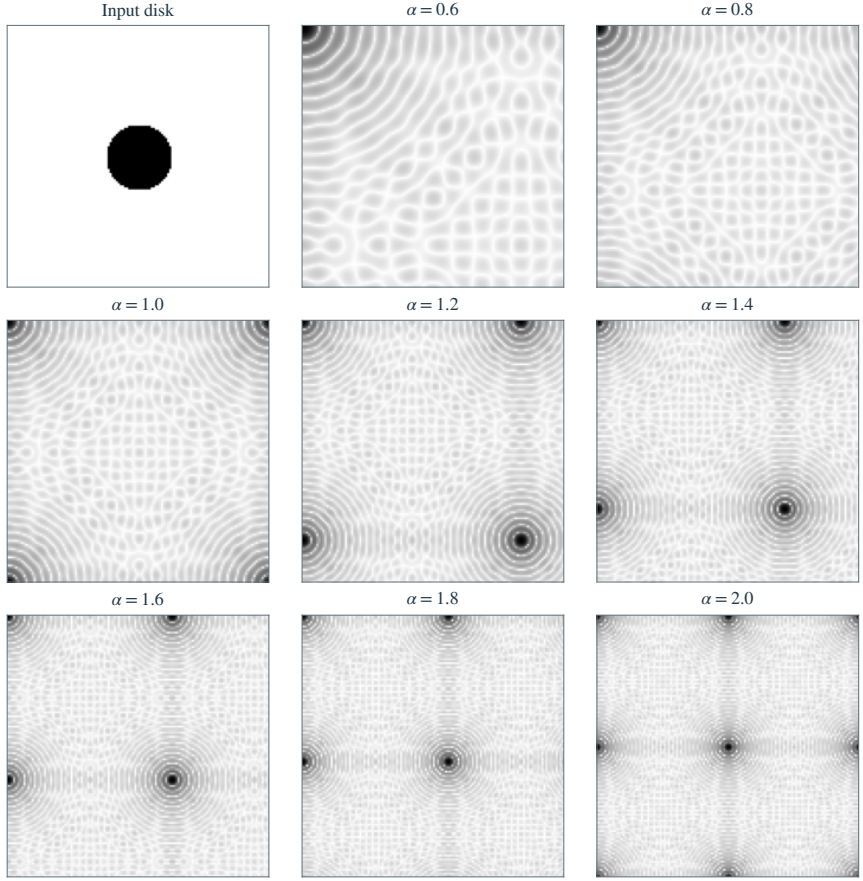
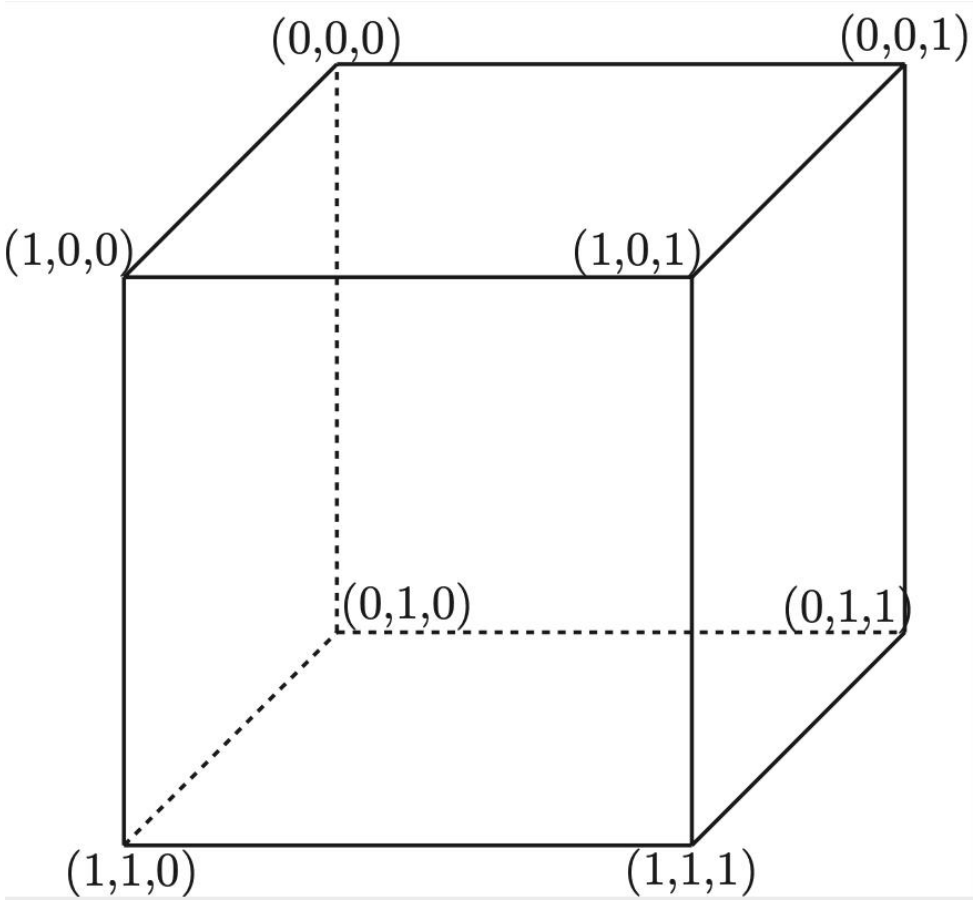


Figure 6.1

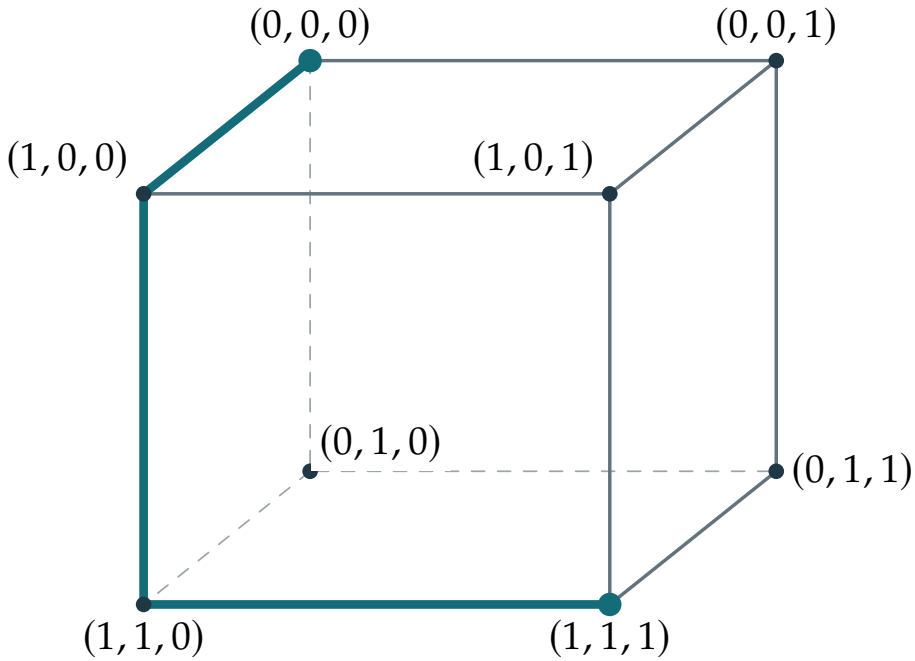
Hamming distance on $\mathbb{F}_2^3$. Each cube edge changes one coordinate; the highlighted shortest path has three edges.

Book page 136; original figure 6.1.

Previous edition



Revised edition



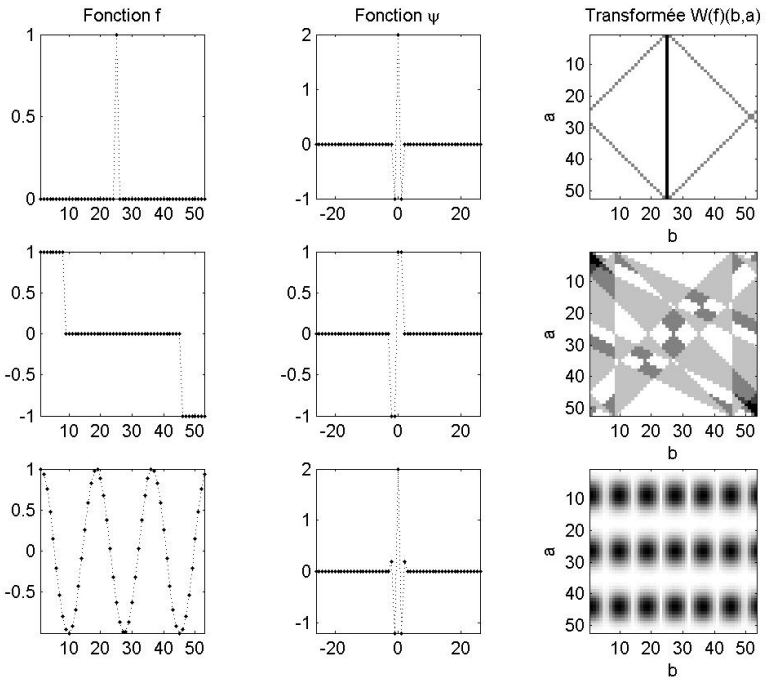
$$d\big((0,0,0),(1,1,1)\big) = 3$$

Figure 7.1

Wavelets on $\mathbb{F}_{53}$: the signal, the centered wavelet, and the magnitude of its affine-group transform. Translation b is horizontal and nonzero dilation a is vertical. Each row has its own indicated magnitude scale. The first signal has nonzero mean, which the mean-zero wavelet removes.

Book page 169; original figure 7.1.

Previous edition



Revised edition

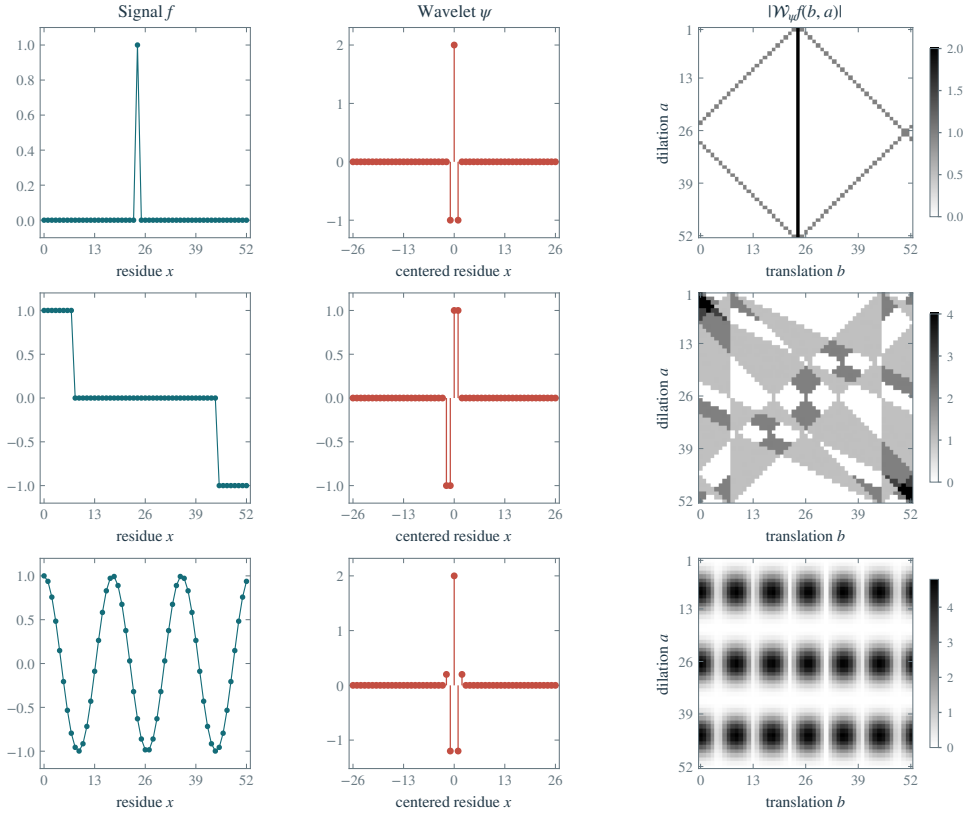
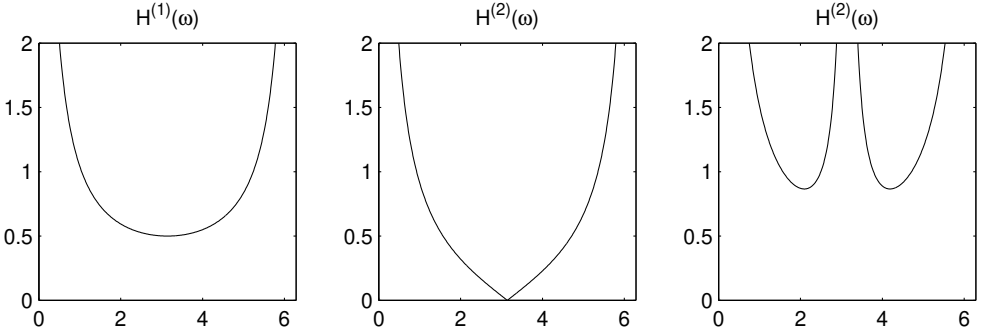


Figure 5.13

Magnitudes of the rectangle, trapezoidal, and Simpson integration filters for $\Delta = 1$. The poles at zero frequency and the additional Simpson pole at the Nyquist frequency are excluded from the plotted grid.

Book page 127; original figure A.1.

Previous edition



Revised edition

